Probabilistic Combinatorics

Kon Yi

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Preface

These are my notes for *Probabilistic Combinatorics* (MATH 5390), taught by Shagnik Das at National Taiwan University in Fall 2026.

The course is about the *probabilistic method* in extremal combinatorics. The idea is simple: to show that a structure with some desired property exists, build a probability space over all candidate structures and prove that a random one has the property with positive probability. Such an argument rarely says *which* structure works, only that one must; yet with sharper tools from probability theory it becomes remarkably powerful, and it has settled countless problems in combinatorics. Rather than develop the probability first, the course builds the toolkit gradually, introducing each technique through applications from different corners of combinatorics. Roughly, the topics fall into three groups.

- **Basic probabilistic method**: quick applications, linearity of expectation, the method of alterations, the second moment method, and the Rödl nibble.
- **Advanced methods**: the Lovász Local Lemma, martingales and concentration inequalities, Janson's inequalities, and quasirandomness.
- **Modern advances** (time permitting): hypergraph containers, spread distributions and sunflowers, and fractional versus expectation thresholds.

Prerequisites. The course assumes Combinatorics I and II or Graph Theory I and II, together with undergraduate probability and calculus.

Sources. Most of the course follows Alon and Spencer, *The Probabilistic Method*. These notes are based on the lectures; any errors in them are mine.

Assessment. Homework is assigned every two weeks and done in pairs. There is a closed-book midterm (Week 8) and final (Week 16), each asking both for definitions, statements and proofs of theorems from lectures and for solutions to problems in the style of the homework. The grade is the higher of 30% homework + 30% midterm + 40% final and 30% homework + 70% final.

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1. Extremal Combinatorics

Lecture 1

§1.1. Mantel's Theorem

15 Sep.

Question 1.1.1. How large can a structure be if it satisfies certain properties?

Example 1.1.2 (Mantel, 1907)**.** How many edges can a graph in n vertices have if it doesn't contain a triangle?

To show the answer is $f(n)$, we need to do two things.

(1) Show there is a triangle-free graph on n vertices with $f(n)$ edges.

(2) Prove that any n-vertex graph with $> f(n)$ edges must have a triangle.

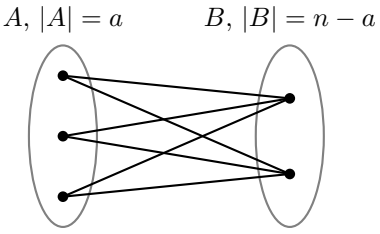
Theorem 1.1.3 (Mantel, 1907)

The largest n-vertex triangle-free graph has $\left\lfloor \frac{n^2}{4} \right\rfloor$ edges.

Remark 1.1.4. The complete graph K_n on n vertices has $\binom{n}{2} \approx \frac{1}{2}n^2$ edges.

proof of Mantel's theorem.

(1) Bipartite graphs are triangle-free. Thus, we can split the vertices into two parts, take all edges between the parts. Suppose first part is of a vertices and the other one is of $n-a$ vertices.



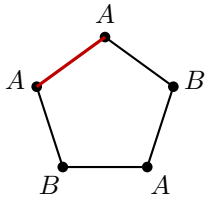
$K_{a,n-a}$: all edges between the two parts

Note that this construction is triangle-free becuase given any three vertices, two fall in same part, and thus there is no edge between them. Note that the number of edges is $a(n-a) \leq \left\lfloor \frac{n^2}{4} \right\rfloor$, where the equality holds iff $a = \lfloor \frac{n}{2} \rfloor$ or $\lceil \frac{n}{2} \rceil$.

Note 1.1.5. Every bipartite graph is triangle-free, but a triangle-free graph need not be bipartite. For example, C_5 consists of the five edges of a pentagon, with no diagonals, so it contains no triangle.
To see why C_5 is not bipartite, suppose that its vertices could be partitioned into two sets A and B, with every edge joining different sets. Put the top vertex in A; we can always exchange the names of the two sets to do this. Moving clockwise around the pentagon, each vertex must belong to the opposite set from the previous vertex. Thus the five vertices are forced to belong to

$$A, \quad B, \quad A, \quad B, \quad A.$$

But the last vertex is adjacent to the first vertex, and both are in A, a contradiction. Hence no bipartition exists.



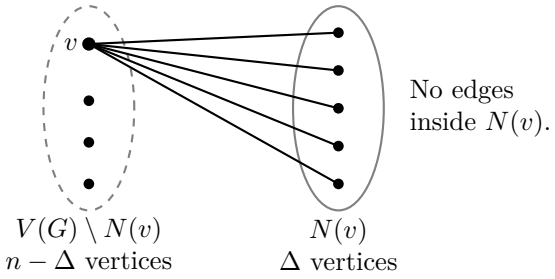
The highlighted edge joins two vertices forced into A.

The labels A, B indicate the proposed parts, not vertex names. The red edge is an ordinary edge of C_5, highlighted only to show the contradiction; this is not an edge-coloring problem.

(2)

Claim 1.1.1. If G is an n-vertex, triangle-free graph, then $e(G) \leq \left\lfloor \frac{n^2}{4} \right\rfloor$.

proof of claim. Let $v \in V(G)$ be a vertex of maximum degree Δ. Since G is triangle-free, the neighborhood of v is independent (no edges).



Thus, every edge contains at least one vertex of $V(G) \setminus N(v)$, where this set includes v itself. Thus,

$$e(G) \leq \sum_{u:u\nsim v} \#\{\text{edges containing } u\} = \sum_{u:u\nsim v} \deg(u) \leq \sum_{u:u\nsim v} \Delta = \Delta(n-\Delta) \leq \left\lfloor \frac{n^2}{4} \right\rfloor.$$

■

Remark 1.1.6. Up to isomorphism, the unique n-vertex triangle-free graph with $\lfloor n^2/4 \rfloor$ edges is the balanced complete bipartite graph

$$K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}.$$

To prove uniqueness, suppose that G attains this bound. As in the proof above, choose a vertex v of maximum degree Δ, and write

$$A = N(v), \qquad B = V(G) \setminus N(v).$$

Then $|A| = \Delta$, $|B| = n - \Delta$, and A is independent because G is triangle-free. The inequalities in the proof become

$$\left\lfloor \frac{n^2}{4} \right\rfloor = e(G) \leq \sum_{u \in B} \deg(u) \leq \Delta(n - \Delta) \leq \left\lfloor \frac{n^2}{4} \right\rfloor.$$

Since the first and last quantities agree, every inequality must be an equality. Each equality gives a structural restriction:

(i) **There are no edges inside B.** Let $e(B)$ denote the number of edges with both endpoints in B. Since there are no edges inside A, the degree sum over B counts every edge between A and B once and every edge inside B twice. Thus

$$\sum_{u \in B} \deg(u) = e(G) + e(B).$$

Equality $e(G) = \sum_{u \in B} \deg(u)$ forces $e(B) = 0$. Both A and B are therefore independent, so G is bipartite with these parts.

(ii) **Every possible edge between the parts is present.** Each $u \in B$ has $\deg(u) \leq \Delta$. Equality

$$\sum_{u \in B} \deg(u) = |B|\Delta$$

forces $\deg(u) = \Delta$ for every $u \in B$; otherwise the sum would be strictly smaller. All neighbors of u lie in A, which has exactly Δ vertices. Hence every vertex of B is adjacent to every vertex of A, giving $G = K_{\Delta, n-\Delta}$.

(iii) **The two parts are as equal in size as possible.** Finally, $\Delta(n - \Delta) = \lfloor n^2/4 \rfloor$. Using

$$\Delta(n - \Delta) = \frac{n^2 - (2\Delta - n)^2}{4},$$

we obtain $2\Delta - n = 0$ when n is even, and $2\Delta - n = \pm 1$ when n is odd. Thus the part sizes are $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$.

Conversely, this balanced complete bipartite graph is triangle-free and has exactly $\lfloor n^2/4 \rfloor$ edges, as shown in the construction. This proves both extremality and uniqueness up to relabeling vertices.

■

§1.2. Ramsey Numbers

Definition 1.2.1. The **Ramsey number** $R(k)$ is the smallest n s.t. every red-/blue-coloring of the edges of K_n contains a monochromatic copy of K_k.

Theorem 1.2.2 (Ramsey, 1930)
$R(k)$ is finite.

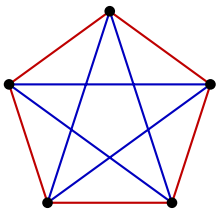
We can simply show $R(1) = 1, R(2) = 2$, and for $R(3)$, it is a little harder to show $R(3) = 6$.

Claim 1.2.1. $R(1) = 1$, $R(2) = 2$, and $R(3) = 6$.



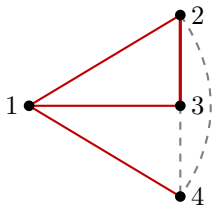
K_1 K_2: one monochromatic edge K_3: need not be monochromatic

We can first show $R(3) > 5$ by giving an example:

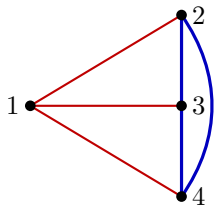


Red: the outer 5-cycle.
Blue: the diagonal 5-cycle.
Neither color contains a triangle.

We can also show $R(3) \leq 6$ by pigeonhole principle: Consider K_6 with vertices indexed by $1, 2, 3, 4, 5, 6$, we know there exists 3 edges incident to vertex 1 are of same colors, WLOG, say vertices 2, 3, 4 are of red. Then we can show that no matter how we color $\{2, 3\}, \{3, 4\}, \{2, 4\}$, there must be a monochromatic triangle.



A red edge among $2, 3, 4$ gives a red triangle with 1.



Otherwise, $2, 3, 4$ form a blue triangle.

Only the relevant edges are shown; vertices $5, 6$ and all other edges can be ignored. Hence every red-/blue-coloring of K_6 has a monochromatic triangle, so $R(3) = 6$.

Theorem 1.2.3 (Erdős–Szekeres, 1935)

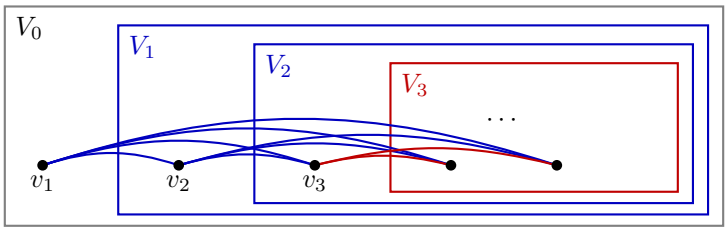
$$R(k) \leq 4^k.$$

Proof. Let $k \geq 2$, and consider any red-/blue-coloring of $K_{2^{2k}}$. Set $V_0 = V(K_{2^{2k}})$. We successively choose vertices $v_1, v_2, \ldots$, colors $c_1, c_2, \ldots$, and nested sets $V_0 \supseteq V_1 \supseteq V_2 \supseteq \cdots$.

At step i, choose $v_i \in V_{i-1}$. Among the edges from v_i to $V_{i-1} \setminus \{v_i\}$, at least half have the same color, say c_i. Let V_i be the set of endpoints of these edges. Then every edge from v_i to V_i has color c_i, and

$$|V_i| \geq \left\lceil \frac{|V_{i-1}| - 1}{2} \right\rceil \geq 2^{2k-i}.$$

The last inequality follows inductively from $|V_0| = 2^{2k}$; the ceiling accounts for removing v_i before keeping a color class.



Keep a monochromatic neighborhood at each step.

In particular, for $j < \ell \leq i$, we have $v_\ell \in V_j$, so the edge $v_j v_\ell$ has color c_j. Also, every edge from v_j to V_i has color c_j.

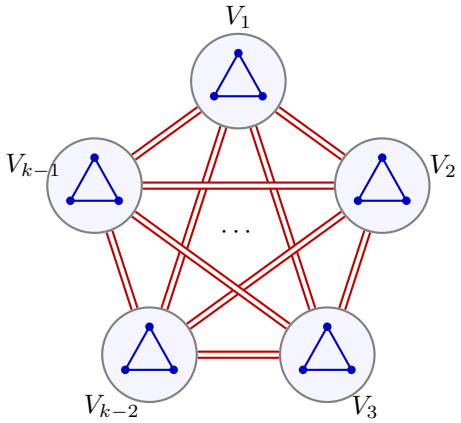
Repeat until $i = 2k$. This is possible since $|V_{2k-1}| \geq 2$, and hence V_{2k} still contains at least one vertex. We obtain

vertex	v_1	v_2	v_3	$\cdots$	v_{2k-1}	v_{2k}
color	c_1	c_2	c_3	$\cdots$	c_{2k-1}	c_{2k}

By the pigeonhole principle, at least k of the colors $c_1, \ldots, c_{2k}$ are equal. Choose indices $i_1 < \cdots < i_k$ with this common color. For any $r < s$, the edge $v_{i_r} v_{i_s}$ has color c_{i_r}, so these k vertices form a monochromatic K_k. Thus $R(k) \leq 2^{2k} = 4^k$. The case $k = 1$ is immediate. ∎

Remark 1.2.4. More careful counting gives $R(k) \leq \binom{2k-2}{k-1}$.

For the lower bound, we can use a smart construction to show $R(k) > (k-1)^2$: For $k \geq 2$, partition the vertices of $K_{(k-1)^2}$ into $k-1$ parts $V_1, \ldots, V_{k-1}$, each of size $k-1$. Color all edges inside a part blue and all edges between different parts red.



$k-1$ parts, each of size $k-1$.
Blue: all edges inside each part.
Red: all edges between parts.

The diagram is schematic: each circle represents an entire blue K_{k-1}, and each red connection represents all edges between two parts. A blue clique must lie in a single part, so it has at most $k-1$ vertices. A red clique contains at most one vertex from each part, so it also has at most $k-1$ vertices. Thus this coloring has no monochromatic K_k, proving $R(k) > (k-1)^2$.

Theorem 1.2.5 (Erdős, 1947)

$$R(k) > (\sqrt{2} - o(1))^k = 2^{\left(\frac{1}{2} - o(1)\right)k}.$$

Proof. Take the complete graph K_n on n vertices. Coloring each edge red or blue with probability $\frac{1}{2}$, independently.

The bad events, say E_S, is the event that set S of k vertices whose edges are all the same color. Then

$$\mathbb{P}(E_S) = 2 \cdot \left(\frac{1}{2}\right)^{\binom{k}{2}}.$$

Then

$$\begin{aligned}\mathbb{P}\left(\text{coloring is bad}\right) = \mathbb{P}\left(\bigcup_{S \in \binom{V}{k}} E_S\right) &\leq \sum_{S \in \binom{V}{k}} \mathbb{P}(E_S) = \binom{n}{k} 2 \left(\frac{1}{2}\right)^{\binom{k}{2}} < 2n^k 2^{-\frac{k(k-1)}{2}} \\ &= 2\left(\frac{n}{2^{\frac{k-1}{2}}}\right)^k.\end{aligned}$$

If $n < (1-c)2^{\frac{k-1}{2}}$ for some constant $c > 0$, then

$$\mathbb{P}\left(\text{bad coloring}\right) \leq 2(1-c)^k \to 0,$$

and thus

$$\mathbb{P}\left(\text{coloring has no monochromatic } K_k\right) > 0,$$

i.e. $\exists$ a coloring of K_n without monochromatic K_k. Hence, we must have $R(k) > n$. ■

Theorem 1.2.6

$$2^{\frac{1}{2}k} \lesssim R(k) \leq 2^{2k}.$$

§1.3. Two Recurring Patterns

Definition 1.3.1 (Extremal problem)**.** An **extremal problem** is a problem to find the size of a largest structure with a desired property.

To solve such problem, we should

(1) show a structure of large size with given property. (lower bound)

(2) prove that larger structures can't have property. (upper bound)

Definition 1.3.2 (Probabilistic method)**.** To use **probabilistic method** to solve a problem, you should construct a probability space on the set of structures, then show that the desired property has positive parts, which shows the existence of a structure with such property.

Lecture 2

22 Sep.

Recall . Erdős's lower bound on $R(k)$ colored the edges of K_n red and blue uniformly at random and applied the union bound over all k-sets of vertices: if

$$\binom{n}{k} 2^{1-\binom{k}{2}} < 1,$$

then with positive probability no copy of K_k is monochromatic, so $R(k) > n$.

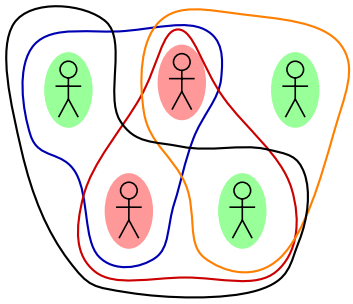
That argument never used that the colored objects are the edges of a complete graph. All it needed was a family of sets, the edge sets of the copies of K_k, none of which may end up monochromatic. This lecture studies that coloring problem in its own right.

§1.4. Property B

§1.4.i. Two-Colorability of Hypergraphs

Example 1.4.1 (Chess Olympiad)**.** A national team travels to the Chess Olympiad. Its members fill the following roles.

- Team manager
- Team captains
- Women
- Endgame specialists
- Opening experts
- e4 players
- d4 players
- Sport psychologist



A member can hold several roles, e.g. a team captain who is also an endgame specialist, and a role is usually held by several members. The e4 and d4 players are those whose usual first move is 1.e4 and 1.d4, respectively. The team travels on two flights, and each flight should carry someone in every role, so that the team is complete even if the other flight is delayed.

Question 1.4.2. How can we assign the members to the 2 flights so that each flight has a person with every role?

The picture shows five members and the four underlined roles. Each closed curve encloses the members holding the role underlined in the same color, and the shading behind each member is a flight assignment: the members shaded red take one flight, those shaded green take the other. This answers the question, since each of the four curves contains both a red and a green member.

Abstractly, the data are a set of people with a family of subsets of it, one subset per role, and a flight assignment colors the people with two colors. This is the following structure.

Definition 1.4.3 (Hypergraph)**.** A **hypergraph** is a pair $H = (V, E)$ of vertices V and edges $E \subseteq 2^V$, where an edge is a subset of vertices. It is **k-uniform** if $|e| = k$ for all $e \in E$.

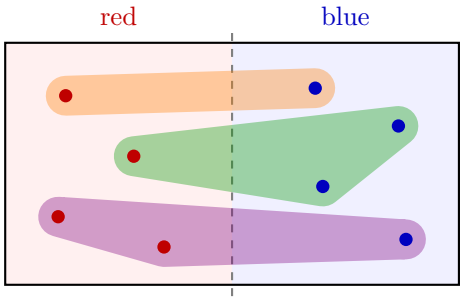
Notation . 2^V is the power set of V, and $V(H)$, $E(H)$ are the vertex and edge sets of H; all hypergraphs here are finite. For a set X, $\binom{X}{k}$ is the family of all k-element subsets of

X, and $[N] = \{1, \dots, N\}$.

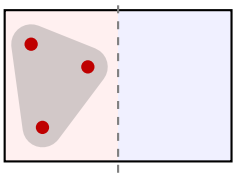
Remark 1.4.4. A graph is a 2-uniform hypergraph: each of its edges is a set $\{u, v\}$ of two vertices.

Definition 1.4.5 (Property B)**.** A hypergraph has **Property B**, or is **two-colorable**, if we can two-color the vertices of it s.t. every edge has vertices of both 2 colors.

Picture the two colors as the two halves of a box: put the red vertices in the left half and the blue vertices in the right half. Property B asks for a way to place every vertex so that each edge crosses the dividing line, i.e. has a vertex on both sides.



Property B: every edge crosses the line



× not allowed: an edge on one side is monochromatic

Equivalently, there is a coloring $\chi\colon V \to \{\text{red}, \text{blue}\}$ under which no edge is **monochromatic**, i.e. has all of its vertices of one color. The name refers to Felix Bernstein, who studied this property in 1908.

In Example 1.4.1, the vertices are the members, each role is an edge, and a flight assignment is a two-coloring, so the question asks whether this hypergraph has Property B. The shading shows that the drawn part, a 3-uniform hypergraph with five vertices and four edges, does.

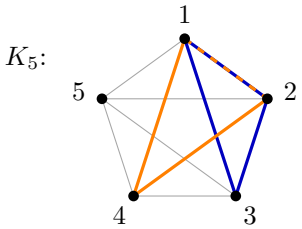
§1.4.ii. Connection to Ramsey Numbers

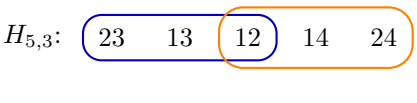
The Ramsey problem is a question about Property B for one particular family of hypergraphs. Given n and k, build the hypergraph $H_{n,k}$ with

$$V(H_{n,k}) = E(K_n), \qquad E(H_{n,k}) = \{\, E(K) : K \text{ is a copy of } K_k \text{ in } K_n \,\}.$$

Its vertices are the $\binom{n}{2}$ edges of K_n. It has one edge for each k-set S of vertices of K_n, consisting of the $\binom{k}{2}$ edges of K_n inside S. Thus $H_{n,k}$ is a $\binom{k}{2}$-uniform hypergraph with $\binom{n}{k}$ edges.

Example 1.4.6 ($n = 5$, $k = 3$)**.** Write ij for the edge of K_5 between the vertices i and j.





The ten vertices of $H_{5,3}$ are the ten edges of K_5, and its ten edges are the triangles of K_5, each a set of three edges of K_5. Only two triangles are drawn. The blue triangle 123 is the edge $\{12, 13, 23\}$ of $H_{5,3}$, and the orange triangle 124 is the edge $\{12, 14, 24\}$. These two edges of $H_{5,3}$ share the vertex 12 because the two triangles share the edge 12, drawn in both colors. The colors only mark the two triangles; they are not an edge-coloring.

A red/blue coloring of the edges of K_n is the same thing as a two-coloring of the vertices of $H_{n,k}$, and a copy of K_k is monochromatic exactly when the corresponding edge of $H_{n,k}$ is. Hence K_n has a coloring without a monochromatic K_k iff $H_{n,k}$ has Property B, and

$$R(k) = \min\{\, n : H_{n,k} \text{ does not have Property B} \,\}.$$

Unfolding the definition of Property B, $H_{n,k}$ is without Property B iff every two-coloring of its vertices leaves some edge monochromatic, so

$$R(k) = \min\left\{n : \forall \chi \colon V(H_{n,k}) \to \{\text{red}, \text{blue}\},\ \exists\, e \in E(H_{n,k}) \text{ monochromatic under } \chi\right\}.$$

Since $V(H_{n,k}) = E(K_n)$, such a χ is a red/blue coloring of the edges of K_n, and a monochromatic $e \in E(H_{n,k})$ is a monochromatic copy of K_k; this is word for word the definition of $R(k)$ from Lecture 1.

Remark 1.4.7. For 2-uniform hypergraphs, Property B $\equiv$ bipartite: a 2-uniform hypergraph is a graph, and an edge $\{u, v\}$ sees both colors iff u and v get different colors. So a two-coloring without monochromatic edges is exactly a partition of V into two independent sets, i.e. a bipartition. Property B thus generalizes bipartiteness to hypergraphs; for instance, C_5 from Lecture 1, and every other odd cycle, does not have Property B.

§1.4.iii. The Extremal Problem

Every edge is one more constraint on the coloring, so a hypergraph with few edges should be two-colorable. We measure the size of a hypergraph by its number of edges and leave the number of vertices unrestricted.

Question 1.4.8 (Extremal problem)**.** How small can a k-uniform hypergraph be if it does not have Property B?

Definition 1.4.9.

$$m(k) \coloneqq \min\{\, m : \exists\, k\text{-uniform hypergraph } H = (V, E),\ |E| = m,\ \text{without Property B} \,\}.$$

Unfolding the definition of Property B, H is without Property B iff every two-coloring of V leaves some edge monochromatic, so

$$m(k) = \min\left\{ m : \begin{array}{l} \exists\, H = (V, E) \text{ with } |E| = m \text{ and } |e| = k\ \forall e \in E, \text{ s.t.} \\ \forall \chi \colon V \to \{\text{red}, \text{blue}\},\ \exists\, e \in E \text{ monochromatic under } \chi \end{array} \right\}.$$

Here $\min \varnothing = \infty$, so $m(k) = \infty$ would mean that every k-uniform hypergraph has Property B; Corollary 1.4.13 shows this does not happen. By minimality, every k-uniform hypergraph with fewer than $m(k)$ edges has Property B, while some k-uniform hypergraph with $m(k)$ edges does not.

Proposition 1.4.10 (Erdős, 1963)

$$m(k) \geq 2^{k-1}.$$

Proof. WTS: If $H = (V, E)$ is a k-uniform hypergraph with $m < 2^{k-1}$ edges, then it is two-colorable.

Color the vertices red and blue independently and uniformly. Let $E = \{e_1, e_2, \dots, e_m\}$, and let A_i be the event that the edge e_i is monochromatic. The edge e_i has exactly k vertices with independent colors, and it is monochromatic iff they are all red or all blue, so

$$\mathbb{P}(A_i) = 2\left(\frac{1}{2}\right)^k = \left(\frac{1}{2}\right)^{k-1}.$$

The events A_i need not be independent, since edges may share vertices, but the union bound does not need independence:

$$\mathbb{P}(\exists \text{ monochromatic edge}) = \mathbb{P}\left(\bigcup_{i=1}^{m} A_i\right) \le \sum_{i=1}^{m} \mathbb{P}(A_i) = m \cdot 2^{-(k-1)} < 1 \quad \text{for } m < 2^{k-1}.$$

Taking complements, and using that an event of positive probability contains at least one outcome,

$$\begin{gathered}\mathbb{P}(\nexists \text{ monochromatic edge}) > 0 \\ \Downarrow \\ \exists \text{ a two-coloring of vertices where every edge sees both colors} \\ \Downarrow \\ H \text{ is two-colorable.}\end{gathered}$$

■

Remark 1.4.11. Applied to $H_{n,k}$, which is $\binom{k}{2}$-uniform with $\binom{n}{k}$ edges, Proposition 1.4.10 says: if $\binom{n}{k} < 2^{\binom{k}{2}-1}$, then $H_{n,k}$ has Property B, i.e. $R(k) > n$. This is exactly the condition $\binom{n}{k}2^{1-\binom{k}{2}} < 1$ in Erdős's lower bound on $R(k)$, and the proof above is the same computation with K_n replaced by an arbitrary hypergraph.

§1.4.iv. Upper Bounds on $m(k)$

An upper bound on $m(k)$ comes from exhibiting one k-uniform hypergraph without Property B that has few edges.

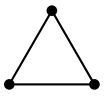
Proposition 1.4.12

$m(1) = 1$, $m(2) = 3$, and for every $k \ge 1$,

$$m(k) \le \binom{2k-1}{k} \sim c\,\frac{2^{2k}}{\sqrt{k}}, \qquad c = \frac{1}{2\sqrt{\pi}}.$$



$k = 1$: one vertex, one edge



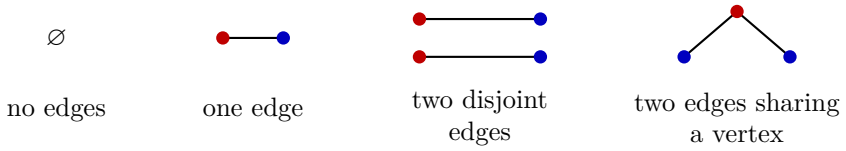
$k = 2$: the triangle K_3

Proof.

(1) $m(1) = 1$: Take one vertex v and the single edge $\{v\}$ (left picture above). This edge has only one vertex, so it is monochromatic under every coloring. A hypergraph with no edges has Property B vacuously, so $m(1) = 1$. In Example 1.4.1, an edge of size 1 is a role held by a single member, e.g. a lone sport psychologist: then no flight assignment works.

(2) $m(2) = 3$: A 2-uniform hypergraph is a graph, and by Remark 1.4.7 it is without Property B iff it is not bipartite. So $m(2)$ is the fewest edges of a non-bipartite graph, and we show two things.

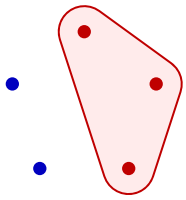
- $m(2) \leq 3$: The triangle K_3 (right picture above) is without Property B. Take any two-coloring of its three vertices. By the pigeonhole principle, two of the three vertices get the same color, and since every two vertices of K_3 are adjacent, the edge between them is monochromatic.
- $m(2) \geq 3$: Every graph G with at most two edges has Property B. We color the vertices of the edges red and blue as below, and all other vertices, which lie in no edge, arbitrarily.



With one edge $\{u, v\}$, color u red and v blue. With two disjoint edges $\{a, b\}$, $\{c, d\}$, color a, c red and b, d blue. With two edges $\{a, b\}$, $\{a, c\}$ sharing the vertex a, color a red and b, c blue. In every case each edge has one red and one blue endpoint, so no edge is monochromatic.

Hence $m(2) = 3$. Note that Proposition 1.4.10 alone only gives $m(2) \geq 2^{2-1} = 2$, so the case check above is needed for the lower bound.

(3) $m(k) \leq \binom{2k-1}{k}$: Consider $2k-1$ vertices and let any k of them form an edge, i.e. take the hypergraph $\left([2k-1], \binom{[2k-1]}{k}\right)$. In a two-coloring of $2k-1$ vertices, by the pigeonhole principle, some color is used on at least k vertices, and any k of these form a monochromatic edge.



$k = 3$: five vertices, any three of which form an edge.
In this coloring three vertices are red,
and they form a monochromatic edge.

So this hypergraph does not have Property B, and $m(k) \leq \binom{2k-1}{k}$. For $k = 1, 2$, it is exactly the edge $\{v\}$ and the triangle above, so the bound is tight there.

(4) Asymptotics: By Pascal's rule and $\binom{2k-1}{k-1} = \binom{2k-1}{k}$,

$$\binom{2k}{k} = \binom{2k-1}{k-1} + \binom{2k-1}{k} = 2\binom{2k-1}{k},$$

and Stirling's formula $N! \sim \sqrt{2\pi N}(N/e)^N$ gives

$$\binom{2k}{k} = \frac{(2k)!}{(k!)^2} \sim \frac{\sqrt{4\pi k}\,(2k/e)^{2k}}{2\pi k\,(k/e)^{2k}} = \frac{4^k}{\sqrt{\pi k}}.$$

Hence $\binom{2k-1}{k} = \frac{1}{2}\binom{2k}{k} \sim \frac{1}{2\sqrt{\pi}} \cdot \frac{2^{2k}}{\sqrt{k}}$. ■

Corollary 1.4.13

$m(k)$ is finite for every $k \geq 1$.

Proof. The hypergraph $\left([2k-1], \binom{[2k-1]}{k}\right)$ in the proof of Proposition 1.4.12 is k-uniform and does not have Property B, so $m(k) \le \binom{2k-1}{k} < \infty$. ∎

Remark 1.4.14. Together, Proposition 1.4.10 and Proposition 1.4.12 give

$$2^{k-1} \le m(k) \le (1+o(1))\frac{4^k}{2\sqrt{\pi k}}.$$

Both bounds are exponential in k, with bases 2 and 4, the same kind of gap as $\sqrt{2}^k \lesssim R(k) \le 4^k$ for Ramsey numbers. Unlike for Ramsey numbers, this gap can be closed up to a polynomial factor: the base 2 of the lower bound is the right one. The explicit hypergraph $\left([2k-1], \binom{[2k-1]}{k}\right)$ is wasteful because it uses every k-set of its vertices, and the next subsection shows that a *random* choice of far fewer k-sets on a larger vertex set already fails to have Property B.

§1.4.v. A Random Construction

To show that $m(k) \le m$ we must exhibit a k-uniform hypergraph with m edges and without Property B. Instead of constructing one, we pick one at random and show that it fails to have Property B with positive probability.

Theorem 1.4.15 (Erdős, 1964)
$m(k) = O(k^2 2^k)$, i.e. there is a constant $C > 0$ s.t. $m(k) \le Ck^2 2^k$ for all large enough k.

Proof. Idea: take a random hypergraph H.

- Fix a set V of n vertices.
- Choose m edges $e_1, e_2, \dots, e_m$ uniformly at random from $\binom{V}{k}$, independently of each other.

The numbers n and m will be chosen later. The same k-set may be picked twice; then H has fewer than m distinct edges, and deleting repeated copies of an edge does not affect Property B. So it suffices to show

$$\mathbb{P}(H \text{ is not two-colorable}) > 0 \Rightarrow m(k) \le |E(H)| \le m.$$

Now the random object is the hypergraph, and the bad event is that H *is* two-colorable, i.e. for some two-coloring $\chi\colon V \to \{\text{red}, \text{blue}\}$ there is no monochromatic edge. There are 2^n colorings, so we fix one, bound the probability that it works, and take a union bound over all of them. This is the reverse of Proposition 1.4.10, where the hypergraph was fixed and the coloring was random.

Let χ be a two-coloring of V, and let A_χ be the event that none of $e_1, e_2, \dots, e_m$ is monochromatic under χ. Consider one edge e_i. Since e_i is uniformly random,

$$\mathbb{P}(e_i \text{ is monochromatic under } \chi) = \frac{\#\text{ monochromatic } k\text{-sets}}{\#\text{ possible } k\text{-sets}} = \frac{\binom{r}{k} + \binom{n-r}{k}}{\binom{n}{k}},$$

where r is the number of red vertices under χ. The function $x \mapsto \binom{x}{k} = \frac{x(x-1)\cdots(x-k+1)}{k!}$, extended by 0 for $x < k-1$, is convex on $[0,\infty)$.[1] By convexity,

$$\frac{1}{2}\left[\binom{r}{k} + \binom{n-r}{k}\right] \ge \binom{\frac{1}{2}(r+n-r)}{k} \Rightarrow \binom{r}{k} + \binom{n-r}{k} \ge 2\binom{n/2}{k},$$

[1] On $[k-1,\infty)$ it is a polynomial all of whose roots are $\le k-1$, so by Rolle's theorem all of its derivatives are positive there; gluing it to the constant 0 at $x = k-1$, where both vanish and the right derivative is ≥ 0, keeps it convex.

that is, a balanced coloring has the fewest monochromatic k-sets. Hence, for every χ,

$$\mathbb{P}(e_i \text{ is monochromatic under } \chi) \geq \frac{2\binom{n/2}{k}}{\binom{n}{k}} =: p,$$

where $\binom{n/2}{k}$ is the polynomial above evaluated at $x = n/2$, so n need not be even. The edges are chosen independently, so

$$\begin{aligned}\mathbb{P}(A_\chi) = \mathbb{P}(\nexists \text{ monochromatic edge under } \chi) &= \prod_{i=1}^{m}\big(1 - \mathbb{P}(e_i \text{ is monochromatic under } \chi)\big)\\ &\leq (1-p)^m,\end{aligned}$$

and by the union bound over the 2^n colorings,

$$\mathbb{P}(H \text{ is two-colorable}) = \mathbb{P}\Big(\bigcup_{\chi} A_\chi\Big) \leq \sum_{\chi} \mathbb{P}(A_\chi) \leq 2^n(1-p)^m.$$

We must choose n and m so that $2^n(1-p)^m < 1$, where $p = 2\binom{n/2}{k}/\binom{n}{k}$. Using $1 - x < e^{-x}$ for $x > 0$,

$$2^n(1-p)^m < 2^n e^{-pm} = e^{n\ln 2 - pm}, \qquad \text{where } e^{n\ln 2 - pm} \leq 1 \Leftrightarrow m \geq \frac{n\ln 2}{p}.$$

We will take $m = \lfloor \frac{n\ln 2}{p} \rfloor + 1$, and choose n to minimize it. For this we need p precisely as a function of n and k. Writing out both binomial coefficients, the $i = 0$ factor equals $\frac{1}{2}$ and cancels the 2:

$$\begin{aligned}p = \frac{2\binom{n/2}{k}}{\binom{n}{k}} &= 2\prod_{i=0}^{k-1}\frac{\frac{n}{2}-i}{n-i} = \prod_{i=1}^{k-1}\frac{n-2i}{2(n-i)} = \prod_{i=1}^{k-1}\left(\frac{1}{2} - \frac{i}{2(n-i)}\right) = \frac{1}{2^{k-1}}\prod_{i=1}^{k-1}\left(1 - \frac{i}{n-i}\right)\\ &= \frac{1}{2^{k-1}}\prod_{i=1}^{k-1}\left(1 - \frac{i}{n} + O\Big(\frac{i^2}{n^2}\Big)\right) = \frac{1}{2^{k-1}}\prod_{i=1}^{k-1} e^{-\frac{i}{n} + O\left(\frac{i^2}{n^2}\right)}\\ &= \frac{1}{2^{k-1}}\, e^{-\sum_{i=1}^{k-1}\left(\frac{i}{n} + O\left(\frac{i^2}{n^2}\right)\right)} = \frac{1}{2^{k-1}}\, e^{-\frac{k(k-1)}{2n} + O\left(\frac{k^3}{n^2}\right)}.\end{aligned}$$

Here we assumed $n \geq 2k$, so that $\frac{i}{n-i} \leq 1$ and $\frac{i}{n-i} = \frac{i}{n} + O(\frac{i^2}{n^2})$, and used $1 - x = e^{-x+O(x^2)}$ for small $x \geq 0$. Now, since

$$m \approx \frac{n\ln 2}{p} = 2^{k-1}\, n\, e^{\frac{k(k-1)}{2n} + O\left(\frac{k^3}{n^2}\right)} \ln 2,$$

we minimize $ne^{k^2/(2n)}$ over n: its derivative in n is $e^{k^2/(2n)}(1 - \frac{k^2}{2n})$, which vanishes at $n = \frac{k^2}{2}$. The optimal choice is thus $n = ck^2$ with $c = \frac{1}{2}$, rounded to an integer. Then $\frac{k(k-1)}{2n} = 1 + O(\frac{1}{k})$ and $\frac{k^3}{n^2} = O(\frac{1}{k})$, so

$$m(k) \leq m \leq 2^{k-1} \cdot \frac{k^2}{2} \cdot e^{1+O(1/k)} \ln 2 + 1 = (1 + o(1))\, \frac{e\ln 2}{4}\, k^2 2^k \leq Ck^2 2^k. \qquad \blacksquare$$

Remark 1.4.16 (Estimating binomial coefficients)**.** Exact computation of p is not needed for $O(k^2 2^k)$, only for the constant. The standard estimates

$$\begin{aligned}\frac{(n-k)^k}{k!} \leq \binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!} \leq \frac{n^k}{k!},\\ \left(\frac{n}{k}\right)^k \leq \binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k(k-1)\cdots(k-k+1)} \leq \left(\frac{ne}{k}\right)^k\end{aligned}$$

follow from bounding each factor $n-i$ between $n-k$ and n, from $\frac{n-i}{k-i} \geq \frac{n}{k}$, and from $k! \geq (k/e)^k$. The first one already gives

$$p = \frac{2\binom{n/2}{k}}{\binom{n}{k}} \geq 2 \cdot \frac{(n/2-k)^k/k!}{n^k/k!} = \frac{1}{2^{k-1}}\left(1-\frac{2k}{n}\right)^k \geq \frac{1}{2^{k-1}} e^{-4k^2/n} \quad \text{for } n \geq 4k,$$

using $1-x \geq e^{-2x}$ for $0 \leq x \leq \frac{1}{2}$. With $n = 4k^2$ this yields $m(k) \leq (1+o(1))\, 2e \ln 2 \cdot k^2 2^k$, the same order with a worse constant.

§1.4.vi. Better Lower Bounds: Random Greedy Coloring

Combining Proposition 1.4.10 and Theorem 1.4.15,

Theorem 1.4.17 (Erdős, 1963–64)
$2^{k-1} \leq m(k) \leq Ck^2 2^k$ for some constant $C > 0$.

Since then the lower bound has been improved by polynomial factors in k, while the upper bound $O(k^2 2^k)$ is still the best known.

Theorem 1.4.18 (Radhakrishnan–Srinivasan, 2000)
$m(k) \geq c\sqrt{\frac{k}{\ln k}}\, 2^k$ for some constant $c > 0$.

Theorem 1.4.19 (Pluhár, 2009)
$m(k) \geq c\, k^{1/4} 2^k$ for some constant $c > 0$.

We now prove Theorem 1.4.19. The proof has two ideas: first a deterministic greedy coloring, whose failures we can describe exactly, and then a random order of the vertices, which makes such failures unlikely.

Proof. The proof of Proposition 1.4.10 colored every vertex independently at random, which ignores the structure of H entirely. Instead:

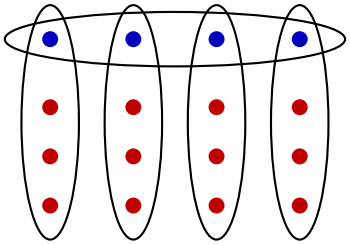
Idea 1: Color the vertices greedily. Order the vertices $v_1, v_2, \ldots, v_n$, and consider the vertices one by one. Color v_i red, unless that creates a monochromatic (red) edge, i.e. unless some edge contains v_i and all of its other vertices are already colored red, in which case we color v_i blue. Colors are never changed afterwards.

Observation .

(1) No monochromatic red edge: when the last vertex of an edge is reached and all the others are red, it is colored blue.

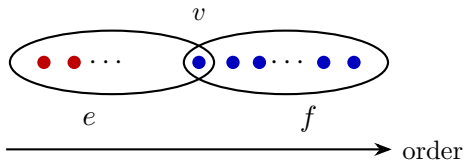
(2) Monochromatic blue edges can occur. However, a vertex v is colored blue only because coloring it red would have completed a red edge. So every blue vertex v lies in some edge e such that all vertices of e other than v are red and come before v in the order.

For example, in the 4-uniform hypergraph below, order the twelve lower vertices first and the four top vertices last. The lower vertices are all colored red, since each vertical edge still has an uncolored top vertex. Each top vertex is then the last vertex of a vertical edge whose other vertices are red, so it is colored blue, and the horizontal edge ends up monochromatic blue.



Each blue vertex lies in a vertical edge
that would otherwise be red.
The horizontal edge is monochromatic blue.

We say the greedy coloring *fails* if the coloring it produces is not a proper two-coloring, i.e. some edge is monochromatic. By (1), this happens exactly when some edge is monochromatic blue, as in the example. What must H and the order look like for this to happen? Suppose f is a monochromatic blue edge, and let v be the *first* vertex of f in the order. By (2), v lies in an edge e whose other vertices are red and come before v. Since all of f is blue, $e \cap f = \{v\}$; and since v is the last vertex of e and the first vertex of f, all of e comes before all of f:



We record what this argument shows. Note that it works for every order of the vertices.

Claim 1.4.1. For any order of the vertices, if the greedy coloring produces a monochromatic edge, then there are edges $e, f \in E(H)$ with $|e \cap f| = 1$ such that all of e comes before all of f in the order.

Here $e = f$ is possible only if $f = \{v\}$ is an edge of size 1: then $e = f = \{v\}$ and $|e \cap f| = 1$. Contrapositively, if no two edges meet in exactly one vertex, the greedy coloring cannot fail, whatever the order:

Corollary 1.4.20

If H is a hypergraph with $|e \cap f| \neq 1$ for all $e, f \in E(H)$, then H is two-colorable.

Proof. Fix any order $v_1, \ldots, v_n$ of the vertices and run the greedy coloring. It colors every vertex red or blue, so it gives a two-coloring χ of V; we show that no edge is monochromatic under χ.

- No edge is monochromatic red, by (1) of the observrvation.
- Suppose some edge f is monochromatic blue. By Claim 1.4.1 there would be edges e, f with $|e \cap f| = 1$, contradicting the assumption on H.

So every edge contains both colors, and H is two-colorable. ∎

Idea 2: Order the vertices uniformly at random. Let H be a k-uniform hypergraph with m edges. Choose an order σ of V uniformly at random among all $n!$ orders, run the greedy coloring along σ, and let χ_σ be the resulting two-coloring. We will show that

$$\mathbb{P}(\chi_\sigma \text{ is not a proper two-coloring}) < 1 \quad \text{whenever } m < c\,k^{1/4}2^k.$$

Then some order σ makes χ_σ proper, so H is two-colorable, and hence $m(k) \geq c\,k^{1/4}2^k$.

Write $e \prec f$ if all of e comes before all of f in σ. By (1), χ_σ is not proper exactly when it has a monochromatic blue edge, and by Claim 1.4.1 this forces a pair $e \prec f$ with $|e \cap f| = 1$. Hence, by the union bound,

$$\begin{aligned}\mathbb{P}(\chi_\sigma \text{ is not proper}) &= \mathbb{P}(\exists \text{ monochromatic blue edge}) \\ &\le \mathbb{P}\big(\exists\, e, f \in E(H) : |e \cap f| = 1,\ e \prec f\big) \\ &\le \sum_{\substack{(e,f) \\ |e\cap f|=1}} \mathbb{P}(e \prec f).\end{aligned}$$

Fix an ordered pair (e, f) with $|e \cap f| = 1$, so $|e \cup f| = 2k - 1$. Since σ is uniform, the order it induces on $e \cup f$ is uniform among all $(2k-1)!$ orders of $e \cup f$. In an order with $e \prec f$, the $k-1$ vertices of $e \setminus f$ come first, in any of $(k-1)!$ orders, then the common vertex, then the $k-1$ vertices of $f \setminus e$, in any of $(k-1)!$ orders. Thus

$$\mathbb{P}(e \prec f) = \frac{\overbrace{(k-1)!}^{\substack{\text{ordering } e\setminus f \\ \text{at front}}}\ \overbrace{(k-1)!}^{\substack{\text{ordering } f\setminus e \\ \text{at back}}}}{\underbrace{(2k-1)!}_{\substack{\text{total \# of} \\ \text{orderings of } e\cup f}}}.$$

There are at most m^2 ordered pairs (e, f), so

$$\mathbb{P}(\chi_\sigma \text{ is not proper}) \le m^2\, \frac{(k-1)!\,(k-1)!}{(2k-1)!}.$$

By Stirling's approximation $N! \sim \sqrt{2\pi N}(N/e)^N$, in the form $\binom{2N}{N} \sim \frac{4^N}{\sqrt{\pi N}}$ from the proof of Proposition 1.4.12,

$$m^2\, \frac{(k-1)!\,(k-1)!}{(2k-1)!} = \frac{m^2}{(2k-1)\binom{2k-2}{k-1}} \sim \frac{m^2\sqrt{\pi k}}{2k \cdot 4^{k-1}} = \frac{2\sqrt{\pi}\, m^2}{\sqrt{k}\, 2^{2k}},$$

which is < 1 if $m^2 < c'\sqrt{k}\, 2^{2k}$, i.e. if $m < c\, k^{1/4} 2^k$ (any $c < (2\sqrt{\pi})^{-1/2}$ works for large k). This is what we wanted to show. ∎

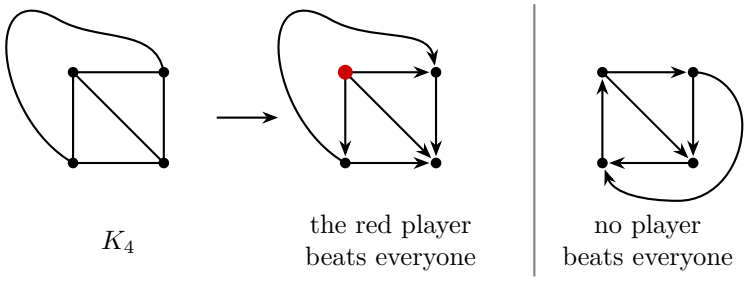
Theorem 1.4.21 (Cherkashin–Kozik, 2014)
$m(k) \ge c\sqrt{\frac{k}{\ln k}}\, 2^k$ for some constant $c > 0$.

Proof idea. Exactly the same random greedy coloring of Pluhár, analyzed a little better. This recovers the bound of Radhakrishnan–Srinivasan with a much simpler proof. ∎

§1.5. Schütte Tournaments

Definition 1.5.1 (Tournament)**.** A **tournament** is an orientation of the edges of the complete graph K_n. **Orientation** means: for each edge $\{u, v\}$, we choose to orient the edge either from u to v, giving the arc (u, v), or from v to u, giving the arc (v, u). We say u **beats** v if $(u, v) \in E(T)$.

Think of a round-robin tournament: every two of the n players play one game, and the arc points from the winner to the loser. Sometimes the results determine a winner, a player who beats everyone else, and sometimes they do not.

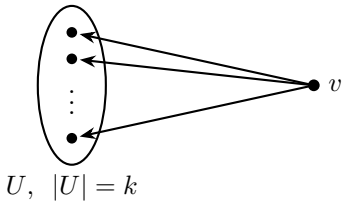


In the right tournament, the top-left player beats two others but loses to the bottom-left one, who in turn loses to both right players; every player loses to someone, so this condition cannot decide a winner. A winner is a single vertex u s.t. every other vertex is beaten by u. When there is no winner, we can relax the requirement: instead of one player who beats everyone, look for a small group of k players who together beat everyone else. This leads to the following question.

Question 1.5.2. Given a tournament T on n vertices, can we find a set U of k vertices s.t. for every vertex $v \notin U$, $\exists\, u \in U$ that beats v, i.e. $(u, v) \in E(T)$?

For $k = 1$ this asks for a winner. Schütte asked whether for every k there are tournaments in which no k players jointly beat everyone else.

Definition 1.5.3 (Schütte property)**.** We say a tournament has the ***k*-Schütte property** if $\nexists$ such a set of k vertices, i.e. for any set U of k vertices, $\exists\, v \notin U$ s.t. v beats every $u \in U$.



Example 1.5.4. For $k = 1$, the 1-Schütte property says that there is no winner: every vertex is beaten by some other vertex. The right tournament above has it, and so does the cyclic triangle $a \to b \to c \to a$. No tournament on $n \leq 2$ vertices has it, so 3 is the fewest vertices for $k = 1$.

Question 1.5.5. How small can a tournament with the k-Schütte property be?

Lecture 3

29 Sep.

Recall . A tournament T is an orientation of K_n, and it has the k-Schütte property if no k of its vertices jointly beat everyone else: for every $U \subseteq V(T)$ with $|U| = k$ there is a vertex $v \notin U$ beating every $u \in U$. For $k = 1$ this says that there is no winner.

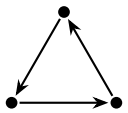
Section 1.5 ended by asking how small a tournament with the k-Schütte property can be. This lecture answers that question up to a polynomial factor, by the same pair of techniques used for $m(k)$ in Section 1.4: a deterministic greedy argument for the lower bound, and a random construction with a union bound for the upper bound.

§1.5.i. The Extremal Function

Definition 1.5.6. Let $f_S(k)$ be the smallest n s.t. there is an n-vertex tournament with the k-Schütte property.[2]

Since adding a vertex to a tournament with the k-Schütte property preserves the property (an old witness v for an old k-set U still works), tournaments with the k-Schütte property exist on every $n \geq f_S(k)$, and $f_S(k)$ is the threshold above which they appear.

Example 1.5.7. $f_S(1) = 3$. No tournament on 1 or 2 vertices has the 1-Schütte property: on one vertex v, the set $U = \{v\}$ has no vertex outside it; on two vertices, one of them beats the other and is therefore a winner. The cyclic triangle has it, since each of its three vertices is beaten by another one.

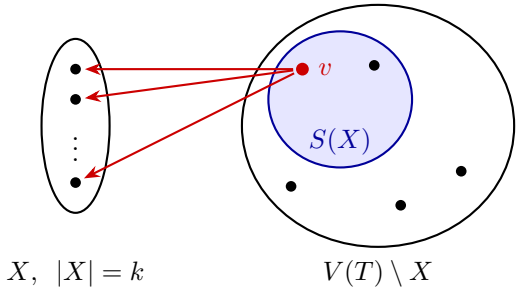


To work with the k-Schütte property we name the set of vertices that a given set fails to dominate.

Notation . Given a tournament T and a subset $X \subseteq V(T)$ of its vertices, let

$$S(X) \coloneqq \{v \in V(T) : (v, x) \in E(T)\ \forall\, x \in X\},$$

i.e. $S(X)$ is the set of vertices that beat everything in X.



A vertex v lies in $S(X)$ exactly when all k arcs between v and X point away from v, as drawn in red. Note that $S(\emptyset) = V(T)$, that S is order-reversing, i.e. $X \subseteq Y \Rightarrow S(Y) \subseteq S(X)$, and that $S(X) \cap X = \emptyset$ for $X \neq \emptyset$, since no vertex beats itself. The k-Schütte property is now a statement about S.

Lemma 1.5.8

T has the k-Schütte property $\Leftrightarrow$ $S(X) \neq \emptyset$ for all $X \subseteq V(T)$ with $|X| = k$.

Both bounds below are proved through this reformulation. In particular, to show $f_S(k) > n$ we must rule out all n-vertex tournaments at once:

$$\begin{aligned} f_S(k) > n &\Leftrightarrow \text{no } n\text{-vertex tournament has the } k\text{-Schütte property} \\ &\Leftrightarrow \text{every } n\text{-vertex tournament has a set } X \text{ of size } k \text{ with } S(X) = \emptyset. \end{aligned}$$

So for the lower bound we are handed an arbitrary tournament and must *find* a bad k-set in it, which is what the next subsection does greedily.

[2] Here $n > k$ is understood: if $n < k$ there is no k-set of vertices at all and the condition holds vacuously, and if $n = k$ the only k-set is $V(T)$, which has no vertex outside it.

§1.5.ii. Lower Bound: A Greedy Argument

The whole argument rests on the following averaging observation.

Lemma 1.5.9

In any tournament, there is a vertex that wins at least half of its games.

Proof. Count the total number of wins in a tournament T on n vertices. Every game produces exactly one winner, and there are $\binom{n}{2}$ games, so

$$\sum_{v \in V(T)} \#\{\text{wins of } v\} = \binom{n}{2}.$$

Since there are n vertices, the average number of wins per vertex is

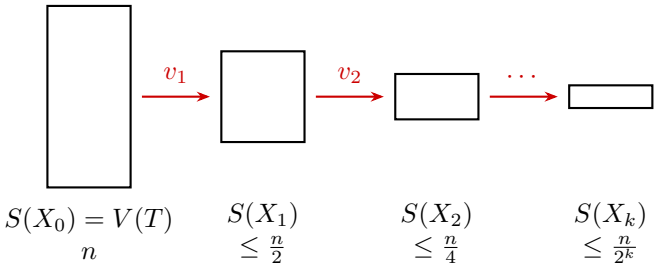
$$\frac{\binom{n}{2}}{n} = \frac{n-1}{2},$$

and some vertex is at least average, i.e. there is a vertex winning at least $\frac{n-1}{2}$ of the $n-1$ games it plays. ∎

Proposition 1.5.10

$f_S(k) \geq 2^k$.

The idea is to build a bad k-set X one vertex at a time, choosing at each step a vertex that halves $S(X)$. Since $S(\emptyset) = V(T)$ has n elements, after k steps at most $n/2^k$ vertices are left, and if $n < 2^k$ there are none.



Proof. Let T be any tournament on $n < 2^k$ vertices; by Lemma 1.5.8 it suffices to produce a set X of k vertices with $S(X) = \emptyset$. Start with $X_0 = \emptyset$, so that $S(X_0) = V(T)$, and repeat the following for $i = 1, 2, \ldots, k$.

If $S(X_{i-1}) = \emptyset$ already, stop. Otherwise apply Lemma 1.5.9 to the tournament $T[S(X_{i-1})]$ induced by T on $S(X_{i-1})$:

- find a vertex $v_i \in S(X_{i-1})$ beating at least half of the other vertices of $S(X_{i-1})$;
- add it to the set, $X_i := X_{i-1} \cup \{v_i\}$;
- this shrinks S to those vertices of $S(X_{i-1})$ that beat v_i, since

$$S(X_i) = S(X_{i-1}) \cap \{u : u \text{ beats } v_i\}.$$

Writing $s = |S(X_{i-1})|$, the vertex v_i beats at least $\frac{s-1}{2}$ of the remaining $s-1$ vertices, so at most $\frac{s-1}{2}$ of them beat v_i, and $v_i \notin S(X_i)$. Hence $S(X)$ shrinks by a factor of at least $\frac{1}{2}$ every round:

$$|S(X_i)| \leq \frac{|S(X_{i-1})| - 1}{2} \leq \frac{|S(X_{i-1})|}{2}.$$

After k rounds, $|X_k| = k$ and

$$|S(X_k)| \le \frac{|V(T)|}{2^k} = \frac{n}{2^k} < 1,$$

so $S(X_k) = \emptyset$. If instead the process stopped early, at some X_i with $i < k$ and $S(X_i) = \emptyset$, extend X_i to any k-set $X \supseteq X_i$; as S is order-reversing, $S(X) \subseteq S(X_i) = \emptyset$. Either way T fails the k-Schütte property, so $f_S(k) \ge 2^k$. ■

§1.5.iii. Upper Bound: A Random Tournament

For the upper bound we must exhibit one tournament with the k-Schütte property. As in Theorem 1.4.15, we do not construct it: we choose a tournament at random and show that it has the property with positive probability.

Theorem 1.5.11 (Erdős)
If $\binom{n}{k}\left(1-2^{-k}\right)^{n-k} < 1$, then $f_S(k) \le n$.

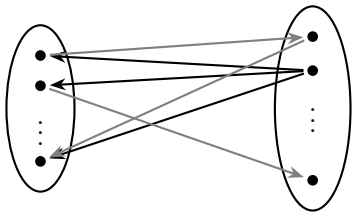
Proof. We consider a uniformly random tournament on the vertices $[n] = \{1, 2, \dots, n\}$, i.e. for every pair $\{i, j\}$ we choose the arc (i, j) or (j, i) uniformly and independently of all other pairs. By Lemma 1.5.8, the bad events are indexed by the k-sets X, the event for X being that $S(X) = \emptyset$; we bound each one and take a union bound.

Fix X with $|X| = k$ and a vertex $v \in V \setminus X$. The vertex v lies in $S(X)$ iff all k arcs between v and X point away from v, and these k coin flips are independent, so

$$\mathbb{P}(v \in S(X)) = \mathbb{P}(v \text{ beats all of } X) = \left(\frac{1}{2}\right)^k \Rightarrow \mathbb{P}(v \notin S(X)) = 1 - \frac{1}{2^k}.$$

For distinct $v \notin X$, the events $\{v \in S(X)\}$ involve disjoint sets of pairs, namely the k pairs joining v to X, hence they are independent. Since $|V \setminus X| = n - k$,

$$\mathbb{P}(S(X) = \emptyset) = \mathbb{P}\big(\forall\, v \notin X,\ v \notin S(X)\big) = \prod_{v \notin X} \mathbb{P}(v \notin S(X)) = \prod_{v \notin X} \left(1 - \frac{1}{2^k}\right) = \left(1 - \frac{1}{2^k}\right)^{n-k}.$$



each $v \notin X$ sees its own k arcs,
so the $n - k$ events are independent

$X,\ |X| = k$ $\qquad$ $V \setminus X,\ n - k$

Now take the union bound over all $\binom{n}{k}$ choices of X:

$$\mathbb{P}(T \text{ is not } k\text{-Schütte}) = \mathbb{P}\big(\exists\, X : S(X) = \emptyset\big) = \mathbb{P}\left(\bigcup_{X \in \binom{V}{k}} \{S(X) = \emptyset\}\right)$$

$$\le \sum_{X \in \binom{V}{k}} \mathbb{P}(S(X) = \emptyset) = \binom{n}{k}\left(1 - \frac{1}{2^k}\right)^{n-k} < 1$$

by assumption. Therefore $\mathbb{P}(T \text{ is } k\text{-Schütte}) > 0$, so there is an n-vertex tournament with the k-Schütte property, and $f_S(k) \le n$. ■

It remains to find how small an n satisfies the hypothesis of Theorem 1.5.11.

Corollary 1.5.12
$f_S(k) = O(k^2 2^k)$.

Proof. We estimate the two factors by

$$\binom{n}{k} \le n^k, \qquad \left(1 - 2^{-k}\right)^{n-k} \le e^{-(n-k)2^{-k}},$$

the second from $1 - x \le e^{-x}$. So it suffices to have

$$n^k e^{-(n-k)2^{-k}} < 1 \Leftrightarrow k \ln n - (n-k)2^{-k} < 0 \Leftrightarrow n > k2^k \ln n + k.$$

The right-hand side grows only logarithmically in n, so the inequality holds as soon as n is a little larger than $k2^k \ln n$. Taking $n = \lceil k^2 2^k \rceil$ gives $\ln n = k \ln 2 + 2 \ln k + O(1) = (1 + o(1))k \ln 2$, hence

$$k2^k \ln n + k = (1 + o(1))\,(\ln 2)\, k^2 2^k < k^2 2^k \le n$$

for large k, as $\ln 2 < 1$. Thus $n = \Theta(k^2 2^k)$ suffices, and $f_S(k) = O(k^2 2^k)$. ∎

Putting Proposition 1.5.10 and Corollary 1.5.12 together:

Theorem 1.5.13
$2^k \le f_S(k) \le Ck^2 2^k$ for some constant $C > 0$.

Remark 1.5.14. This is the same picture as for Property B: an easy greedy lower bound 2^k, a random upper bound $O(k^2 2^k)$, and a gap of a polynomial factor $O(k^2)$ between them. Compare $2^{k-1} \le m(k) \le Ck^2 2^k$ from Section 1.4, where the lower bound has since been improved to $c\sqrt{k/\ln k}\, 2^k$.

Exercise 1.5.15. Show that $f_S(k) \ge c\,k2^k$ for some constant $c > 0$, improving Proposition 1.5.10 by a factor of k.

§1.6. The Erdős–Ko–Rado Theorem

§1.6.i. Intersecting Families

Property B asked how to *split* a family of sets; we now ask how large a family can be if no two of its members can be split apart at all, i.e. if they pairwise overlap.

Definition 1.6.1 (Intersecting family)**.** A family $\mathcal{F} \subseteq 2^{[n]}$ is **intersecting** if $A \cap B \ne \emptyset$ for any two members $A, B \in \mathcal{F}$.

Question 1.6.2. How large can an intersecting family be?

There are two natural ways to build a large one: take only *big* sets, which are forced to overlap for lack of room, or take only sets through one fixed element.

Example 1.6.3 (Large sets)**.** For $n \ge 3$, the family $\mathcal{F} = \{[n]\} \cup \big\{[n] \setminus \{x\} : x \in [n]\big\}$ is intersecting, since any two of its members miss at most two elements between them. More generally, take every set of size more than $\frac{n}{2}$,

$$\mathcal{F} = \binom{[n]}{n} \cup \binom{[n]}{n-1} \cup \cdots \cup \binom{[n]}{\lceil \frac{n+1}{2} \rceil}.$$

If $|A|, |B| > \frac{n}{2}$ then $|A| + |B| > n$, so there is not enough space in $[n]$ for A and B to be disjoint. For odd n, exactly half of all subsets have size more than $\frac{n}{2}$, so $|\mathcal{F}| = 2^{n-1}$.

Example 1.6.4 (Stars)**.** Fix $x \in [n]$ and let $\mathcal{F}_x = \{F \subseteq [n] : x \in F\}$ be the **star** at x, i.e. the family of *all* subsets containing x. Any two members share x, so $\mathcal{F}_x$ is intersecting, and $|\mathcal{F}_x| = 2^{n-1}$, since a subset containing x is determined by its intersection with $[n] \setminus \{x\}$.

Both examples reach 2^{n-1}, and nothing does better.

Proposition 1.6.5

If $\mathcal{F} \subseteq 2^{[n]}$ is intersecting, then $|\mathcal{F}| \leq 2^{n-1}$.

Proof. Partition all 2^n subsets into the 2^{n-1} complementary pairs $\{F, [n] \setminus F\}$. The two members of a pair are disjoint, so an intersecting family contains at most one set from each pair, whence $|\mathcal{F}| \leq 2^{n-1}$. ∎

Example 1.6.3 is really a cheat: it wins by using only large sets. The question becomes interesting once all sets are required to have the same size.

Question 1.6.6 (Erdős–Ko–Rado)**.** What is the largest intersecting family $\mathcal{F} \subseteq \binom{[n]}{k}$?

If $n < 2k$, then any two k-sets $A, B \subseteq [n]$ satisfy $|A| + |B| = 2k > n$ and hence intersect, so $\binom{[n]}{k}$ is itself intersecting and the answer is $\binom{n}{k}$. **From now on we assume $n \geq 2k$.** The star of Example 1.6.4, cut down to k-sets, still works.

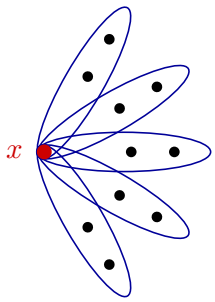
Example 1.6.7 (The k-uniform star)**.** Fix $x \in [n]$. The **star** at x is

$$\mathcal{F}_x^{(k)} := \left\{F \in \binom{[n]}{k} : x \in F\right\},$$

the family consisting of *every* k-set that contains x, and of nothing else. It is intersecting, as any two of its members share x. Choosing a k-set through x amounts to choosing its other $k-1$ elements from $[n] \setminus \{x\}$, so

$$|\mathcal{F}_x^{(k)}| = \binom{n-1}{k-1}, \qquad \frac{|\mathcal{F}_x^{(k)}|}{\left|\binom{[n]}{k}\right|} = \frac{k}{n}.$$

In the picture, each ellipse is one member of the star, and they meet exactly in the red element x.



More generally, call a family $\mathcal{F} \subseteq \binom{[n]}{k}$ **trivially intersecting** if all of its members share a common element, i.e. $\bigcap_{F \in \mathcal{F}} F \neq \emptyset$. Such a family is certainly intersecting. The two notions are related but not identical: $\mathcal{F}$ is trivially intersecting exactly when $\mathcal{F} \subseteq \mathcal{F}_x^{(k)}$ for some x, so the trivially intersecting families are precisely the subfamilies of stars, and the stars are the largest of them. In particular a trivially intersecting family of the full size $\binom{n-1}{k-1}$ *is* a star.

The Erdős–Ko–Rado theorem says that for $n \geq 2k$ a star is already optimal.

Theorem 1.6.8 (Erdős–Ko–Rado, 1961)
If $n \geq 2k$ and $\mathcal{F} \subseteq \binom{[n]}{k}$ is intersecting, then $|\mathcal{F}| \leq \binom{n-1}{k-1}$.

§1.6.ii. Katona's Cycle Method

The proof is Katona's. We build a random k-set A that is uniform on $\binom{[n]}{k}$, so that

$$\mathbb{P}(A \in \mathcal{F}) = \frac{|\mathcal{F}|}{\binom{n}{k}},$$

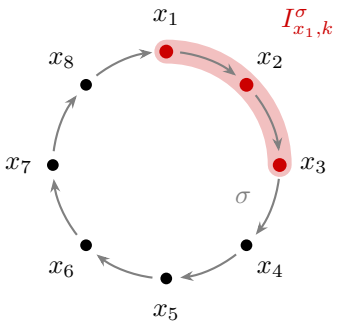
and then bound the same probability by $\frac{k}{n}$ by first conditioning on a cyclic permutation σ of $[n]$, which arranges $[n]$ on a circle.

Definition 1.6.9 (Cyclic permutation)**.** A permutation σ of $[n]$ is **cyclic** if its cycle representation is a single n-cycle $(x_1\ x_2\ \cdots\ x_n)$, i.e. $\sigma(x_j) = x_{j+1}$ with indices mod n. Each of the $n!$ orderings $(x_1, \ldots, x_n)$ of $[n]$ gives a cyclic permutation, and an n-cycle can be written in exactly n ways, one for each choice of starting element. So there are $n!/n = (n-1)!$ cyclic permutations.

Place $x_1, \ldots, x_n$ around a circle in this order, so that σ moves every element one step clockwise. For $x \in [n]$, the **interval of length k** of σ starting at x is

$$I^{\sigma}_{x,k} \coloneqq \{x, \sigma(x), \sigma^2(x), \ldots, \sigma^{k-1}(x)\},$$

the set of k consecutive elements on the circle beginning at x. For $k < n$, x is the only element of $I^{\sigma}_{x,k}$ whose predecessor $\sigma^{-1}(x)$ is not in $I^{\sigma}_{x,k}$. So distinct starting points give distinct intervals, and σ has exactly n of them.



A circle has very little room for pairwise intersecting intervals.

Lemma 1.6.10 (Katona)
Let $n \geq 2k$ and let $\mathcal{F} \subseteq \binom{[n]}{k}$ be intersecting. Then for every cyclic permutation σ of $[n]$, at most k of the n intervals $I^{\sigma}_{x,k}$ belong to $\mathcal{F}$.

Proof. If no interval of σ lies in $\mathcal{F}$ there is nothing to prove, so fix one, $F \in \mathcal{F}$. Write $\sigma = (x_1\ x_2\ \cdots\ x_n)$ starting at the first element of F, so that $F = \{x_1, \ldots, x_k\}$ and, with indices mod n, $I^{\sigma}_{x_j,k} = \{x_j, \ldots, x_{j+k-1}\}$.

Every interval of $\mathcal{F}$ other than F meets F, and an interval $I^{\sigma}_{x_j,k}$ meets F iff $j \in \{2-k, \ldots, k\}$. Excluding $j = 1$ (that is F), these are the $2k-2$ intervals that

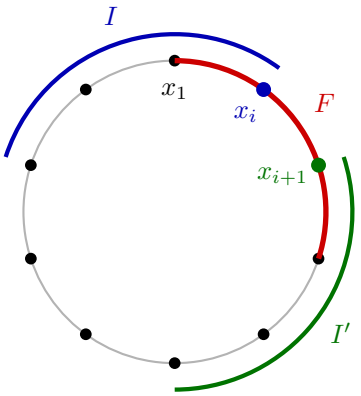
- start at x_{i+1} for some $1 \leq i \leq k-1$, or

- end at x_i for some $1 \le i \le k-1$ (i.e. start at x_{i-k+1}).

They are distinct because their starting indices $2-k, \dots, 0, 2, \dots, k$ lie in a window of $2k-1 \le n$ consecutive integers. For each $1 \le i \le k-1$, pair

$$\underbrace{\{x_{i-k+1}, \dots, x_i\}}_{\text{ends at } x_i} \quad \text{with} \quad \underbrace{\{x_{i+1}, \dots, x_{i+k}\}}_{\text{starts at } x_{i+1}}.$$

These two intervals occupy the $2k$ consecutive indices $i-k+1, \dots, i+k$, which are distinct mod n since $n \ge 2k$, so the two intervals are *disjoint*.



I ends at x_i, I' starts at x_{i+1}; together they cover $2k \le n$ consecutive elements, so $I \cap I' = \emptyset$.

As i runs over $1, \dots, k-1$, every one of the $2k-2$ intervals meeting F appears in exactly one pair, so they split into $k-1$ disjoint pairs. An intersecting family contains at most one interval of each pair, so $\mathcal{F}$ contains at most $(k-1)+1 = k$ intervals of σ, counting F. ∎

Proof of Theorem 1.6.8. Let $n \ge 2k$ and let $\mathcal{F} \subseteq \binom{[n]}{k}$ be an intersecting family; we show $|\mathcal{F}| \le \binom{n-1}{k-1}$ by computing $\mathbb{P}(A \in \mathcal{F})$ in two ways for the following random k-set A, which does not depend on $\mathcal{F}$.

(1) Choose a uniformly random cyclic permutation σ of $[n]$.

(2) Choose $x \in [n]$ uniformly at random, independently of σ, and set

$$A = I_{x,k}^{\sigma} = \{x, \sigma(x), \dots, \sigma^{k-1}(x)\}.$$

A is uniform on $\binom{[n]}{k}$. Fix $S \in \binom{[n]}{k}$ and count the pairs (σ, x) with $I_{x,k}^{\sigma} = S$. Writing σ as a cycle that starts at x, the condition says that the cycle lists

- first $x \in S$, in k ways;
- then the other $k-1$ elements of S in some order, in $(k-1)!$ ways;
- then the $n-k$ elements of $[n] \setminus S$ in some order, in $(n-k)!$ ways.

A cycle written from a given starting element is unique, so these choices give distinct pairs, $k \cdot (k-1)!\,(n-k)! = k!\,(n-k)!$ in total. As (σ, x) is uniform among all $(n-1)! \cdot n = n!$ pairs,

$$\mathbb{P}(A = S) = \frac{k!\,(n-k)!}{n!} = \frac{1}{\binom{n}{k}} \quad \text{for every } S,$$

and therefore

$$\mathbb{P}(A \in \mathcal{F}) = \frac{|\mathcal{F}|}{\binom{n}{k}}. \tag{1}$$

Conditioning on σ. Given σ, x is still uniform on $[n]$, and distinct x give distinct intervals, so A is a uniformly random one of the n intervals of length k of σ, and

$$\mathbb{P}(A \in \mathcal{F} \mid \sigma) = \frac{\#\{\text{intervals of length } k \text{ of } \sigma \text{ in } \mathcal{F}\}}{\#\{\text{intervals of length } k \text{ of } \sigma\}} = \frac{\#\{x \in [n] : I^{\sigma}_{x,k} \in \mathcal{F}\}}{n} \le \frac{k}{n} \quad \text{for } \textit{every } \sigma$$

by Lemma 1.6.10. Averaging over σ,

$$\mathbb{P}(A \in \mathcal{F}) = \sum_{\sigma} \mathbb{P}(\sigma)\,\mathbb{P}(A \in \mathcal{F} \mid \sigma) \le \frac{k}{n}. \tag{2}$$

Comparing (1) and (2),

$$\frac{|\mathcal{F}|}{\binom{n}{k}} \le \frac{k}{n} \Rightarrow |\mathcal{F}| \le \frac{k}{n}\binom{n}{k} = \binom{n-1}{k-1}. \qquad \blacksquare$$

Theorem 1.6.8 says that no intersecting family beats a star. In fact nothing else even ties with one.

Corollary 1.6.11 (Uniqueness in Erdős–Ko–Rado)

Let $n > 2k$ and let $\mathcal{F} \subseteq \binom{[n]}{k}$ be intersecting with $|\mathcal{F}| = \binom{n-1}{k-1}$. Then $\mathcal{F} = \mathcal{F}_x^{(k)}$ for some $x \in [n]$, i.e. the extremal families are exactly the stars.

Proof sketch. It is enough to show that $\mathcal{F}$ is trivially intersecting: then $\mathcal{F} \subseteq \mathcal{F}_x^{(k)}$ for some x, and as $\left|\mathcal{F}_x^{(k)}\right| = \binom{n-1}{k-1} = |\mathcal{F}|$ the two families coincide.

For $k = 1$ an intersecting family of singletons has a single member $\{x\}$, which is the star at x; so assume $k \ge 2$ and suppose $\mathcal{F}$ were *not* trivially intersecting. Theorem 1.6.13 below, whose proof is independent of this corollary, then gives

$$|\mathcal{F}| \le \binom{n-1}{k-1} - \binom{n-k-1}{k-1} + 1.$$

Since $n > 2k$ we have $n - k - 1 \ge k$, so $\binom{n-k-1}{k-1} \ge \binom{k}{k-1} = k \ge 2$, and therefore $|\mathcal{F}| < \binom{n-1}{k-1}$, contradicting the hypothesis. $\blacksquare$

Remark 1.6.12. The strict inequality $n > 2k$ is needed. For $n = 2k$ the k-sets fall into $\frac{1}{2}\binom{2k}{k}$ complementary pairs $\{A, [n] \setminus A\}$, and any two non-complementary k-sets meet; so choosing one set out of each pair gives an intersecting family of size $\frac{1}{2}\binom{2k}{k} = \binom{2k-1}{k-1}$, which is extremal. Choosing inconsistently leaves it with no common element, so for $n = 2k$ the extremal families are far from unique.

If we forbid the star, the maximum drops sharply.

Theorem 1.6.13 (Hilton–Milner, 1967)

If $n > 2k$ and $\mathcal{F} \subseteq \binom{[n]}{k}$ is non-trivially intersecting, i.e. $\bigcap_{F \in \mathcal{F}} F = \emptyset$, then

$$|\mathcal{F}| \le \binom{n-1}{k-1} - \binom{n-k-1}{k-1} + 1.$$

Remark 1.6.14. For $n \gg k^2$ we have $\binom{n-k-1}{k-1} \approx \binom{n-1}{k-1}$, so the Hilton–Milner bound is a vanishing fraction of the Erdős–Ko–Rado bound: once a common element is forbidden, an intersecting family must be much smaller.

§1.7. Bollobás Set-Pair Systems

Katona's proof averaged over random orderings of a circle. The next theorem averages over random orderings of a line, and it is one of the most useful counting tools in extremal set theory.

Definition 1.7.1 (Set-pair system)**.** A **Bollobás set-pair system** is a sequence of pairs of sets $\{(A_i, B_i) : 1 \le i \le m\}$ s.t.

(1) $A_i \cap B_i = \emptyset$ for every i;

(2) $A_i \cap B_j \neq \emptyset$ for every $i \neq j$.

So each B_i is a set that A_i avoids but that every other A_j hits: B_i is a certificate distinguishing the index i from all the others. The theorem says that such certificates cannot be too numerous if they are small.

Theorem 1.7.2 (Bollobás, 1965)
Let $\{(A_i, B_i) : 1 \le i \le m\}$ be a set-pair system with $|A_i| = k$ and $|B_i| = \ell$ for all i. Then

$$m \le \binom{k+\ell}{k}.$$

Proof. Let $X = \bigcup_i (A_i \cup B_i)$ be the ground set.

(1) Take a uniformly random ordering of the elements of X.

(2) For each i, define the event

$$E_i = \{\text{all elements of } A_i \text{ come before all elements of } B_i\}.$$

Since $A_i \cap B_i = \emptyset$, the set $A_i \cup B_i$ has exactly $k+\ell$ elements, and the random ordering restricted to it is uniform among its $(k+\ell)!$ orderings. Of these, $k!\,\ell!$ put all of A_i first, so

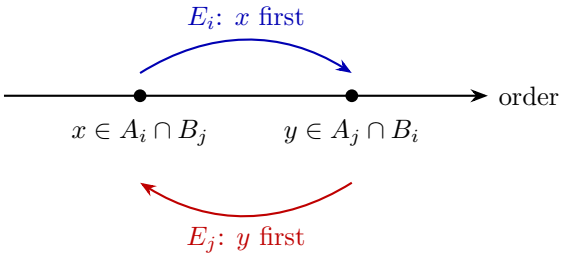
$$\mathbb{P}(E_i) = \frac{k!\,\ell!}{(k+\ell)!} = \frac{1}{\binom{k+\ell}{k}}.$$

Claim 1.7.1. The events $E_1, \dots, E_m$ are pairwise disjoint.

Proof. Suppose E_i and E_j both occur for some $i \neq j$. By condition (2) of Definition 1.7.1 applied both ways, we may pick

$$x \in A_i \cap B_j \qquad \text{and} \qquad y \in A_j \cap B_i.$$

These are distinct elements: $x \in A_i$ and $y \in B_i$, while $A_i \cap B_i = \emptyset$ by condition (1) of Definition 1.7.1, so $x \neq y$, and it makes sense to ask which comes first. Since E_i occurs and $x \in A_i$, $y \in B_i$, the element x comes before y. Since E_j occurs and $y \in A_j$, $x \in B_j$, the element y comes before x. This is a contradiction.



■

Being pairwise disjoint, the probabilities of the E_i add up, and they cannot exceed 1:

$$1 \geq \mathbb{P}\Big(\bigcup_{i=1}^{m} E_i\Big) = \sum_{i=1}^{m} \mathbb{P}(E_i) = \sum_{i=1}^{m} \frac{1}{\binom{k+\ell}{k}} = \frac{m}{\binom{k+\ell}{k}},$$

which is exactly $m \leq \binom{k+\ell}{k}$. ■

Remark 1.7.3 (The bound is attained)**.** Take the ground set $[k+\ell]$, let $A_1, \ldots, A_m$ list *all* of its k-element subsets, so that $m = \binom{k+\ell}{k}$, and let $B_i = [k+\ell] \setminus A_i$ be the complement of A_i. This is a set-pair system with the required sizes:

- $|A_i| = k$ and $|B_i| = (k+\ell) - k = \ell$;
- $A_i \cap B_i = \emptyset$, as B_i is the complement of A_i;
- for $i \neq j$, $A_i \cap B_j = A_i \setminus A_j \neq \emptyset$, because A_i and A_j are distinct sets of the same size, so neither can contain the other.

So Theorem 1.7.2 is sharp for every k and ℓ.

§1.8. Application: Saturated Graphs

Throughout this section H is a fixed graph, thought of as a pattern we want to avoid.

Definition 1.8.1 (H-free)**.** A **copy of H** in a graph G is a subgraph of G isomorphic to H, i.e. an injection $\varphi\colon V(H) \to V(G)$ with $\varphi(u)\varphi(v) \in E(G)$ for every edge $uv \in E(H)$. If G contains no copy of H, we call G **H-free**.

Remark 1.8.2. A copy need not be *induced*: G may have further edges among the vertices $\varphi(V(H))$, and they do not stop φ from being a copy. So G is K_k-free exactly when no k vertices of G are pairwise adjacent.

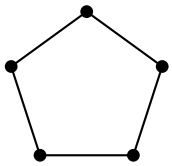
Classical extremal graph theory asks for the *maximum* number of edges of an H-free graph.

Question 1.8.3 (Mantel)**.** How many edges can an n-vertex graph have if it does not contain a triangle?

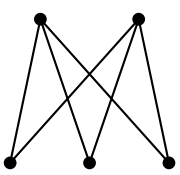
Question 1.8.4 (Turán, 1941)**.** How many edges can an n-vertex graph have if it does not contain K_k? Write $\mathrm{ex}(n, H)$ for this maximum, the **Turán number** of H.

One might try to construct such a graph greedily: consider the edges of K_n one by one, and add an edge to our graph if it does not create a copy of K_k. This certainly produces a K_k-free graph, but it need not produce one with the maximum number of edges: the process only guarantees that no further edge *can* be added.

Example 1.8.5. Take $n = 5$ and $k = 3$. Both graphs below are triangle-free, and in both of them adding any missing edge creates a triangle, so either can be the output of the greedy algorithm. They have a different number of edges.



C_5: 5 edges



$K_{2,3}$: 6 edges

So the greedy algorithm computes a *maximal* H-free graph rather than a maximum one, and it is natural to ask how few edges a maximal H-free graph can have.

Definition 1.8.6 (Saturated graph)**.** Given a fixed graph H, a graph G is **H-saturated** if

(1) G is H-free, and

(2) adding any missing edge to G creates a copy of H,

i.e. G is maximally H-free.

Observation . G is H-saturated $\Leftrightarrow$ G is the output of the H-free greedy algorithm for some ordering of the edges of K_n.

Proof. ($\Leftarrow$) The output G is H-free by construction. If f is a missing edge, it was rejected at its turn because adding it to the graph G' built so far created a copy of H. As $G' \subseteq G$, that copy is still present in $G + f$.

($\Rightarrow$) Order the edges of K_n so that those of G come first. Every prefix of $E(G)$ is a subgraph of the H-free graph G, so the algorithm accepts all of them. Afterwards the current graph is G, and each remaining edge f is rejected since $G + f$ contains a copy of H by saturation. The output is G. ■

Definition 1.8.7 (Saturation number)**.** Given a graph H, the **saturation number** $\mathrm{sat}(n, H)$ is the minimum number of edges in an n-vertex H-saturated graph.

Remark 1.8.8 (Why the minimum)**.** The maximum gives nothing new: it is the Turán number $\mathrm{ex}(n, H)$ of Question 1.8.4 again. An H-saturated graph is H-free, so has at most $\mathrm{ex}(n, H)$ edges; conversely an H-free graph with $\mathrm{ex}(n, H)$ edges is saturated, as adding any edge leaves too many for an H-free graph. So every n-vertex H-saturated graph G satisfies $\mathrm{sat}(n, H) \leq e(G) \leq \mathrm{ex}(n, H)$, with both ends attained.

Notation (Subscripted O)**.** A subscript on O records which parameters the implied constant is allowed to depend on. Thus $f(n) = O_k(n)$ means: for each fixed k there is a constant C_k, depending on k but *not* on n, with $|f(n)| \leq C_k\, n$ for all n. Plain $O(\cdot)$, as used earlier, allows no such dependence. Dually, $g(n) = \Omega(n^2)$ means $g(n) \geq c\, n^2$ for some constant $c > 0$ and all large n.

Theorem 1.8.9 (Bollobás, 1965)

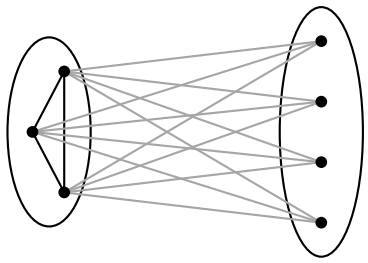
For every $k \geq 2$,

$$\mathrm{sat}(n, K_k) = \binom{n}{2} - \binom{n-k+2}{2} = (k-2)n - \binom{k-1}{2} = O_k(n).$$

Here $\mathrm{sat}(n, K_k) = (k-2)n - \binom{k-1}{2} \leq (k-2)n$, so $C_k = k - 2$ works: for each fixed k the saturation number grows only linearly in n, though the slope $k - 2$ grows with k.

Remark 1.8.10. Note how far this is from the Turán problem: for $k \geq 3$ the largest K_k-free graphs have $\Omega(n^2)$ edges, whereas the smallest maximal ones have only $O_k(n)$.

Remark 1.8.11 (The bound is best possible)**.** Join a clique K_{k-2} to an independent set of $n - k + 2$ vertices, i.e. put in all edges inside the clique and all edges between the two parts, and none inside the independent set.



This graph is K_k-free, since a clique may use at most one vertex of the independent set and so has at most $k-1$ vertices; and adding any missing edge, which joins two vertices of the independent set, completes a K_k. Its number of edges is

$$\binom{k-2}{2} + (k-2)(n-k+2) = \binom{n}{2} - \binom{n-k+2}{2},$$

so

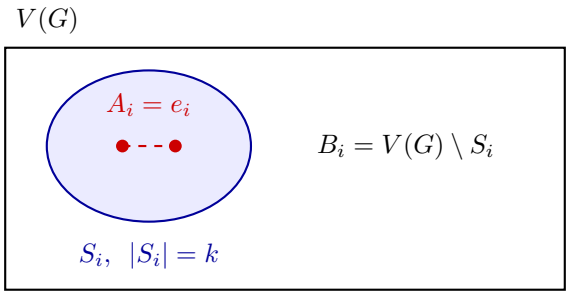
$$\mathrm{sat}(n, K_k) \le \binom{n}{2} - \binom{n-k+2}{2}.$$

It remains to prove the matching lower bound, and this is where set pairs enter: the missing edges of a saturated graph, together with the cliques they complete, form a set-pair system.

Proof of the lower bound in Theorem 1.8.9. Let G be any K_k-saturated graph on n vertices, and let $e_1, e_2, \dots, e_m$ be its missing edges, so that

$$m = \binom{n}{2} - e(G).$$

We construct a set-pair system. Put $A_i = e_i$, a set of two vertices. By saturation, adding e_i to G creates a k-clique, say on the vertex set S_i with $|S_i| = k$; this clique must use the new edge, since G itself is K_k-free, so $e_i \subseteq S_i$. Put $B_i = V(G) \setminus S_i$, a set of $n-k$ vertices.



We check the two conditions of Definition 1.7.1.

- $A_i \cap B_i = e_i \cap (V(G) \setminus S_i) = \emptyset$, since $e_i \subseteq S_i$.
- Let $i \neq j$ and suppose $A_i \cap B_j = \emptyset$, i.e. $e_i \subseteq S_j$. Adding e_j to G makes S_j a clique, so every pair inside S_j lies in $E(G) \cup \{e_j\}$. But e_i is a pair inside S_j that is neither an edge of G nor equal to e_j, so it is still missing and S_j is not a clique in $G + e_j$. Contradiction; hence $A_i \cap B_j \neq \emptyset$.

Thus $\{(A_i, B_i) : 1 \le i \le m\}$ is a set-pair system with $|A_i| = 2$ and $|B_i| = n-k$ for all i, and Theorem 1.7.2 gives

$$m \le \binom{2+(n-k)}{2} = \binom{n-k+2}{2} \Rightarrow e(G) = \binom{n}{2} - m \ge \binom{n}{2} - \binom{n-k+2}{2}.$$

As G was an arbitrary n-vertex K_k-saturated graph, combining this with the upper bound of Remark 1.8.11,

$$\binom{n}{2} - \binom{n-k+2}{2} \leq \mathrm{sat}(n, K_k) \leq \binom{n}{2} - \binom{n-k+2}{2} \Rightarrow \mathrm{sat}(n, K_k) = \binom{n}{2} - \binom{n-k+2}{2}.$$

■

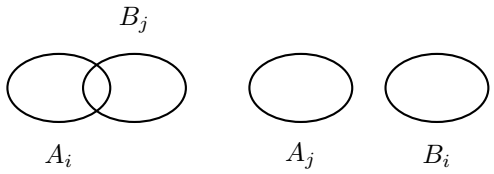
Remark 1.8.12.

(1) The same argument works for complete hypergraphs, with A_i the missing edge and B_i the complement of the clique it completes.

(2) **Weak saturation** asks instead that the missing edges can be added one by one *in some order*, each addition creating a new copy of H. Running the argument above only gives the weaker conditions

$$A_i \cap B_i = \emptyset, \qquad A_i \cap B_j \neq \emptyset \ \text{ for } i < j,$$

i.e. condition (2) of Definition 1.7.1 is required for one of the two orders only. It is still true that $m \leq \binom{k+\ell}{k}$ for such a *skew* set-pair system, but the random-ordering proof breaks down, because the events E_i are no longer pairwise disjoint, and the only known proofs are algebraic.



A_i must meet B_j for $i < j$, but A_j need not meet B_i.

2. Linearity of Expectation

Lecture 4

2 Oct.

So far every probabilistic argument had the same shape: list the bad events, bound the probability of each, and use the union bound to show that with positive probability none of them happens. This chapter replaces probabilities by *expectations*. The basic principle is that a random variable cannot always be above its average, and the tool that makes averages computable is linearity of expectation.

§2.1. Expectation and the First Moment Principle

Throughout, $(\Omega, \mathbb{P})$ is a probability space and $X\colon \Omega \to \mathbb{R}$ is a random variable. In all our applications Ω is finite, and then the **expectation** of X is its weighted average value

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega)\, \mathbb{P}(\omega).$$

§2.1.i. A Random Variable Is Not Always Above Its Mean

Lemma 2.1.1 (First moment principle)
The events $\{X \le \mathbb{E}[X]\}$ and $\{X \ge \mathbb{E}[X]\}$ both have positive probability. In particular

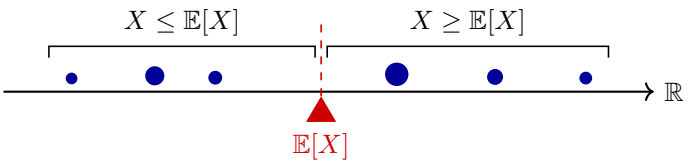
$$\exists\, \omega \in \Omega \text{ s.t. } X(\omega) \le \mathbb{E}[X] \qquad \text{and} \qquad \exists\, \omega' \in \Omega \text{ s.t. } X(\omega') \ge \mathbb{E}[X].$$

Proof. Suppose $\mathbb{P}(X \ge \mathbb{E}[X]) = 0$. Then $X(\omega) < \mathbb{E}[X]$ for every ω with $\mathbb{P}(\omega) > 0$, and averaging over these ω gives

$$\mathbb{E}[X] = \sum_{\omega : \mathbb{P}(\omega) > 0} X(\omega)\, \mathbb{P}(\omega) < \sum_{\omega : \mathbb{P}(\omega) > 0} \mathbb{E}[X]\, \mathbb{P}(\omega) = \mathbb{E}[X],$$

a contradiction. The other event is handled in the same way. ∎

Think of the values of X as weights on the real line: $\mathbb{E}[X]$ is the point where they balance, so there must be weight on each side of it, or on it.



This is how we will use it: to show that some object with a property exists, define a random object and a random variable X measuring the property, compute $\mathbb{E}[X]$, and conclude that some outcome does at least as well as the average.

§2.1.ii. Linearity

Computing $\mathbb{E}[X]$ directly is usually hopeless, since the distribution of X is complicated. The point is that we never need the distribution: we break X into simple pieces.

Theorem 2.1.2 (Linearity of expectation)
If $X = \sum_{i=1}^{n} \alpha_i X_i$, where the X_i are random variables and the α_i are constants, then

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^{n} \alpha_i X_i\right] = \sum_{i=1}^{n} \alpha_i \, \mathbb{E}[X_i].$$

Proof. Exchange the two finite sums:

$$\mathbb{E}[X] = \sum_{\omega} \left(\sum_{i=1}^{n} \alpha_i X_i(\omega)\right) \mathbb{P}(\omega) = \sum_{i=1}^{n} \alpha_i \sum_{\omega} X_i(\omega)\, \mathbb{P}(\omega) = \sum_{i=1}^{n} \alpha_i \, \mathbb{E}[X_i]. \qquad \blacksquare$$

Remark 2.1.3. This holds for *any* random variables X_i; no independence is needed. The proof only ever looks at one outcome ω at a time, so it does not care how the X_i are related to each other. This is what makes linearity so powerful: the pieces X_i can be as dependent as they like, as in both examples below.

The pieces we use are almost always indicators.

Definition 2.1.4 (Indicator)**.** The **indicator random variable** of an event E is

$$\mathbb{1}_E = \begin{cases} 1, & \text{if } E \text{ occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

It satisfies $\mathbb{E}[\mathbb{1}_E] = 1 \cdot \mathbb{P}(E) + 0 \cdot \mathbb{P}(E^c) = \mathbb{P}(E)$.

So if X counts how many of the events $E_1, \ldots, E_m$ occur, then $X = \sum_i \mathbb{1}_{E_i}$, and by Theorem 2.1.2

$$\mathbb{E}[X] = \sum_{i=1}^{m} \mathbb{E}[\mathbb{1}_{E_i}] = \sum_{i=1}^{m} \mathbb{P}(E_i).$$

The expected number of events that occur is the sum of their probabilities, whatever the dependencies between them.

§2.1.iii. Two First Examples

The first example shows that the union bound arguments of the previous chapter are a special case of the first moment principle.

Example 2.1.5 (Union bound)**.** Let $E_1, E_2, \ldots, E_m$ be bad events, e.g. in Erdős's lower bound for $R(k)$ (Theorem 1.2.5), the events that a given k-set of vertices spans a monochromatic clique in a random coloring of K_n. Let

$$X = \#\{\text{bad events that occur}\} = \sum_{i=1}^{m} \mathbb{1}_{E_i} \Rightarrow \mathbb{E}[X] = \sum_{i=1}^{m} \mathbb{E}[\mathbb{1}_{E_i}] = \sum_{i=1}^{m} \mathbb{P}(E_i).$$

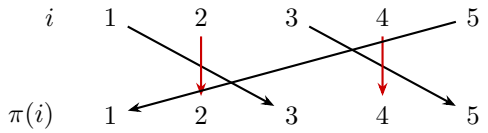
If $\sum_i \mathbb{P}(E_i) < 1$, then $\mathbb{E}[X] < 1$, and by Lemma 2.1.1, with positive probability $X \leq \mathbb{E}[X] < 1$. As X takes values in $\{0, 1, 2, \ldots\}$, $X < 1 \Leftrightarrow X = 0$, i.e. with positive probability no bad event occurs. This is exactly the conclusion of the union bound.

In fact the union bound itself follows. Since X is a non-negative integer, $\mathbb{1}_{\{X\geq 1\}} \leq X$ pointwise, and taking expectations,

$$\mathbb{P}\left(\bigcup_{i=1}^{m} E_i\right) = \mathbb{P}(X \geq 1) = \mathbb{E}\left[\mathbb{1}_{\{X\geq 1\}}\right] \leq \mathbb{E}[X] = \sum_{i=1}^{m} \mathbb{P}(E_i).$$

Example 2.1.6 (Fixed points of permutations)**.** Let $\pi \in S_n$ be a uniformly random permutation of $[n]$, and let

$$X(\pi) = \#\{\text{fixed points of } \pi\} = \#\{i \in [n] : \pi(i) = i\}.$$



$\pi = (3\ 2\ 5\ 4\ 1)$
fixed points $2, 4$ in red, $X(\pi) = 2$

The distribution of X is complicated (already $\mathbb{P}(X = 0)$ needs inclusion–exclusion), but we only want its mean. Write X as a sum of indicators, one per position:

$$X = \sum_{i=1}^{n} \mathbb{1}_{\{i \text{ is a fixed point}\}}.$$

For each i, $\pi(i)$ is uniform on $[n]$, so $\mathbb{P}(\pi(i) = i) = \frac{(n-1)!}{n!} = \frac{1}{n}$. By linearity,

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{P}(i \text{ is a fixed point}) = \sum_{i=1}^{n} \frac{1}{n} = 1,$$

for every n. Note that the events $\{\pi(i) = i\}$ are not independent: if $n - 1$ of them occur, so does the last one. Linearity does not notice.

§2.2. Sum-Free Sets

The first real application of the first moment principle is to a problem in additive combinatorics.

§2.2.i. The Problem

Definition 2.2.1 (Sum-free set)**.** A **sum** is a triple of integers (x, y, z) s.t. $x + y = z$. A set $A \subseteq \mathbb{Z}$ is **sum-free** if there is no sum in A^3, i.e. there are no $x, y, z \in A$ with $x + y = z$.

Here x, y, z need not be distinct: a sum-free set cannot contain both x and $2x$, and it never contains $0 = 0 + 0$. The same definition makes sense in any abelian group, and we will use it in $\mathbb{Z}_p$.

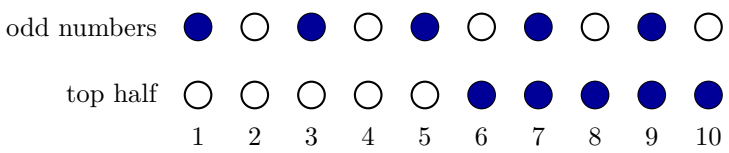
Definition 2.2.2. Given a finite set $S \subseteq \mathbb{Z}$, let

$$f(S) = \max\{|A| : A \subseteq S \text{ is sum-free}\}$$

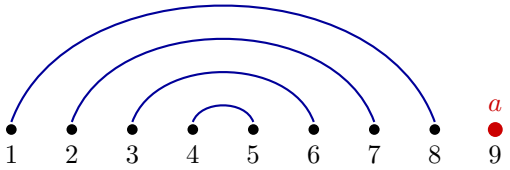
be the size of a largest sum-free subset of S.

Example 2.2.3 (Warm-up: $f([n]) = \lceil \frac{n}{2} \rceil$)**.** *Lower bound.* Two sum-free subsets of $[n]$ of size $\lceil \frac{n}{2} \rceil$ are

- the odd numbers, since odd + odd is even;
- the numbers $> \frac{n}{2}$, since the sum of two of them exceeds n and so is not in the set.



Upper bound. Let $A \subseteq [n]$ be sum-free and $a = \max A$. For every $x < a$, at most one of x and $a - x$ lies in A, since $x + (a - x) = a \in A$. The pairs $\{x, a - x\}$ cover $[a - 1]$:



pairs $\{x, a - x\}$, here $a = 9$:
A takes at most one from each pair

If a is odd there are $\frac{a-1}{2}$ pairs, so $|A| \leq \frac{a-1}{2} + 1 = \frac{a+1}{2}$. If a is even there are $\frac{a-2}{2}$ pairs and the middle element $\frac{a}{2}$ is excluded on its own, as $\frac{a}{2} + \frac{a}{2} = a$, so $|A| \leq \frac{a-2}{2} + 1 = \frac{a}{2}$. Either way $|A| \leq \frac{a+1}{2} \leq \frac{n+1}{2}$, i.e. $|A| \leq \lceil \frac{n}{2} \rceil$.

So an interval always contains a sum-free half. The interval is very structured, though, and both constructions used that structure. What if the numbers are arbitrary?

Question 2.2.4. Does *every* set of n integers contain a large sum-free subset?

Definition 2.2.5. For $n \in \mathbb{N}$, let[1]

$$f(n) = \min\{f(S) : S \subseteq \mathbb{Z} \setminus \{0\},\ |S| = n\}.$$

Remark 2.2.6 (Why the minimum)**.** The maximum is trivially n: take S itself sum-free, e.g. $a,\ b > 2a,\ 2b+1,\ 4b+3, \ldots$ with $a > 0$, each element larger than twice the previous one, hence not a sum of two earlier ones.[2]

Observation . $f(n) \leq f([n]) = \lceil \frac{n}{2} \rceil$.

Now S may be any set of non-zero integers, including negative ones, with no structure to exploit. Yet a constant fraction of it can always be kept.

Theorem 2.2.7 (Erdős, 1965)
$f(n) \geq \frac{n+1}{3}$. That is, every set of n non-zero integers contains a sum-free subset of size at least $\frac{n+1}{3}$.

Proof. Let $S \subseteq \mathbb{Z} \setminus \{0\}$ with $|S| = n$.

Lemma 2.2.8
There are infinitely many primes $p \equiv 2 \pmod 3$.

Proof. Given such primes $p_1, \ldots, p_m$, let $N = 3p_1 \cdots p_m - 1 \equiv 2 \pmod 3$. Then $3 \nmid N$, and not all prime factors of N are $\equiv 1 \pmod 3$, else $N \equiv 1$. So some prime $q \equiv 2 \pmod 3$ divides N, and $q \neq p_i$ since $p_i \mid N + 1$. ∎

[1] We exclude 0, which can never lie in a sum-free set.
[2] $x = y$ is allowed, so beating $a + b$ is not enough: $1 + 1 = 2$.

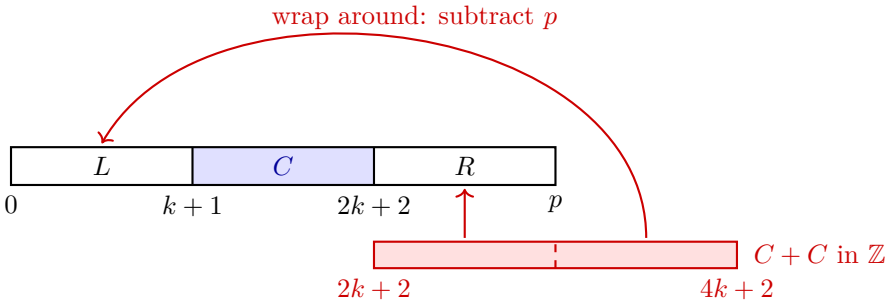
By Lemma 2.2.8, choose a prime $p = 3k + 2$ with $p > \max_{s \in S} |s|$, so $p \nmid s$ for all $s \in S$. We identify $\mathbb{Z}_p$ with $\{0, 1, \ldots, p-1\}$, and for an integer m write $m \bmod p \in \{0, 1, \ldots, p-1\}$ for its remainder. Let

$$C = \{k+1, k+2, \ldots, 2k+1\}.$$

Lemma 2.2.9

C is sum-free in $\mathbb{Z}_p$, and $\dfrac{|C|}{p-1} = \dfrac{k+1}{3k+1} > \dfrac{1}{3}$.

Proof. For $c_1, c_2 \in C$ we have $2k+2 \le c_1+c_2 \le 4k+2$. If $c_1+c_2 \le 3k+1$, then $(c_1+c_2) \bmod p = c_1 + c_2 \in R = \{2k+2, \ldots, 3k+1\}$; otherwise $(c_1 + c_2) \bmod p = c_1 + c_2 - p \in L = \{0, \ldots, k\}$. Either way $(c_1 + c_2) \bmod p \notin C$. Finally $|C| = k + 1$ and $p - 1 = 3k + 1$. ∎



Let x be uniformly random in $\{1, 2, \ldots, p-1\}$, i.e. $\mathbb{P}(x = a) = \frac{1}{p-1}$ for each a, and

$$A_x = \{s \in S : xs \bmod p \in C\} \subseteq S.$$

We show that every A_x is sum-free, and that $|A_x| \ge \frac{n+1}{3}$ for some x; this A_x then proves the theorem.

A_x is sum-free: if $s_1 + s_2 = s_3$ with $s_1, s_2, s_3 \in A_x$, then $xs_1 + xs_2 = xs_3$, so $(xs_1 \bmod p) + (xs_2 \bmod p) \equiv xs_3 \bmod p \pmod p$ with all three remainders in C, contradicting Lemma 2.2.9.

Fix $s \in S$. We claim that $xs \bmod p$ takes each value $1, 2, \ldots, p-1$ with probability exactly $\frac{1}{p-1}$. Consider

$$\varphi_s \colon \{1, \ldots, p-1\} \to \{1, \ldots, p-1\}, \qquad a \mapsto as \bmod p.$$

- It is well defined, i.e. $as \bmod p \neq 0$: p is prime and $p \nmid a$, $p \nmid s$, so $p \nmid as$.
- It is injective: if $as \bmod p = a's \bmod p$, then $p \mid (a - a')s$, so $p \mid a - a'$ as $p \nmid s$; since $|a - a'| < p$, $a = a'$.
- An injection of a finite set into itself is a bijection.

So for each $y \in \{1, \ldots, p-1\}$ there is exactly one a with $as \bmod p = y$, and

$$\mathbb{P}(xs \bmod p = y) = \mathbb{P}(x = \varphi_s^{-1}(y)) = \frac{1}{p-1}.$$

Summing over $y \in C$,

$$\mathbb{P}(xs \bmod p \in C) = \sum_{y \in C} \frac{1}{p-1} = \frac{|C|}{p-1} = \frac{k+1}{3k+1} > \frac{1}{3}.$$

Writing $|A_x| = \sum_{s\in S} \mathbb{1}_{\{xs \bmod p \in C\}}$, linearity of expectation gives

$$\mathbb{E}\big[|A_x|\big] = \sum_{s\in S} \mathbb{P}(xs \bmod p \in C) > \frac{n}{3}.$$

By Lemma 2.1.1 some x has $|A_x| > \frac{n}{3}$, and as $|A_x|$ is an integer, $|A_x| \geq \frac{n+1}{3}$. ■

Remark 2.2.10. So $\frac{n+1}{3} \leq f(n) \leq \lceil \frac{n}{2} \rceil$. Bourgain (1997) improved the lower bound to $\frac{n+2}{3}$, and Eberhard, Green and Manners (2014) showed that the constant $\frac{1}{3}$ cannot be improved: $f(n) \leq \big(\frac{1}{3} + o(1)\big)n$. So for arbitrary sets the random dilation is essentially optimal, and the interval $[n]$, with its sum-free half, is far from the worst case.

Lecture 5

6 Oct.

Recall . A **tournament** T is an orientation of K_n: for every pair of vertices exactly one of the arcs (u, v), (v, u) belongs to $E(T)$, and we say u **beats** v if $(u, v) \in E(T)$. See Section 1.5.

In Section 2.2 the first moment principle produced *one* object of a given kind: a single large sum-free subset. The same principle also produces objects in *quantity*: if the average number of copies of some structure is large, then some object contains at least that many copies. This section carries out the oldest example of this, which is also the oldest application of the probabilistic method at all.

§2.3. Hamilton Paths in Tournaments

§2.3.i. Every Tournament Has One

Definition 2.3.1 (Hamilton path)**.** Given a tournament T on n vertices $v_1, \dots, v_n$, a **Hamilton path** is a directed path that goes through every vertex exactly once, i.e. a sequence

$$v_{\pi(1)}, v_{\pi(2)}, \dots, v_{\pi(n)}, \qquad \text{where } (v_{\pi(i)}, v_{\pi(i+1)}) \in E(T)\ \forall\, 1 \leq i < n \text{ and } \pi \in S_n.$$

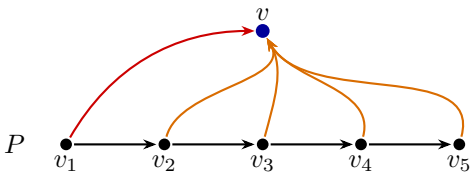
So a Hamilton path is an ordering of the players in which each one beats the next. Not every digraph has one; tournaments always do, because having *all* pairs oriented leaves a vertex nowhere to hide.

Claim 2.3.1. Every tournament has a Hamilton path.

Proof. A tournament is finite, so we may take a longest directed path

$$P = v_1 \to v_2 \to \cdots \to v_t,$$

and it suffices to show $t = n$. Suppose not, and let $v \notin P$.



first v_1 must beat v (red),
then so must every v_i (orange)

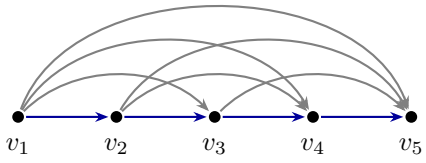
- If $(v, v_1) \in E(T)$, then $v \to v_1 \to v_2 \to \cdots \to v_t$ is a longer path. Hence $(v_1, v) \in E(T)$.

- If $(v_i, v) \in E(T)$ for some $i < t$, then $(v_{i+1}, v) \in E(T)$ as well: otherwise $(v, v_{i+1}) \in E(T)$, and
$$v_1 \to \cdots \to v_i \to v \to v_{i+1} \to \cdots \to v_t$$
is a longer path.

Starting from $(v_1, v) \in E(T)$ and applying the second item repeatedly gives $(v_t, v) \in E(T)$, so $v_1 \to \cdots \to v_t \to v$ is a longer path. ↯ ■

Note that the proof shows more than existence: *every* longest path is already a Hamilton path. The next example shows that the number of Hamilton paths can be as small as 1.

Example 2.3.2 (Transitive tournament)**.** The **transitive tournament** on $v_1, \ldots, v_n$ is the one with
$$(v_i, v_j) \in E(T) \Leftrightarrow i < j,$$
i.e. the players are linearly ranked and the better one always wins. It has a unique Hamilton path, namely $v_1 \to v_2 \to \cdots \to v_n$: if $v_{\pi(1)} \to \cdots \to v_{\pi(n)}$ is a Hamilton path, then $\pi(i) < \pi(i+1)$ for every i, so π is increasing, i.e. $\pi = \mathrm{id}$.



all arcs point to the right, so the only Hamilton path is the left-to-right one

§2.3.ii. Szele's Problem

So one Hamilton path always exists, and sometimes there is exactly one. How many can there be?

Question 2.3.3 (Szele, 1943)**.** What is the maximum number of Hamilton paths an n-vertex tournament can have?

Observation . Since a Hamilton path is specified by the ordering $\pi \in S_n$ in which it visits the vertices, there are at most $n!$ possible paths.

Definition 2.3.4. Let $P(n) = \max\{\#\text{Hamilton paths in } T : T \text{ an } n\text{-vertex tournament}\}$.

By Example 2.3.2 and the observation above we have $1 \leq P(n) \leq n!$, and the transitive tournament shows the lower bound is attained by *some* tournament. The question is how large $P(n)$ is, and this is where a random tournament comes in: no explicit construction is needed, because the average over all tournaments is already large.

Proposition 2.3.5 (Szele, 1943)
$$P(n) \geq \frac{n!}{2^{n-1}}.$$

Proof. Consider a uniformly random tournament T on the vertex set $[n]$: for every pair $i \neq j$ we choose (i, j) or (j, i) uniformly at random, independently for each pair. Let
$$X = \#\{\text{Hamilton paths in } T\}.$$

For every permutation $\pi \in S_n$, let E_π be the event that $\pi(1), \pi(2), \ldots, \pi(n)$ is a Hamilton path of T, and let $X_\pi = \mathbb{1}_{E_\pi}$. Every Hamilton path is counted by exactly one π, so $X = \sum_{\pi \in S_n} X_\pi$ and, by Theorem 2.1.2,

$$\mathbb{E}[X] = \sum_{\pi \in S_n} \mathbb{E}[X_\pi] = \sum_{\pi \in S_n} \mathbb{P}(E_\pi).$$

Now

$$E_\pi = \big\{(\pi(1), \pi(2)) \in E(T)\big\} \cap \big\{(\pi(2), \pi(3)) \in E(T)\big\} \cap \cdots \cap \big\{(\pi(n-1), \pi(n)) \in E(T)\big\}$$

is an intersection of $n-1$ events about $n-1$ *distinct* pairs, which are independent and each of probability $\frac{1}{2}$, so $\mathbb{P}(E_\pi) = \left(\frac{1}{2}\right)^{n-1}$. Hence

$$\mathbb{E}[X] = \sum_{\pi \in S_n} \mathbb{P}(E_\pi) = n! \cdot \left(\frac{1}{2}\right)^{n-1} = \frac{n!}{2^{n-1}},$$

and by Lemma 2.1.1 some tournament T has $X(T) \geq \mathbb{E}[X] = \frac{n!}{2^{n-1}}$ Hamilton paths. ∎

Remark 2.3.6.

(1) The events E_π are heavily dependent — two permutations can share many pairs — and their number $n!$ is enormous. Linearity does not care, exactly as in Example 2.1.6.

(2) Recall Stirling's formula $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$. So

$$\frac{n!}{2^{n-1}} \sim 2\sqrt{2\pi n}\left(\frac{n}{2e}\right)^n,$$

which is still superexponentially large: a positive proportion $2^{-(n-1)}$ of all $n!$ conceivable orderings are realised simultaneously by one tournament.

(3) This is the first published use of the probabilistic method, predating Erdős's lower bound for $R(k)$ by four years. Note how little is proved about the good tournament T: we never see it.

Szele also proved an upper bound, $P(n) \leq \frac{n!}{2^{\frac{3}{4}(n-1)}}$, leaving a gap in the exponent between $\frac{3}{4}(n-1)$ and $n-1$, and conjectured that the lower bound has the truth:

$$P(n) = \frac{n!}{2^{(1-o(1))(n-1)}}, \qquad \text{equivalently} \qquad \lim_{n\to\infty}\left(\frac{P(n)}{n!}\right)^{1/n} = \frac{1}{2}.$$

This was settled almost fifty years later, and in a strong form: the random tournament is optimal up to a polynomial factor.

Theorem 2.3.7 (Alon, 1990)

There exists a constant $c > 0$ s.t.

$$P(n) \leq c\, n^{3/2} \cdot \frac{n!}{2^{n-1}}.$$

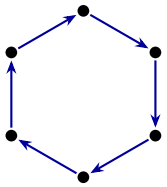
§2.3.iii. From Paths to Cycles to Permanents

The proof of Theorem 2.3.7 replaces the quantity we want to bound by one that algebra can see. Hamilton paths themselves are awkward: they have two loose ends, and no matrix identity counts them. Hamilton *cycles* are better, and *cycle covers* are better still, because they are exactly the terms of a permanent.

Definition 2.3.8. Given an n-vertex tournament T, let

- $P(T)$ be the number of Hamilton paths of T;
- $C(T)$ be the number of Hamilton cycles of T, where a **Hamilton cycle** is a cyclic ordering $(v_1, \dots, v_n)$ of $V(T)$ with $(v_i, v_{i+1}) \in E(T)$ for all $1 \le i \le n$, indices mod n;
- $F(T)$ be the number of spanning subgraphs of T in which every vertex has in-degree and out-degree 1.

A subgraph counted by $F(T)$ keeps all n vertices and uses exactly n arcs, one leaving and one entering each vertex; following the arcs from any vertex must return to it, so such a subgraph is precisely a disjoint union of directed cycles covering all vertices, a **cycle cover**. A Hamilton cycle is the special case of a cycle cover consisting of a single cycle.



a Hamilton cycle:
one cycle



a cycle cover:
several disjoint cycles

Observation .

(1) $C(T) \le F(T)$.

(2) $n \cdot C(T) \le P(T)$.

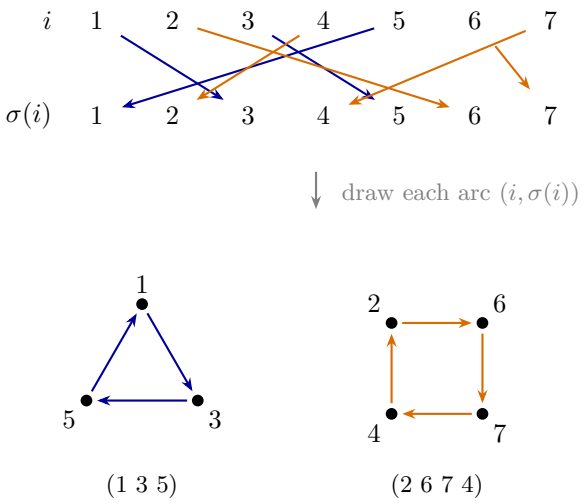
Proof. (1) Every Hamilton cycle is a cycle cover, by the discussion above, and distinct Hamilton cycles are distinct subgraphs.

(2) Given a Hamilton cycle and one of its n arcs, deleting that arc leaves a Hamilton path. The map is injective: from the resulting path $v_1 \to \cdots \to v_n$ we recover the cycle by adding back the arc (v_n, v_1), and that arc is the one deleted. So the $n \cdot C(T)$ pairs give $n \cdot C(T)$ distinct Hamilton paths. ∎

The plan is therefore to bound $F(T)$, and this is possible because a cycle cover is a permutation drawn as a picture: draw the arc $(i, \sigma(i))$ for every i, and the cycle representation of σ turns into disjoint directed cycles covering $[n]$.

Example 2.3.9. Let $\sigma = (1\ 3\ 5)(2\ 6\ 7\ 4) \in S_7$, i.e.

$$\sigma(1) = 3, \quad \sigma(2) = 6, \quad \sigma(3) = 5, \quad \sigma(4) = 2, \quad \sigma(5) = 1, \quad \sigma(6) = 7, \quad \sigma(7) = 4.$$



Lemma 2.3.10 (Permutations are cycle covers)

For $\sigma \in S_n$, let D_σ be the digraph on the vertex set $[n]$ with arc set $\{(i, \sigma(i)) : i \in [n]\}$. Then

$$\sigma \longmapsto D_\sigma$$

is a bijection from S_n onto the set of digraphs on $[n]$ in which every vertex has in-degree and out-degree 1, and the directed cycles of D_σ are exactly the cycles of σ.

Proof. In D_σ, the only arc leaving i is $(i, \sigma(i))$ and the only arc entering j is $(\sigma^{-1}(j), j)$, so all degrees are 1, and the directed cycles of D_σ are the cycles of σ read as paths $i \to \sigma(i) \to \sigma^2(i) \to \cdots$. Conversely, given H with all in- and out-degrees 1, let $\sigma_H(i)$ be the unique out-neighbour of i in H; it is injective, as $\sigma_H(i) = \sigma_H(i')$ would give two arcs into that vertex, hence $\sigma_H \in S_n$. The two constructions are inverse to each other. ∎

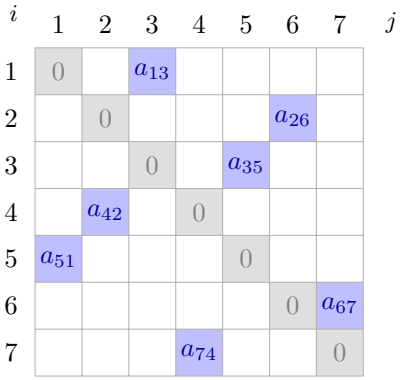
So cycle covers of $[n]$ *are* the permutations of $[n]$, there being $n!$ of them in total. Counting $F(T)$ is therefore counting those σ all of whose n arcs happen to lie in T, and that condition is read off from a matrix.

Definition 2.3.11 (Adjacency matrix)**.** The **adjacency matrix** of a tournament T on $[n]$ is the $n \times n$ matrix $A = (a_{ij})$ over $\{0, 1\}$ with

$$a_{ij} = \begin{cases} 1, & \text{if } (i,j) \in E(T), \\ 0, & \text{if } (i,j) \notin E(T). \end{cases}$$

In particular $a_{ii} = 0$ and $a_{ij} + a_{ji} = 1$ for $i \neq j$.

In the matrix, choosing a permutation σ means choosing n cells $(i, \sigma(i))$, one in each row and one in each column; the arc $(i, \sigma(i))$ lies in T exactly when the chosen cell holds a 1. For the σ of Example 2.3.9 the chosen cells are



one cell per row and per column;
the shaded diagonal is forced to 0,
so no cell on it is ever chosen

$D_\sigma \subseteq T \Leftrightarrow$ all seven
chosen cells are 1
$\Leftrightarrow \prod_i a_{i\sigma(i)} = 1$

Summing this indicator over all σ is exactly the following classical quantity.

Definition 2.3.12 (Permanent)**.** For an $n \times n$ matrix A,

$$\operatorname{per}(A) = \sum_{\sigma \in S_n} \prod_{i=1}^{n} a_{i\sigma(i)}.$$

Lemma 2.3.13
$F(T) = \operatorname{per}(A)$.

Proof. By Lemma 2.3.10, the subgraphs counted by $F(T)$ are exactly the D_σ with $\sigma \in S_n$ that happen to be subgraphs of T, each arising from one σ. Since every $a_{ij} \in \{0, 1\}$,

$$D_\sigma \subseteq T \Leftrightarrow a_{i\sigma(i)} = 1 \ \forall\, i \Leftrightarrow \prod_{i=1}^{n} a_{i\sigma(i)} = 1,$$

and the product is 0 for every other σ. So the product is the indicator of $\{D_\sigma \subseteq T\}$, and summing it over S_n counts the cycle covers of T:

$$F(T) = \sum_{\sigma \in S_n} \prod_{i=1}^{n} a_{i\sigma(i)} = \operatorname{per}(A). \qquad \blacksquare$$

Remark 2.3.14. The permutations contributing to $\operatorname{per}(A)$ have no fixed points, since $a_{ii} = 0$, and no 2-cycles, since $a_{ij}a_{ji} = 0$ for $i \neq j$ in a tournament. So all cycles of a cycle cover have length ≥ 3, as they must.

The permanent differs from the determinant

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i\sigma(i)}$$

only in the signs, yet the two behave completely differently: $\det(A)$ is computable in $O(n^3)$ steps by Gaussian elimination, because row operations are available, while computing $\operatorname{per}(A)$ exactly is #P-hard even for $\{0, 1\}$ matrices (Valiant, 1979). So we should not expect a formula for $F(T)$. Fortunately we do not need one: an upper bound on $\operatorname{per}(A)$ in terms of the row sums of A will do, and those row sums are the out-degrees of T, which sum to $\binom{n}{2}$.

§2.3.iv. Brégman's Theorem

So far the chain of inequalities reads

$$C(T) \le F(T) = \operatorname{per}(A), \qquad A = A_T \text{ the adjacency matrix of } T,$$

and the tournament has disappeared: what is left is a question about $\{0,1\}$ matrices.

Question 2.3.15. How large can the permanent of an $n \times n$ matrix over $\{0,1\}$ be?

Let $r_i = \sum_{j=1}^{n} a_{ij}$ be the number of 1s in the i-th row. A permutation σ contributing to $\operatorname{per}(A)$ must pick a 1 in every row, and there are r_i choices in row i, so

$$\operatorname{per}(A) \le r_1 r_2 \cdots r_n.$$

This overcounts badly, because the choices in different rows may land in the same column, and then they do not form a permutation at all. The following theorem quantifies exactly how much is lost.

Theorem 2.3.16 (Minc's conjecture, 1963; Brégman, 1973)
For any $n \times n$ matrix A over $\{0,1\}$ with row sums $r_i = \sum_{j=1}^{n} a_{ij}$,

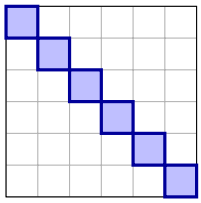
$$\operatorname{per}(A) \le \prod_{i=1}^{n} (r_i!)^{1/r_i}.$$

Remark 2.3.17.

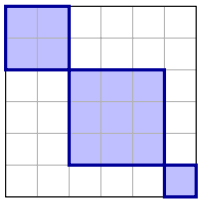
(1) This is sharper than the trivial bound: $r_i! \le r_i^{r_i}$ gives $(r_i!)^{1/r_i} \le r_i$. The gain per row is roughly a factor e, since $(r!)^{1/r} \approx r/e$.

(2) A row with $r_i = 0$ forces $\operatorname{per}(A) = 0$, so we may assume $r_i \ge 1$ throughout.

(3) The bound is best possible: let A be block diagonal with all-1 blocks of sizes $d_1, \dots, d_m$, where $d_1 + \cdots + d_m = n$. A permutation contributes iff it preserves each block, so $\operatorname{per}(A) = \prod_j d_j!$; and every row inside the j-th block has $r_i = d_j$, so $\prod_i (r_i!)^{1/r_i} = \prod_j \big((d_j!)^{1/d_j}\big)^{d_j} = \prod_j d_j!$ as well.



sizes $(1,1,1,1,1,1)$
per $= 1$



sizes $(2,3,1)$
per $= 2!\,3!\,1! = 12$

§2.3.v. Balancing the Out-Degrees

We apply Theorem 2.3.16 to the adjacency matrix of a tournament. There r_i is the out-degree of the i-th vertex, and since every edge of K_n is oriented exactly once,

$$\sum_{i=1}^{n} r_i = e(T) = \binom{n}{2}.$$

This is the only information we have about the r_i, so we must maximise the Brégman bound subject to it.

Question 2.3.18. Given n and S, if $r_1 + \cdots + r_n = S$ with all $r_i \geq 1$, how big can $\prod_{i=1}^{n} (r_i!)^{1/r_i}$ be?

Claim 2.3.2. The product is maximised when the r_i are as equal as possible, i.e. when every r_i is $\lfloor \frac{S}{n} \rfloor$ or $\lceil \frac{S}{n} \rceil$.

Proof. Write $g(x) = (x!)^{1/x}$ for $x \geq 1$. It suffices to show that if two of the r_i are a and b with $a \leq b-2$, then replacing them by $a+1$ and $b-1$ strictly increases the product, i.e.

$$g(a)\, g(b) < g(a+1)\, g(b-1) \Leftrightarrow \frac{g(a)}{g(a+1)} < \frac{g(b-1)}{g(b)}.$$

So with $f(x) \coloneqq \dfrac{g(x)}{g(x+1)}$ it is enough that f is strictly increasing, since $a \leq b-2 < b-1$.

Fix $x \geq 2$. Then

$$f(x-1) < f(x) \Leftrightarrow g(x-1)\, g(x+1) < g(x)^2 \Leftrightarrow \big((x-1)!\big)^{\frac{1}{x-1}} \big((x+1)!\big)^{\frac{1}{x+1}} < (x!)^{\frac{2}{x}}.$$

Raise both sides to the power $x(x-1)(x+1) > 0$ and substitute $(x-1)! = \frac{x!}{x}$ and $(x+1)! = (x+1) \cdot x!$:

$$\Big(\frac{x!}{x}\Big)^{x(x+1)} \big((x+1)\, x!\big)^{x(x-1)} < (x!)^{2(x^2-1)} \Leftrightarrow (x!)^{2x^2} \frac{(x+1)^{x(x-1)}}{x^{x(x+1)}} < (x!)^{2x^2-2},$$

i.e.

$$\Big(\frac{x+1}{x}\Big)^{x(x-1)} < \Big(\frac{x^x}{x!}\Big)^2. \tag{$*$}$$

Now AM–GM applied to $x! = 1 \cdot 2 \cdots x$ gives

$$x! \leq \Big(\frac{1+2+\cdots+x}{x}\Big)^x = \Big(\frac{x+1}{2}\Big)^x \Rightarrow \Big(\frac{x^x}{x!}\Big)^2 \geq \Big(\frac{2x}{x+1}\Big)^{2x} = 4^x \Big(\frac{x}{x+1}\Big)^{2x},$$

so the ratio of the two sides of $(*)$ is

$$\frac{(x^x/x!)^2}{\big(\frac{x+1}{x}\big)^{x(x-1)}} \;\geq\; 4^x \Big(\frac{x}{x+1}\Big)^{2x+x(x-1)} = 4^x \Big(\frac{x}{x+1}\Big)^{x(x+1)} = \Big[4\Big(1 - \frac{1}{x+1}\Big)^{x+1}\Big]^x.$$

As $x \geq 2$ we have $x+1 \geq 3$, and $m \mapsto (1 - \frac{1}{m})^m$ is increasing,[3] so $(1 - \frac{1}{x+1})^{x+1} \geq (\frac{2}{3})^3 = \frac{8}{27} > \frac{1}{4}$ and the bracket exceeds 1. Hence $(*)$ holds and f is strictly increasing. ∎

Lemma 2.3.19

There is an absolute constant $c_1 > 0$ s.t. every n-vertex tournament T satisfies

$$C(T) \leq F(T) = \operatorname{per}(A) \leq c_1 \sqrt{n} \cdot \frac{n!}{2^n}.$$

Proof. The out-degrees satisfy $\sum_i r_i = \binom{n}{2}$, with average $\frac{n-1}{2}$, so by Claim 2.3.2 the Brégman bound is largest when every r_i is $\lfloor \frac{n-1}{2} \rfloor$ or $\lceil \frac{n-1}{2} \rceil$. Also $g(x) = (x!)^{1/x}$ is increasing, since

$$g(x)^{x(x+1)} = (x!)^{x+1} \leq (x!)^x (x+1)^x = \big((x+1)!\big)^x = g(x+1)^{x(x+1)},$$

[3] It increases to e^{-1}, so it is always *below* e^{-1} and e^{-1} cannot be used as a lower bound here; any constant beating $\frac{1}{4}$ suffices, and $\frac{8}{27}$ does.

so bounding every factor by the larger one gives, with $R = \lceil \frac{n-1}{2} \rceil \le \frac{n}{2}$,

$$\operatorname{per}(A) \le \prod_{i=1}^{n} (r_i!)^{1/r_i} \le \big((R!)^{1/R}\big)^n.$$

Stirling's formula in the form $R! = \sqrt{2\pi R}\,(\frac{R}{e})^R e^{\theta/(12R)}$ with $0 < \theta < 1$ gives

$$\big((R!)^{1/R}\big)^n = \left(\frac{R}{e}\right)^n (2\pi R)^{\frac{n}{2R}} e^{\frac{\theta n}{12R^2}}.$$

Here $\frac{n}{2R} \le \frac{n}{n-1} \le 2$ for $n \ge 2$, so $(2\pi R)^{n/(2R)} \le (\pi n)^{n/(n-1)} = O(n)$, and $e^{\theta n/(12R^2)} = e^{O(1/n)} = O(1)$. Hence

$$\operatorname{per}(A) \le O(n) \cdot \left(\frac{n}{2e}\right)^n,$$

and Stirling once more, now for $n!$, turns $\left(\frac{n}{2e}\right)^n = \frac{1}{2^n}\left(\frac{n}{e}\right)^n = (1 + o(1))\frac{n!}{\sqrt{2\pi n}\,2^n}$ into

$$\operatorname{per}(A) \le O(n) \cdot O\Big(\frac{n!}{\sqrt{n}\,2^n}\Big) = O\Big(\sqrt{n} \cdot \frac{n!}{2^n}\Big). \qquad \blacksquare$$

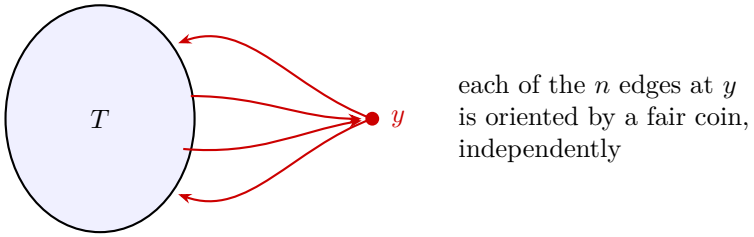
§2.3.vi. Proof of Alon's Theorem

Collecting what we have, every n-vertex tournament satisfies

$$C(T) \le F(T) = \operatorname{per}(A_T) \le c_1\sqrt{n} \cdot \frac{n!}{2^n} \qquad \text{and} \qquad n \cdot C(T) \le P(T).$$

The second inequality points the wrong way: it bounds cycles by paths, whereas we want to bound paths. The fix is to turn the paths of T into cycles by adding one vertex, and this is done at random, exactly as in Proposition 2.3.5.

Proof of Theorem 2.3.7. Let T be an n-vertex tournament. Construct a random tournament $T^* \supseteq T$ on $n+1$ vertices by adding a new vertex y and orienting each of the n edges at y uniformly at random, independently of each other.



Every Hamilton cycle of T^* passes through y, and deleting y from it leaves a Hamilton path of T. Conversely, a Hamilton path $v_1 \to \cdots \to v_n$ of T closes up into a Hamilton cycle of T^* exactly when $(v_n, y) \in E(T^*)$ and $(y, v_1) \in E(T^*)$, which are two of the n independent coin flips, so this happens with probability $\frac{1}{4}$. Writing $C(T^*)$ as a sum of indicators over the $P(T)$ Hamilton paths of T, linearity of expectation gives

$$\mathbb{E}\big[C(T^*)\big] = \sum_{\text{Hamilton paths of } T} \mathbb{P}(\text{the path closes up}) = \frac{P(T)}{4}.$$

By Lemma 2.1.1 some outcome T^* has $C(T^*) \ge \frac{P(T)}{4}$. Applying Lemma 2.3.19 to this $(n+1)$-vertex tournament,

$$\frac{P(T)}{4} \le C(T^*) \le c_1\sqrt{n+1} \cdot \frac{(n+1)!}{2^{n+1}} \Rightarrow P(T) \le 2c_1(n+1)^{3/2} \cdot \frac{n!}{2^n} = c_1(n+1)^{3/2} \cdot \frac{n!}{2^{n-1}},$$

using $(n+1)! = (n+1) \cdot n!$. As $(n+1)^{3/2} \le (2n)^{3/2}$, this is at most $c\,n^{3/2}\frac{n!}{2^{n-1}}$ with $c = 2^{3/2}c_1$, and taking the maximum over T gives the theorem. $\blacksquare$

Remark 2.3.20. It is worth seeing where the polynomial factor $n^{3/2}$ comes from: $\sqrt{n}$ is the Stirling factor in Lemma 2.3.19, and the extra n is the price of the one added vertex, $(n+1)! = (n+1)\cdot n!$. So Szele's conjecture is settled: by Proposition 2.3.5 and Theorem 2.3.7,

$$\frac{n!}{2^{n-1}} \le P(n) \le c\, n^{3/2} \cdot \frac{n!}{2^{n-1}},$$

and the random tournament of Proposition 2.3.5 is optimal up to the factor $n^{3/2}$.

§2.4. Max-Cut

The next application of linearity is shorter, and the random object is a partition of the vertex set rather than a random graph.

Question 2.4.1. Given a graph G, how large a bipartite subgraph can we find inside it?

Definition 2.4.2 (Cut)**.** A **cut** of G is a partition $V(G) = V_1 \cup V_2$. An edge is **crossing** if its two endpoints lie in different parts, and the **size** of the cut is the number of crossing edges.

The crossing edges of a cut form a bipartite subgraph of G, and conversely every bipartite subgraph arises this way, so Question 2.4.1 asks for the largest cut, the **max-cut** of G.

§2.4.i. Half of the Edges

Proposition 2.4.3

Every graph G contains a bipartite subgraph with at least $\frac{1}{2}e(G)$ edges.

Proof. Take a uniformly random partition $V(G) = V_1 \cup V_2$, i.e. put each vertex into V_1 or V_2 by an independent fair coin, and keep all crossing edges. Let X be the number of crossing edges, so that

$$X = \sum_{e \in E(G)} \mathbb{1}_{\{e \text{ is crossing}\}}.$$

For a fixed edge $e = uv$, the four placements of u and v are equally likely and two of them separate the endpoints, so $\mathbb{P}(e \text{ is crossing}) = \frac{1}{2}$. By Theorem 2.1.2,

$$\mathbb{E}[X] = \sum_{e \in E(G)} \mathbb{P}(e \text{ is crossing}) = \sum_{e \in E(G)} \frac{1}{2} = \frac{1}{2}e(G),$$

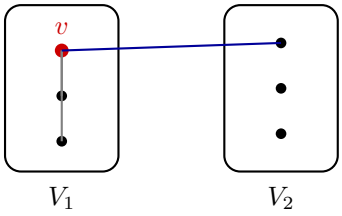
and by Lemma 2.1.1 some partition achieves at least this many crossing edges. ∎

The same bound has a second proof, with no randomness at all, which is worth comparing: instead of averaging over all cuts, it starts at the best cut and perturbs it.

Second proof of Proposition 2.4.3. Take a *maximum* cut $V = V_1 \cup V_2$.

Claim 2.4.1. Every vertex sends at least half of its edges across the cut.

Proof. If some v sent more than half of its edges inside its own part, moving v to the other part would turn all of those into crossing edges and only lose the ones it already had crossing, strictly increasing the size of the cut, contradicting maximality. ∎



v has 2 edges inside and 1 across, so moving v to V_2 gains 2 and loses 1: this cut is not maximum

Summing the claim over all vertices, each crossing edge being counted twice,

$$2 \cdot \#\{\text{crossing edges}\} = \sum_{v \in V} \#\{\text{crossing edges at } v\} \geq \sum_{v \in V} \frac{d(v)}{2} = e(G). \qquad \blacksquare$$

§2.4.ii. Beating One Half

Remark 2.4.4. The constant $\frac{1}{2}$ cannot be improved, as $G = K_n$ shows. Its largest bipartite subgraph is $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$, with $\lfloor \frac{n^2}{4} \rfloor$ edges, while $e(K_n) = \binom{n}{2} = \frac{n(n-1)}{2}$, and

$$\frac{\lfloor n^2/4 \rfloor}{\binom{n}{2}} \longrightarrow \frac{1}{2}.$$

The ratio tends to $\frac{1}{2}$ but is never equal to it: for $n = 2k$ it is $\frac{k^2}{k(2k-1)} = \frac{k}{2k-1}$. So there should be room for an improvement of Proposition 2.4.3 by a term depending on n, and there is. The proof is the same one-line computation, with one change: the random partition is taken to be *balanced*, which makes an edge slightly more likely to cross.

Theorem 2.4.5

Let G be a graph on n vertices. Then G has a bipartite subgraph with at least

$$e(G) \cdot \frac{k}{2k-1} \quad \text{edges if } n = 2k, \qquad e(G) \cdot \frac{k+1}{2k+1} \quad \text{edges if } n = 2k+1.$$

Proof. Take $V = V_1 \cup V_2$ where V_1 is a uniformly random subset of exactly k vertices, and let X be the number of crossing edges. For a fixed edge $e = uv$, the pair (u, v) is equally likely to be any ordered pair of distinct vertices, so

$$\mathbb{P}(e \text{ is crossing}) = 2 \cdot \frac{|V_1|}{n} \cdot \frac{|V_2|}{n-1} = \begin{cases} 2 \cdot \dfrac{k}{2k} \cdot \dfrac{k}{2k-1} = \dfrac{k}{2k-1}, & n = 2k, \\ 2 \cdot \dfrac{k}{2k+1} \cdot \dfrac{k+1}{2k} = \dfrac{k+1}{2k+1}, & n = 2k+1. \end{cases}$$

By Theorem 2.1.2, $\mathbb{E}[X] = e(G) \cdot \mathbb{P}(e \text{ is crossing})$, and Lemma 2.1.1 gives a partition attaining it. $\blacksquare$

Remark 2.4.6. Both bounds are exactly tight for K_n, by Remark 2.4.4: for $n = 2k$, $e(K_n) \cdot \frac{k}{2k-1} = k(2k-1) \cdot \frac{k}{2k-1} = k^2$, and for $n = 2k+1$, $e(K_n) \cdot \frac{k+1}{2k+1} = k(2k+1) \cdot \frac{k+1}{2k+1} = k(k+1)$. Writing the two factors as $\frac{1}{2} + \frac{1}{2(n-1)}$ and $\frac{1}{2} + \frac{1}{2n}$ shows how much is gained over Proposition 2.4.3, and that the gain vanishes as n grows.

§2.5. Ramsey Theory Revisited

The last application of this chapter returns to where the course began, the Ramsey number $R(k)$ of Definition 1.2.1. Erdős's lower bound Theorem 1.2.5 was proved by a union bound; we now redo it as a first moment computation, which gives a stronger conclusion for free, and then ask what happens for hypergraphs.

§2.5.i. Counting Monochromatic Cliques

Lemma 2.5.1

For any n and t, there is a red/blue coloring of $E(K_n)$ with at most $\binom{n}{t}2^{1-\binom{t}{2}}$ monochromatic copies of K_t.

Proof. Color the edges uniformly at random, independently, and let X be the number of monochromatic copies of K_t. Indexing by the vertex set of the copy,

$$X = \sum_{S \in \binom{V}{t}} \mathbb{1}_{\{S \text{ spans a monochromatic clique}\}}.$$

A fixed S spans $\binom{t}{2}$ edges, all of which must receive the same one of two colors, so

$$\mathbb{P}(S \text{ is monochromatic}) = 2 \cdot 2^{-\binom{t}{2}}.$$

By Theorem 2.1.2,

$$\mathbb{E}[X] = \sum_{S \in \binom{V}{t}} \mathbb{P}(S \text{ is monochromatic}) = \binom{n}{t} 2^{1-\binom{t}{2}},$$

and by Lemma 2.1.1 some coloring has at most this many monochromatic copies. ∎

Corollary 2.5.2

If $\binom{n}{t}2^{1-\binom{t}{2}} < 1$, then $R(t) > n$. Consequently $R(t) \geq \sqrt{2}^{(1-o(1))t}$.

Proof. The number of monochromatic copies is an integer, so the coloring of Lemma 2.5.1 has none at all, and K_n has a coloring without a monochromatic K_t. The estimate $\binom{n}{t}2^{1-\binom{t}{2}} < 2n^t 2^{-t(t-1)/2}$ is < 1 as soon as $n \leq (1-c)2^{(t-1)/2}$, exactly as in Theorem 1.2.5. ∎

Remark 2.5.3. Lemma 2.5.1 says strictly more than the union bound of Theorem 1.2.5: it gives a coloring with *few* monochromatic cliques even when the expectation is much larger than 1, whereas the union bound is silent as soon as the sum of probabilities reaches 1. Compare Example 2.1.5.

§2.5.ii. Hypergraph Ramsey Numbers

Question 2.5.4. What happens for k-uniform hypergraphs?

Two-color the k-element edges $\binom{[n]}{k}$ of the complete k-uniform hypergraph $K_n^{(k)}$, and look for a set S of t vertices all of whose $\binom{t}{k}$ edges have the same color.

Definition 2.5.5. The ***k*-uniform Ramsey number** $R^{(k)}(t)$ is the smallest n s.t. every red/blue coloring of $\binom{[n]}{k}$ contains a monochromatic set of t vertices. So $R^{(2)}(t) = R(t)$.

Remark 2.5.6. A t-set of vertices spans $\binom{t}{2}$ graph edges but $\binom{t}{k} \gg \binom{t}{2}$ hypergraph edges, so there are far more constraints to violate and it is correspondingly easier to make a t-set non-monochromatic. We should therefore expect $R^{(k)}(t) \gg R(t)$, and indeed each extra level of uniformity costs an exponential.

What is known is a tower of exponentials, of height $k-2$ from below and $k-1$ from above, for some constant $c > 0$:

$$\sqrt{2}^{\,t} \le R(t) \le 4^t, \qquad c^{\,t} \le R^{(3)}(t) \le 2^{c^{\,t}}, \qquad 2^{c^{\,t}} \le R^{(4)}(t) \le 2^{2^{c^{\,t}}}, \qquad \dots$$

Already for $k = 3$ the gap between the two sides is an entire exponential, and closing it is a well-known open problem. The monochromatic sets we can guarantee are therefore tiny. The next lecture asks for something weaker — not that *every* edge inside a set has one color, but only that one color is *dominant* there — and shows that this is enough to find very much larger sets.

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Recall . $R^{(k)}(t)$ is the least n s.t. every red/blue coloring of $\binom{[n]}{k}$ has a monochromatic t-set, and it grows like a tower of exponentials in t. Equivalently, in a coloring of $K_n^{(k)}$ the largest monochromatic set we can guarantee has size only $O(\underbrace{\log\log\cdots\log}_{k-2 \text{ times}} n)$; see Section 2.5.

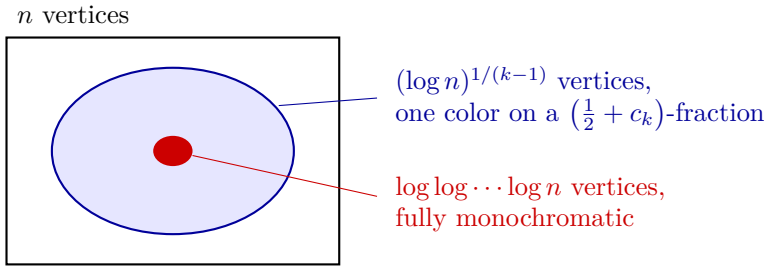
§2.6. Finding a Dominant Color

Demanding that *all* $\binom{t}{k}$ edges inside a t-set receive one color is a very strong requirement, and the previous section showed how badly it limits t. This lecture proves that a much weaker requirement — one color on a fixed proportion above a half — is satisfied by sets that are larger by a tower of exponentials.

Theorem 2.6.1

Suppose we two-color the edges of a complete k-uniform hypergraph on n vertices. Then there is a set S of $\Omega\big((\log n)^{1/(k-1)}\big)$ vertices s.t. a $\big(\frac{1}{2} + c_k\big)$-proportion of its edges receive the same color, for some constant $c_k > 0$.

So the two conclusions live on completely different scales: insisting on all edges of S having one color forces S to be of iterated logarithmic size, while asking only for a fixed advantage over $\frac{1}{2}$ allows S to be polylogarithmic in n.



The proof has two ingredients: a structural theorem of Erdős, which supplies a configuration on which one color is already guaranteed, and an analytic lemma about polynomials, which is what converts that configuration into a dominant color.

§2.6.i. Erdős's Transversal Theorem

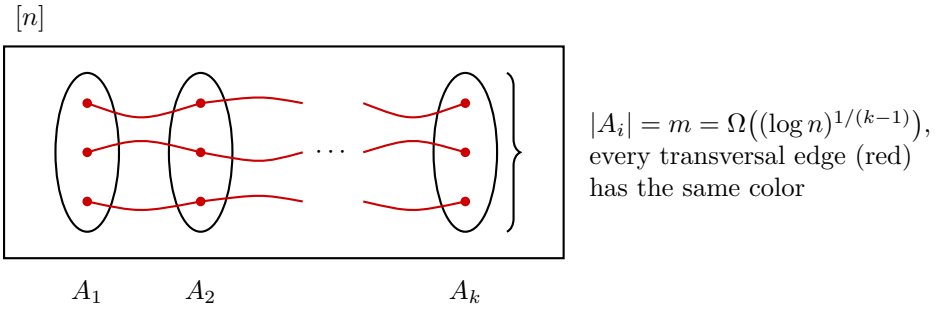
Theorem 2.6.2 (Erdős)

Given any red/blue coloring of $E(K_n^{(k)})$, there exist k pairwise disjoint subsets $A_1, \dots, A_k$ of $[n]$, each of size

$$m = \Omega\big((\log n)^{1/(k-1)}\big),$$

s.t. every edge e with $|e \cap A_i| = 1$ for all i receives the same color.

Call an edge meeting each A_i exactly once a **transversal** edge; there are m^k of them, one for each choice of a vertex from each part, and the theorem says they are monochromatic as a family.



Remark 2.6.3. Nothing is claimed about the other edges, those meeting some A_i in 0 or in ≥ 2 vertices; they may be colored arbitrarily. This is exactly why Theorem 2.6.2 can afford parts of size $(\log n)^{1/(k-1)}$, so much larger than the monochromatic sets of Section 2.5: a transversal family is a far weaker demand than a clique.

§2.6.ii. A Polynomial Lemma

The second ingredient looks unrelated at first, and we state it before explaining where it is used.

Lemma 2.6.4

Let $k \in \mathbb{N}$, and let $\mathcal{P}_k$ be the set of homogeneous polynomials in $x_1, x_2, \dots, x_k$ of degree k whose coefficients lie in $[-1, 1]$ and whose coefficient of $x_1 x_2 \cdots x_k$ is 1. Then there is a constant $c_k > 0$ s.t.

$$\forall\, f \in \mathcal{P}_k \quad \exists\, (x_1, \dots, x_k) \in [0,1]^k \ \text{ s.t. } \ |f(x_1, \dots, x_k)| \geq c_k.$$

The content is the uniformity in f: for each single f the existence of such a point is obvious, since $f \not\equiv 0$; the lemma says one constant works for the whole family.

Proof. Given $f \in \mathcal{P}_k$, let

$$M(f) = \max_{x_1,\dots,x_k \in [0,1]} |f(x_1, \dots, x_k)|,$$

which exists because f is continuous and $[0,1]^k$ is compact. Since the coefficient of $x_1 \cdots x_k$ is 1 we have $f \not\equiv 0$, and a non-zero polynomial cannot vanish on all of $[0,1]^k$,[4] so $M(f) > 0$.

Next, $\mathcal{P}_k$ is compact. Writing $D = \binom{2k-1}{k}$ for the number of degree-k monomials in $x_1, \dots, x_k$, a member of $\mathcal{P}_k$ is determined by its $D - 1$ free coefficients, the coefficient of $x_1 \cdots x_k$ being fixed to 1. This gives a map

$$\Phi\colon [-1,1]^{D-1} \longrightarrow C\big([0,1]^k\big), \qquad \text{coefficients} \longmapsto \text{the polynomial they define},$$

[4] Write $f = \sum_j g_j(x_1, \dots, x_{k-1})\, x_k^j$. By induction on k there are values of $x_1, \dots, x_{k-1}$ in $[0,1]$ at which some g_j is non-zero; fixing them turns f into a non-zero polynomial in x_k alone, which vanishes at only finitely many x_k. The "finitely many roots" step is available for one variable only: in $k \geq 2$ variables the zero set is a hypersurface, e.g. $x_1 - x_2$ vanishes along a diagonal of $[0,1]^2$.

whose image is exactly $\mathcal{P}_k$. It is continuous for the supremum norm, since every monomial is bounded by 1 on $[0,1]^k$ and hence $\|\Phi(a)-\Phi(b)\|_\infty \le D\max_i |a_i-b_i|$. A continuous image of a compact set is compact, so $\mathcal{P}_k$ is compact; and $M=\|\cdot\|_\infty$ is continuous.

So M attains its minimum on $\mathcal{P}_k$, i.e. there is $f_0\in\mathcal{P}_k$ with

$$\min_{f\in\mathcal{P}_k} M(f) = M(f_0) =: c_k > 0,$$

and this c_k works for every $f\in\mathcal{P}_k$. ■

§2.6.iii. Proof of the Theorem

Proof of Theorem 2.6.1. By Theorem 2.6.2 there are pairwise disjoint $A_1,\dots,A_k$ of size $m=\Omega\big((\log n)^{1/(k-1)}\big)$ s.t. every edge e with $|e\cap A_i|=1$ for all i is, without loss of generality, colored blue.

The coloring as a ±1 function. Think of the coloring as a map $\chi\colon E\to\{\pm1\}$,

$$\chi(e)=\begin{cases}+1, & \text{if } e \text{ is blue},\\ -1, & \text{if } e \text{ is red},\end{cases}$$

and for a set S of vertices put

$$\chi(S)=\sum_{e\in\binom{S}{k}}\chi(e) = \#\{\text{blue edges inside } S\} - \#\{\text{red edges inside } S\}.$$

> **Observation .** If the two colors were equally distributed over the $\binom{|S|}{k}$ edges inside S, then $\chi(S)=0$. So a dominant color on S is exactly the statement that $|\chi(S)|$ is a positive proportion of $\binom{|S|}{k}$, and we want to find a large S with $|\chi(S)|$ big.

The random set. Choose $S\subseteq A_1\cup\cdots\cup A_k$ at random: if $v\in A_i$, let $v\in S$ with probability p_i, independently over all vertices, where $p_1,\dots,p_k\in[0,1]$ are to be determined. For an edge $e\subseteq A_1\cup\cdots\cup A_k$ set

$$X_e=\begin{cases}\chi(e), & \text{if } e\subseteq S,\\ 0, & \text{if } e\not\subseteq S,\end{cases}\qquad\text{so that}\qquad \chi(S)=\sum_{e\in\binom{S}{k}}\chi(e)=\sum_{e\in\binom{A_1\cup\cdots\cup A_k}{k}}X_e.$$

By Theorem 2.1.2,

$$\mathbb{E}\big[\chi(S)\big]=\sum_{e\in\binom{A_1\cup\cdots\cup A_k}{k}}\mathbb{E}[X_e]=\sum_{e\in\binom{A_1\cup\cdots\cup A_k}{k}}\chi(e)\,\mathbb{P}(e\subseteq S).$$

Grouping the edges by type. Say that e is of **type** $(a_1,a_2,\dots,a_k)$ if $|e\cap A_i|=a_i$ for all i; the types are the tuples of non-negative integers with $a_1+\cdots+a_k=k$. All the vertices of e are chosen independently, so an edge of type $(a_1,\dots,a_k)$ satisfies $\mathbb{P}(e\subseteq S)=p_1^{a_1}p_2^{a_2}\cdots p_k^{a_k}$, and this depends only on the type. Hence

$$\mathbb{E}\big[\chi(S)\big]=\sum_{\substack{(a_1,\dots,a_k)\\ a_i\ge0,\ \sum_i a_i=k}}\left(\sum_{\substack{e\in\binom{A_1\cup\cdots\cup A_k}{k}\\ e \text{ of type } (a_1,\dots,a_k)}}\chi(e)\right)p_1^{a_1}p_2^{a_2}\cdots p_k^{a_k}.$$

This is a homogeneous polynomial of degree k in $p_1,\dots,p_k$, and we now show that, after dividing by m^k, it is a member of the family $\mathcal{P}_k$ of Lemma 2.6.4.

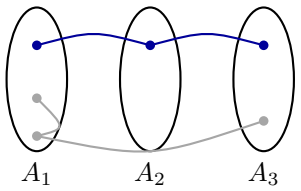
The transversal type dominates. The number of edges of type $(a_1,\dots,a_k)$ is $\prod_i\binom{m}{a_i}$.

- For the transversal type $(1,1,\dots,1)$ there are exactly m^k edges, and $\chi(e)=+1$ for every one of them, since they are all blue. So the inner sum is exactly m^k.
- For any other type some $a_i \geq 2$, and then

$$\prod_{i=1}^{k}\binom{m}{a_i} \leq \prod_{i=1}^{k}\frac{m^{a_i}}{a_i!} = \frac{m^k}{\prod_i a_i!} \leq \frac{m^k}{2}.$$

As $|\chi(e)|=1$ for every edge, the inner sum is at most this in absolute value:

$$\left|\sum_{\substack{e\in\binom{A_1\cup\cdots\cup A_k}{k}\\ e \text{ of type } (a_1,\dots,a_k)}}\chi(e)\right| \leq \frac{m^k}{2} < m^k.$$



type $(1,1,1)$: blue, m^3 edges

type $(2,0,1)$: any color, fewer edges

Applying the lemma. Dividing through by m^k,

$$\mathbb{E}\big[\chi(S)\big] = m^k \sum_{\substack{(a_1,\dots,a_k)\\ a_i\geq 0,\ \sum_i a_i = k}} \left(\frac{1}{m^k}\sum_{\substack{e\in\binom{A_1\cup\cdots\cup A_k}{k}\\ e \text{ of type } (a_1,\dots,a_k)}}\chi(e)\right) p_1^{a_1}\cdots p_k^{a_k} = m^k f_\chi(p_1,\dots,p_k)$$

for a polynomial f_χ which is homogeneous of degree k, has all coefficients in $[-1,1]$, and has the coefficient of $p_1p_2\cdots p_k$ equal to 1. That is, $f_\chi \in \mathcal{P}_k$. By Lemma 2.6.4 there is a choice of $(p_1,\dots,p_k)\in[0,1]^k$ with $|f_\chi(p_1,\dots,p_k)|\geq c_k$, and we run the random experiment with these probabilities. Then

$$\mathbb{E}\big[|\chi(S)|\big] \geq \big|\mathbb{E}[\chi(S)]\big| \geq c_k m^k,$$

so by Lemma 2.1.1 some set $S\subseteq A_1\cup\cdots\cup A_k$ satisfies $|\chi(S)|\geq c_k m^k$. Fix this S.

Reading off the conclusion. First, S is large: since $|\chi(S)|$ is a sum of $\binom{|S|}{k}$ terms of absolute value 1,

$$c_k m^k \leq |\chi(S)| \leq \binom{|S|}{k} \leq |S|^k \Rightarrow |S| \geq c_k^{1/k} m = \Omega(m) = \Omega\big((\log n)^{1/(k-1)}\big).$$

Second, one color dominates on S. Write B and R for the numbers of blue and red edges inside S, so $B+R=\binom{|S|}{k} =: N$ and $|B-R| = |\chi(S)| \geq c_k m^k$. As $S\subseteq A_1\cup\cdots\cup A_k$ we have $|S|\leq km$ and hence $N\leq\binom{km}{k}\leq\frac{(km)^k}{k!}$. Therefore the majority color occupies a proportion

$$\frac{\max(B,R)}{N} = \frac{1}{2} + \frac{|B-R|}{2N} \geq \frac{1}{2} + \frac{c_k m^k}{2(km)^k/k!} = \frac{1}{2} + \frac{c_k\, k!}{2k^k}$$

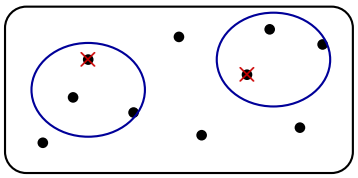
of the edges inside S. The last quantity is a positive constant depending only on k, which is the constant in the statement. ∎

3. The Method of Alterations

The first moment principle finds an object that is good *on average*. Often the average object is not quite good enough: a random coloring has a few monochromatic cliques, a random set of vertices spans a few edges, a random set of points contains a few thin triangles. The method of this chapter is to accept those defects and then repair them by hand.

Choose a random structure which is close to having the desired property, and then make a small alteration — usually deleting one element from each defect — to establish the property outright.

The alteration costs something, but if the defects are few the loss is lower order, and we end up with more than the pure first moment gives.



the random object has a few defects (blue), delete one element from each (red) and what remains has the property

§3.1. Ramsey Numbers Again

§3.1.i. What the Union Bound Gives

We first redo the computation behind Corollary 2.5.2 keeping track of the constants, so that there is something to improve on. Recall from Lemma 2.5.1 that a uniformly random coloring of $E(K_n)$ has

$$\mathbb{E}[X] = \binom{n}{k} 2^{1-\binom{k}{2}}$$

monochromatic copies of K_k on average, and that $R(k) > n$ as soon as this is < 1.

Proposition 3.1.1

$R(k) \geq (1 - o(1)) \dfrac{k}{e\sqrt{2}}\, 2^{k/2}$.

Proof. Put $x := \frac{ne}{k\, 2^{(k-1)/2}}$. Using $\binom{n}{k} \leq \left(\frac{ne}{k}\right)^k$,

$$\binom{n}{k} 2^{1-\binom{k}{2}} \leq 2\left(\frac{ne}{k}\right)^k 2^{-\frac{k(k-1)}{2}} = 2x^k. \tag{3.1}$$

By (3.1), $\mathbb{E}[X] < 1$ whenever $2x^k < 1$, i.e. $x < 2^{-1/k}$, i.e.

$$n < y := 2^{-1/k} \cdot \frac{k}{e\sqrt{2}}\, 2^{k/2},$$

and by Corollary 2.5.2 every such integer n satisfies $R(k) > n$. Taking $n = \lceil y \rceil - 1$ gives $R(k) \geq \lceil y \rceil \geq y$, that is,

$$R(k) \geq 2^{-1/k} \cdot \frac{k}{e\sqrt{2}}\, 2^{k/2} \qquad \text{for every } k.$$

This is the statement: the $1-o(1)$ there is exactly the factor $2^{-1/k} = e^{-(\ln 2)/k} \geq 1 - \frac{\ln 2}{k}$, which tends to 1. ■

§3.1.ii. Deleting a Vertex from Each Bad Clique

The union bound throws away everything as soon as $\mathbb{E}[X] \geq 1$. But a coloring with a handful of monochromatic cliques is nearly as good as one with none: destroy them.

Theorem 3.1.2

For every k and every n,

$$R(k) > n - \binom{n}{k} 2^{1-\binom{k}{2}}.$$

Proof. Let X be the number of monochromatic copies of K_k in a uniformly random red/blue coloring of $E(K_n)$. Delete one vertex from each monochromatic clique; this removes at most X vertices, and the coloring induced on the remaining $\geq n - X$ vertices has no monochromatic K_k left, since every one of them lost a vertex. Hence

$$R(k) > n - X.$$

By Lemma 2.5.1 there is a coloring with $X \leq \mathbb{E}[X] = \binom{n}{k} 2^{1-\binom{k}{2}}$, and using that coloring gives the bound. ■

Now we are free to choose n as large as we like, and the right choice is the one maximising the right-hand side rather than the one making it positive.

Corollary 3.1.3

$R(k) \geq (1 - o(1)) \dfrac{k}{e} 2^{k/2}$.

Proof. Write $c = \left(\frac{e}{k}\right)^k 2^{1-\binom{k}{2}}$, so that Theorem 3.1.2 together with $\binom{n}{k} \leq \left(\frac{ne}{k}\right)^k$ gives $R(k) > n - cn^k$. Differentiating with respect to n,

$$1 - kc\,n^{k-1} = 0 \Leftrightarrow n^{k-1} = \frac{1}{kc} = \frac{k^{k-1}}{e^k} 2^{\binom{k}{2}-1} \Leftrightarrow n = (1 - o(1)) \frac{k}{e} 2^{k/2},$$

where for the last step we take $(k-1)$-th roots factor by factor, using $\binom{k}{2}/(k-1) = \frac{k}{2}$:

$$n = \underbrace{\left(k^{k-1}\right)^{\frac{1}{k-1}}}_{=k} \cdot \underbrace{\left(e^{-k}\right)^{\frac{1}{k-1}}}_{=e^{-1}\,e^{-\frac{1}{k-1}}} \cdot \underbrace{\left(2^{\binom{k}{2}-1}\right)^{\frac{1}{k-1}}}_{=2^{k/2}\,2^{-\frac{1}{k-1}}} = \underbrace{(2e)^{-\frac{1}{k-1}}}_{\to 1} \cdot \frac{k}{e} 2^{k/2}.$$

At this n we have $cn^k = \frac{n}{k}$, so

$$R(k) > n - \frac{n}{k} = \left(1 - \frac{1}{k}\right) n = (1 - o(1)) \frac{k}{e} 2^{k/2}.$$ ■

Remark 3.1.4. The alteration gained exactly a factor $\sqrt{2}$ over Proposition 3.1.1, and nothing more: the exponential rate $2^{k/2}$ is untouched. This is typical. Determining whether the rate itself can be improved is a famous open problem; the best known bounds are still of the form $2^{(\frac{1}{2}+o(1))k} \leq R(k) \leq 2^{(2-o(1))k}$ in the exponent.

§3.2. Independent Sets and Turán's Theorem

Recall . Theorem 1.1.3: an n-vertex graph with more than $\frac{n^2}{4}$ edges contains a triangle.

Turán's theorem is the generalisation of Theorem 1.1.3 from triangles to K_k.

Theorem 3.2.1 (Turán, 1941)
If G is an n-vertex graph with more than $\left(1-\frac{1}{k-1}\right)\frac{n^2}{2}$ edges, then G contains K_k.

Definition 3.2.2 (Independence number)**.** The **independence number** $\alpha(G)$ is the largest number of vertices of G spanning no edge.

Passing to the complement $\overline{G}$ turns cliques into independent sets, and G has more than $\left(1-\frac{1}{k-1}\right)\frac{n^2}{2}$ edges iff $\overline{G}$ has fewer than

$$\binom{n}{2}-\left(1-\frac{1}{k-1}\right)\frac{n^2}{2}=\frac{n^2}{2(k-1)}-\frac{n}{2}$$

edges, which is $\frac{1}{k-1}\binom{n}{2}$ up to lower-order terms. So Theorem 3.2.1 has an equivalent form, which is the one we shall prove.

Theorem 3.2.3 (Turán, 1941; complementary form)
If G is an n-vertex graph with $\lesssim \frac{1}{k-1}\binom{n}{2}$ edges, then $\alpha(G)\geq k$.

Throughout this section G has n vertices and m edges, and the benchmark to keep in mind is

$$m\approx\frac{1}{k-1}\binom{n}{2}\approx\frac{n^2}{2(k-1)},$$

for which we are trying to produce an independent set of size about k. We give three attempts, each stronger than the last.

§3.2.i. Attempt 1: A Random Subset

Let $S\subseteq V(G)$ be a random subset, each vertex included independently with probability p, and let X be the number of edges inside S. Then

$$X=\sum_{e\in E(G)}X_e,\qquad X_e=\begin{cases}1, & \text{if } e\subseteq S,\\ 0, & \text{if } e\not\subseteq S,\end{cases}\qquad \mathbb{P}(e\subseteq S)=p^2,$$

so by Theorem 2.1.2

$$\mathbb{E}[X]=\sum_{e\in E(G)}\mathbb{E}[X_e]=\sum_{e\in E(G)}\mathbb{P}(e\subseteq S)=mp^2.$$

If $mp^2<1$, i.e. $p<\frac{1}{\sqrt{m}}$, then by Lemma 2.1.1 some outcome has $X\leq\mathbb{E}[X]<1$, hence $X=0$, i.e. S is independent. By itself this says nothing about $|S|$: $S=\varnothing$ is also independent. The heuristic is to *pretend* that this independent S has the typical size of $|S|\sim\mathrm{Bin}(n,p)$, namely $\mathbb{E}|S|=np$. Taking p just below $\frac{1}{\sqrt{m}}$ would then give

$$\alpha(G)\geq|S|\approx np\approx\frac{n}{\sqrt{m}},$$

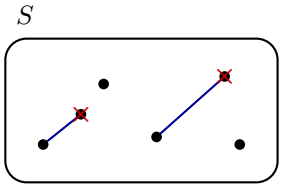
and at the benchmark $m\approx\frac{n^2}{2(k-1)}$, i.e. $\sqrt{m}\approx\frac{n}{\sqrt{2(k-1)}}$,

$$\frac{n}{\sqrt{m}}\approx\sqrt{2(k-1)}.$$

Two things are wrong with this. The conclusion is only of order $\sqrt{k}$ instead of k; and the argument is not even complete, since the pretending step is unjustified: we would need one outcome with both $X = 0$ and $|S| \approx np$, i.e. $\mathbb{P}\big(X = 0,\ |S| \geq (1-\varepsilon)np\big) > 0$, which needs a concentration bound on $|S|$. The next attempt repairs both at once.

§3.2.ii. Attempt 2: Alterations

We do not need S to be independent — it is enough that S has *few* edges, because we can then delete one endpoint from each edge and what remains is independent.



X edges inside S; deleting one endpoint of each leaves $\geq |S| - X$ independent vertices

So for *every* set S, writing $X = X(S)$ for the number of edges inside it,

$$\alpha(G) \geq |S| - X(S),$$

and we may take expectations instead of hoping for a single good outcome:

$$\alpha(G) \geq \mathbb{E}\big[|S| - X\big] = \mathbb{E}\big[|S|\big] - \mathbb{E}[X] = np - mp^2,$$

using Lemma 2.1.1 and Theorem 2.1.2. Now choose p to maximise: differentiating, $n - 2pm = 0$, i.e. $p = \frac{n}{2m}$, which lies in $[0,1]$ provided $m \geq \frac{n}{2}$. Plugging in,

$$\alpha(G) \geq \frac{n^2}{2m} - m \cdot \frac{n^2}{4m^2} = \frac{n^2}{4m}.$$

At the benchmark $m \approx \frac{n^2}{2(k-1)}$ this gives

$$\alpha(G) \geq \frac{k-1}{2},$$

of the right order in k, but still a factor 2 away from Theorem 3.2.3.

§3.2.iii. Attempt 3: A Random Greedy Algorithm

The last factor 2 is recovered by not throwing vertices away at all, but by building the independent set greedily in a random order.

> **Notation .** $\deg(v)$ is the degree of v in G, and $N(v)$ its neighbourhood.

The algorithm is:

(1) order the vertices $v_1, v_2, \ldots, v_n$ uniformly at random;

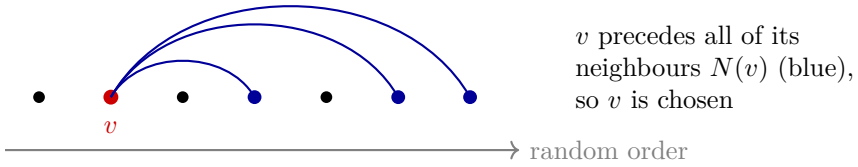
(2) going along the order, add v_i to the independent set I if it is not adjacent to any previously chosen vertex.

By construction I is independent. Let $X = |I|$, and decompose it over the vertices:

$$X = \sum_{v \in V(G)} X_v, \qquad X_v = \begin{cases} 1, & \text{if } v \in I, \\ 0, & \text{if } v \notin I, \end{cases} \qquad \text{so} \qquad \mathbb{E}[X] = \sum_{v \in V(G)} \mathbb{P}(v \in I).$$

Observation . $\mathbb{P}(v \in I) \geq \mathbb{P}(v \text{ comes before all of its neighbours}) = \dfrac{1}{1+\deg(v)}$.

Indeed, if v precedes all of $N(v)$ then none of its neighbours has been chosen when v is examined, so v joins I; and among the $1+\deg(v)$ vertices of $\{v\} \cup N(v)$, each is equally likely to come first in a uniformly random order. The inequality is not an equality, since v can also be chosen when a neighbour precedes it but was itself rejected.



v precedes all of its neighbours $N(v)$ (blue), so v is chosen

Theorem 3.2.4 (Caro–Wei)

For any graph G,

$$\alpha(G) \geq \sum_{v \in V(G)} \frac{1}{1+\deg(v)}.$$

Proof. By the Observation above and Theorem 2.1.2, $\mathbb{E}[X] \geq \sum_v \frac{1}{1+\deg(v)}$, and by Lemma 2.1.1 some order produces an independent set at least this large. ∎

Corollary 3.2.5

If G is an n-vertex graph with m edges, then

$$\alpha(G) \geq \frac{n}{1+\frac{2m}{n}}.$$

Proof. Recall Jensen's inequality: if φ is convex on an interval I and $x_1, \dots, x_n \in I$, then

$$\frac{1}{n}\sum_{i=1}^{n} \varphi(x_i) \geq \varphi\Big(\frac{1}{n}\sum_{i=1}^{n} x_i\Big).$$

Apply it with $\varphi(x) = \frac{1}{1+x}$, which is convex on $[0,\infty)$ since $\varphi''(x) = \frac{2}{(1+x)^3} > 0$, and with the x_i the degrees $\deg(v)$. By the handshake lemma $\sum_v \deg(v) = 2m$, so the average degree is $\frac{2m}{n}$ and

$$\sum_{v \in V(G)} \frac{1}{1+\deg(v)} \geq n\,\varphi\Big(\frac{2m}{n}\Big) = \frac{n}{1+\frac{2m}{n}},$$

which together with Theorem 3.2.4 gives the claim. ∎

At the benchmark $m = \frac{1}{k-1}\binom{n}{2}$ we have $\frac{2m}{n} = \frac{n-1}{k-1}$, and

$$\alpha(G) \geq \frac{n}{1+\frac{n-1}{k-1}} = \frac{n(k-1)}{n+k-2} = (1-o(1))(k-1),$$

which is Theorem 3.2.3.

Remark 3.2.6. Corollary 3.2.5 is sharp. Let $(k-1) \mid n$ and let G be the disjoint union of $k-1$ copies of $K_{n/(k-1)}$. Every vertex has degree $\frac{n}{k-1}-1$, so $\frac{2m}{n} = \frac{n}{k-1}-1$ and the bound of Corollary 3.2.5 reads

$$\alpha(G) \geq \frac{n}{1+\frac{2m}{n}} = \frac{n}{n/(k-1)} = k-1.$$

On the other hand $\alpha(G) \leq k-1$: two vertices in the same clique are adjacent, so an independent set has at most one vertex from each of the $k-1$ cliques. Hence $\alpha(G) = k-1$ and the bound is attained.

The complement $\overline{G}$ is the complete $(k-1)$-partite graph with equal parts, which is K_k-free (a clique in $\overline{G}$ is an independent set in G) and has exactly $\left(1-\frac{1}{k-1}\right)\frac{n^2}{2}$ edges; it is the extremal graph of Theorem 3.2.1.

Comparing the three attempts at the benchmark:

$$\underbrace{\sqrt{2(k-1)}}_{\text{Attempt 1}} < \underbrace{\frac{k-1}{2}}_{\text{Attempt 2}} < \underbrace{k-1}_{\text{Attempt 3}} \quad \text{(for large } k\text{)},$$

and by the example above the last cannot be improved.

§3.3. The Heilbronn Triangle Problem

Question 3.3.1. If we place n points in $[0,1]^2$, how small must the smallest triangle they span be?

Three points taken at random usually span a triangle of constant area, but with many points some triple is bound to be nearly collinear. The question is how nearly, and the alteration method gives a good lower bound with no construction at all.

Proposition 3.3.2

There is a set of n points in $[0,1]^2$ s.t. every triangle spanned by three of them has area $\geq \frac{1}{100n^2}$.

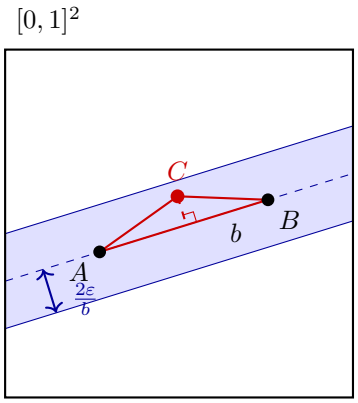
The proof needs one geometric estimate, which is where the areas enter.

Lemma 3.3.3

Let A, B, C be three independent uniformly random points of $[0,1]^2$, and let E be the event that the triangle ABC has area $\leq \varepsilon$. Then $\mathbb{P}(E) \leq 16\pi\varepsilon$.

Proof. Condition on A and B with $|AB| = b$. The area of ABC is $\frac{1}{2}b \cdot \operatorname{dist}(C, AB)$, so E occurs iff C lies in the strip of half-width $\frac{2\varepsilon}{b}$ around the line AB. That strip meets $[0,1]^2$ in a set of length at most $\sqrt{2}$, the diameter of the square, so

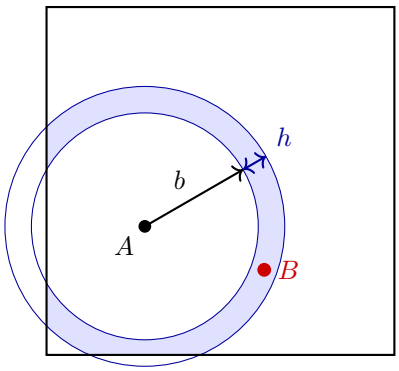
$$\mathbb{P}\big(E \mid |AB| = b\big) \leq \sqrt{2} \cdot \frac{4\varepsilon}{b}.$$



$[\triangle ABC] = \frac{1}{2} b \cdot \operatorname{dist}(C, AB)$,
so $[\triangle ABC] \le \varepsilon$ iff the height
$\operatorname{dist}(C, AB)$ is at most $\frac{2\varepsilon}{b}$,
i.e. iff C lies in the shaded strip

Next we bound the density $f_{|AB|}$ of $|AB|$. Fix A. Then $b < |AB| \le b + h$ iff B lies in the annulus centred at A with radii b and $b + h$. Since B is uniform on $[0,1]^2$, which has area 1, the probability that B lands in a region is the area of that region inside the square, which is at most the area of the whole annulus. This holds for every A, hence also after averaging over A:

$$\mathbb{P}\big(b < |AB| \le b + h\big) \le \pi(b+h)^2 - \pi b^2 = 2\pi b h + \pi h^2.$$



$b < |AB| \le b + h$ iff B is in the ring;
the shaded part (ring $\cap$ square)
has area $\le \pi(b+h)^2 - \pi b^2$

Dividing by h and letting $h \to 0$,

$$f_{|AB|}(b) = \lim_{h \to 0} \frac{\mathbb{P}\big(b < |AB| \le b + h\big)}{h} \le \lim_{h \to 0} (2\pi b + \pi h) = 2\pi b.$$

Since $|AB| \le \sqrt{2}$ always,

$$\mathbb{P}(E) = \int_0^{\sqrt{2}} f_{|AB|}(b)\, \mathbb{P}\big(E \mid |AB| = b\big)\, \mathrm{d}b \le \int_0^{\sqrt{2}} 2\pi b \cdot \frac{4\sqrt{2}\,\varepsilon}{b}\, \mathrm{d}b = 8\sqrt{2}\pi\varepsilon \cdot \sqrt{2} = 16\pi\varepsilon. \qquad \blacksquare$$

Proof of Proposition 3.3.2. Let $P_1, \dots, P_{2n}$ be independent uniformly random points of $[0,1]^2$ — note that we start with twice as many points as we want — and let X be the number of triangles of area at most ε among them:

$$X = \sum_{1 \le i < j < l \le 2n} X_{ijl}, \qquad X_{ijl} = \begin{cases} 1, & \text{if } [\triangle P_i P_j P_l] \le \varepsilon, \\ 0, & \text{otherwise.} \end{cases}$$

By Theorem 2.1.2 and Lemma 3.3.3,

$$\mathbb{E}[X] = \sum_{1 \le i < j < l \le 2n} \mathbb{E}[X_{ijl}] = \binom{2n}{3} \mathbb{P}\big([\triangle P_1 P_2 P_3] \le \varepsilon\big) \le \binom{2n}{3} 16\pi\varepsilon.$$

Choose $\varepsilon = \frac{1}{100n^2}$. Then, as $\binom{2n}{3} \le \frac{4n^3}{3}$,

$$\mathbb{E}[X] \le \frac{4n^3}{3} \cdot \frac{16\pi}{100n^2} = \frac{64\pi}{300}\, n \le n,$$

so by Lemma 2.1.1 there are $2n$ points spanning at most n triangles of area $\le \varepsilon$. Now alter: delete one point from each such triangle. At least $2n - n = n$ points remain, and no three of them span a triangle of area $\le \varepsilon = \frac{1}{100n^2}$. ∎

Remark 3.3.4.

(1) Erdős gave a deterministic construction whose smallest triangle has area $\Omega\big(\frac{1}{n^2}\big)$; see the homework.

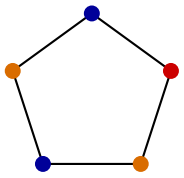
(2) Heilbronn conjectured that $\Theta\big(\frac{1}{n^2}\big)$ is best possible.

(3) This is false: Komlós, Pintz and Szemerédi (1982) produced n points in $[0,1]^2$ in which every triangle has area $\Omega\big(\frac{\log n}{n^2}\big)$, by a probabilistic argument.

§3.4. Graph of Large Chromatic Number and High Girth

Definition 3.4.1 (Chromatic number)**.** Given a graph G, the **chromatic number** $\chi(G)$ is the least number of colors needed to color the vertices so that no edge is monochromatic, i.e. so that adjacent vertices receive different colors.

Equivalently, $\chi(G)$ is the least number of independent sets into which $V(G)$ can be partitioned, the color classes.



$\chi(C_5) = 3$: two colors would force two adjacent vertices to agree somewhere around an odd cycle

Two bounds are immediate.

Proposition 3.4.2

For every graph G,

(1) $\chi(G) \ge \omega(G) \coloneqq \max\{k : K_k \subseteq G\}$, the **clique number**;

(2) $\chi(G) \ge \dfrac{|V(G)|}{\alpha(G)}$.

Proof. (1) The vertices of a clique are pairwise adjacent, so they must all receive different colors.

(2) Every color class is an independent set, hence of size at most $\alpha(G)$, and the classes cover all $|V(G)|$ vertices. ∎

Lecture 7

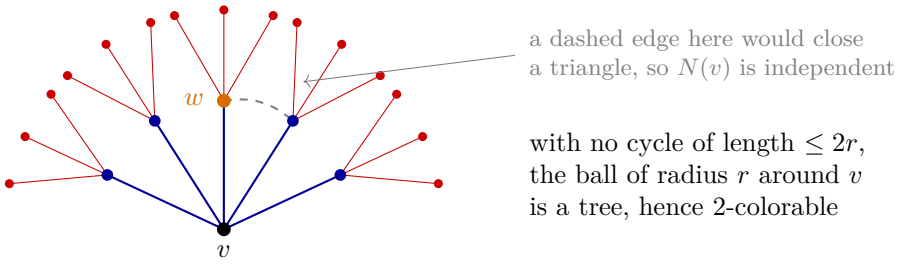
§3.4.i. Is a Large Clique the Only Obstruction?

16 Oct.

The two bounds just proved are of completely different natures. The first is *local*: if $\chi(G)$ is large because G contains a K_k, then there is a small piece of G one can point at and say *this* is

why k colors are needed. The second is *global*: it compares two parameters of the whole graph and says nothing about where the difficulty sits.

It is natural to hope that the local reason is the only one, i.e. that a graph with no small dense substructure can always be colored with few colors. The hope is encouraged by what forbidding short cycles does to a graph locally. If G has no cycle of length 3, then the neighborhood $N(v)$ of any vertex is independent; if moreover G has no cycle of length 4 or 5, then for $w \in N(v)$ the set $N(w)$ is again independent and meets $N(v)$ only in w's own edge back to v. Continuing, a ball of radius r around v is exactly a tree as soon as G has no cycle of length $\leq 2r$. Trees are 2-colorable, so from close up such a graph looks like it ought to need two colors.



Erdős showed that this intuition is entirely wrong, and in the strongest possible sense: one may forbid *all* short cycles and still force arbitrarily many colors. Large chromatic number is a genuinely global phenomenon.

Definition 3.4.3 (Girth)**.** The **girth** $g(G)$ of a graph G is the length of a shortest cycle in G, with $g(G) := \infty$ if G is a forest.

Theorem 3.4.4 (Erdős, 1959)

For all $k, \ell \in \mathbb{N}$ there is a graph G with

$$\chi(G) \geq k \qquad \text{and} \qquad g(G) \geq \ell.$$

Since $g(G) \geq \ell$ forbids every cycle of length $< \ell$, such a G contains no K_3 at all, so $\omega(G) = 2$ and the clique bound is useless. The only tool left is the second bound of Proposition 3.4.2, and that is exactly what the proof uses: we do not try to exhibit a hard-to-color structure, we only make G large and its independent sets small.

Intuition . To get $\chi(G) \geq k$ from $\chi(G) \geq |V(G)|/\alpha(G)$ we need $\alpha(G)$ to be smaller than $|V(G)|/k$, i.e. the graph must have no large independent set. That pushes us to put in *many* edges. High girth pushes the other way: it forbids edges, or at least forbids them from bunching up. The whole proof is the observation that these two demands can be met at once, because a random graph at the right density has very few short cycles — few enough to delete by hand.

§3.4.ii. The Proof Strategy

The argument is the method of the present chapter, applied to two kinds of defect simultaneously.

- Take a random graph $G(n,p)$.
- Show that it has few short cycles, so that *few* is at most $\frac{n}{2}$ of them.
- Show that it has no independent set of size $\frac{n}{2k}$, or at worst few such sets.

- **Alteration**: delete one vertex from every short cycle and from every large independent set.

What is left is a graph of large girth with small independence number and still at least half of the vertices, and $\chi \geq |V|/\alpha$ turns that into a large chromatic number.

A remark on the bookkeeping before we start: the two defects are counted by two different random variables X and Y, but they are repaired by one and the same alteration, so what has to be small is the *sum* $X + Y$. We will therefore aim at $\mathbb{E}[X+Y] \leq \frac{n}{2}$ and let Lemma 2.1.1 produce a single graph in which both counts are simultaneously under control.

Definition 3.4.5 (Binomial random graph)**.** For $n \in \mathbb{N}$ and $p \in [0,1]$, the **binomial random graph** $G(n,p)$ is the random graph on the vertex set $[n]$ in which each of the $\binom{n}{2}$ pairs is an edge with probability p, independently of all the others.

§3.4.iii. Counting Short Cycles

Lemma 3.4.6

Let X be the number of cycles of length $< \ell$ in $G(n,p)$. If $np \geq 1$, then

$$\mathbb{E}[X] \leq \ell\,(np)^{\ell-1}.$$

Proof. For $3 \leq s < \ell$ let $\mathcal{E}_s$ be the set of all s-cycles on the vertex set $[n]$, let $X_s := |\{C \in \mathcal{E}_s : C \subseteq G(n,p)\}|$ be the number of them present, and for $C \in \mathcal{E}_s$ put

$$X_C = \begin{cases} 1, & \text{if all edges of } C \text{ are present,} \\ 0, & \text{otherwise,} \end{cases} \qquad \text{so that} \qquad X_s = \sum_{C \in \mathcal{E}_s} X_C.$$

An s-cycle is determined by an ordering of its s vertices, up to the s rotations and the 2 reflections that give the same cycle, so

$$|\mathcal{E}_s| = \frac{n(n-1)\cdots(n-s+1)}{2s} \leq \frac{n^s}{2s}.$$

A fixed $C \in \mathcal{E}_s$ has s edges, each present independently with probability p, so $\mathbb{P}\big(C \subseteq G(n,p)\big) = p^s$. By Theorem 2.1.2,

$$\mathbb{E}[X_s] = \sum_{C \in \mathcal{E}_s} \mathbb{P}\big(C \subseteq G(n,p)\big) \leq \frac{n^s}{2s}\,p^s = \frac{(np)^s}{2s}.$$

Since $np \geq 1$, the map $s \mapsto (np)^s$ is non-decreasing, so every $3 \leq s \leq \ell - 1$ has $(np)^s \leq (np)^{\ell-1}$ and $2s \geq 6$; there are $\ell - 3$ such s. Summing over s,

$$\mathbb{E}[X] = \sum_{s=3}^{\ell-1} \mathbb{E}[X_s] \leq \sum_{s=3}^{\ell-1} \frac{(np)^s}{2s} \leq (\ell-3)\cdot\frac{(np)^{\ell-1}}{6} \leq \ell\,(np)^{\ell-1}. \qquad \blacksquare$$

So $\mathbb{E}[X] = O\big((np)^{\ell-1}\big)$, and for this to be $o(n)$ — which is what the alteration can afford — we need

$$(np)^{\ell-1} \ll n \Leftrightarrow p \ll n^{\frac{1}{\ell-1}-1} = n^{-\frac{\ell-2}{\ell-1}}.$$

This is an *upper* bound on p: too many edges and the short cycles become too numerous to delete.

§3.4.iv. Counting Large Independent Sets

Lemma 3.4.7

Let $a := \lceil \frac{n}{2k} \rceil$ and let Y be the number of independent sets of size a in $G(n,p)$. If $n \geq 4k$, then

$$\mathbb{E}[Y] \leq 2^n \exp\Big(-\frac{p\,n^2}{16k^2}\Big).$$

Proof. For $T \subseteq V$ with $|T| = a$ put

$$Y_T = \begin{cases} 1, & \text{if } T \text{ is independent,} \\ 0, & \text{if not,} \end{cases} \qquad \text{so that} \qquad Y = \sum_{\substack{T \subseteq V \\ |T|=a}} Y_T.$$

The set T is independent precisely when all $\binom{a}{2}$ pairs inside it are non-edges, so by Theorem 2.1.2,

$$\mathbb{E}[Y] = \sum_T \mathbb{P}(T \text{ independent}) = \binom{n}{a}(1-p)^{\binom{a}{2}}.$$

Now bound both factors. First $\binom{n}{a} \leq 2^n$. Second, $1 - p \leq e^{-p}$, and since $a \geq \frac{n}{2k} \geq 2$ we have $\binom{a}{2} = \frac{a(a-1)}{2} \geq \frac{a^2}{4} \geq \frac{n^2}{16k^2}$, whence

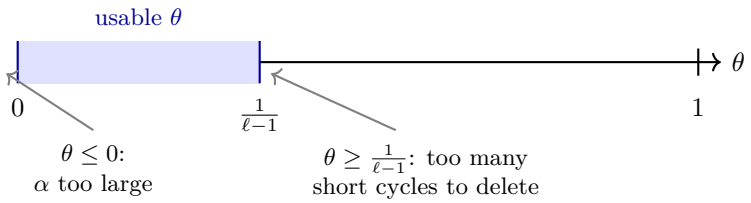
$$(1-p)^{\binom{a}{2}} \leq e^{-p\binom{a}{2}} \leq \exp\Big(-\frac{p\,n^2}{16k^2}\Big). \qquad \blacksquare$$

For $\mathbb{E}[Y]$ to be $o(1)$ we need the exponent to beat $n \log 2$, i.e. $pn^2 \gg n$, i.e. $p \gg \frac{1}{n}$. This is a *lower* bound on p: too few edges and the graph has large independent sets.

Intuition . The proof works because the two constraints

$$\frac{1}{n} \ll p \ll n^{-\frac{\ell-2}{\ell-1}}$$

are compatible: the right-hand exponent is $-1 + \frac{1}{\ell-1} > -1$, so the window is a genuine power of n wide, however large ℓ is. Writing $p = n^{\theta-1}$, the window is simply $0 < \theta < \frac{1}{\ell-1}$.



§3.4.v. Proof of the Theorem

Proof of Theorem 3.4.4. We may assume $\ell \geq 4$ and $k \geq 2$, since otherwise any K_k, resp. any graph with a cycle, already works. Fix

$$\theta := \frac{1}{2(\ell-1)} \in \Big(0, \tfrac{1}{\ell-1}\Big), \qquad p := n^{\theta-1},$$

and consider $G(n,p)$ for n large. Note $np = n^\theta \geq 1$, so Lemma 3.4.6 applies.

Let X be the number of cycles of length $< \ell$, let $a = \lceil \frac{n}{2k} \rceil$ and let Y be the number of independent sets of size a.

Step 1: $\mathbb{E}[X+Y] \le \frac{n}{2}$. By Lemma 3.4.6,

$$\mathbb{E}[X] \le \ell\,(np)^{\ell-1} = \ell\, n^{\theta(\ell-1)} = \ell\, n^{1/2} = o(n),$$

the point being that $\theta(\ell-1) = \frac{1}{2} < 1$. By Lemma 3.4.7 and $pn^2 = n^{1+\theta}$,

$$\mathbb{E}[Y] \le 2^n \exp\Big(-\frac{n^{1+\theta}}{16k^2}\Big) = \exp\Big(n\log 2 - \frac{n^{1+\theta}}{16k^2}\Big) = o(1),$$

since $n^{1+\theta}$ grows faster than n and k is a constant. Hence $\mathbb{E}[X+Y] = o(n) \le \frac{n}{2}$ for n large enough.

Step 2: the alteration. By Lemma 2.1.1 there is a graph G_0 on $[n]$ with $X + Y \le \frac{n}{2}$: it has at most $\frac{n}{2}$ short cycles and large independent sets in total. Delete one vertex from each cycle of length $< \ell$ and one vertex from each independent set of size a, and let G be the induced subgraph on what remains. Then:

- $|V(G)| \ge n - (X+Y) \ge \frac{n}{2}$, as we deleted at most one vertex per defect;
- $g(G) \ge \ell$, since every cycle of G_0 of length $< \ell$ lost a vertex, and passing to an induced subgraph creates no new cycles;
- $\alpha(G) < \frac{n}{2k}$: every a-set of G_0 that was independent lost a vertex, so no a-set of G is independent; as any independent set of size $\ge a$ contains an independent a-set, this gives $\alpha(G) \le a - 1 < \frac{n}{2k}$.

Step 3: conclusion. By Proposition 3.4.2,

$$\chi(G) \ge \frac{|V(G)|}{\alpha(G)} > \frac{n/2}{n/(2k)} = k,$$

and $g(G) \ge \ell$, which is what we wanted. ∎

Remark 3.4.8.

(1) The case $\ell = 4$ already says something striking: χ cannot be bounded by any function of ω, i.e. there is no $f\colon \mathbb{N} \to \mathbb{N}$ with $\chi(G) \le f(\omega(G))$ for all G. Indeed, for each k Theorem 3.4.4 with $\ell = 4$ gives a G_k with $g(G_k) \ge 4$; as K_3 is a 3-cycle, G_k is triangle-free, so $\omega(G_k) \le 2$, while $\chi(G_k) \ge k$. Such an f would force

$$k \le \chi(G_k) \le f(\omega(G_k)) \le \max\{f(1), f(2)\} \quad \text{for every } k,$$

which is absurd. So the lower bound $\chi \ge \omega$ of Proposition 3.4.2 has no converse: a graph can need many colors with no large clique to blame.

(2) Nothing in the proof identifies a graph; it is one of the first results for which the probabilistic argument came long before any construction. Explicit constructions of graphs with large girth and large chromatic number do exist — Lovász gave one in 1968 — but they are considerably harder than the argument above.

(3) The choice $\theta = \frac{1}{2(\ell-1)}$ was for definiteness only; any $\theta \in \big(0, \frac{1}{\ell-1}\big)$ works, and pushing θ towards the upper end is what the lecture's choice $p \approx n^{-\frac{\ell-2}{\ell-1}}$ does. Nothing is gained in the statement, since k and ℓ are fixed constants and n is free.

(4) The two defects were deleted against a single budget $\frac{n}{2}$. This is why we needed $\mathbb{E}[X] = o(n)$ rather than merely $\mathbb{E}[X] < 1$: the first moment principle is applied once, to $X + Y$, and produces one graph in which both counts are small at the same time.

4. The Second Moment

Lecture 8

Everything so far has been an existence argument. The union bound and Lemma 2.1.1 both end with the words *there is an ω with…*, and the method of alterations repaired a single such ω by hand. None of this says anything about how a *typical* outcome looks. 20 Oct.

This chapter changes that. Knowing $\mathbb{E}[X]$ alone, one cannot tell whether X is usually close to its mean or wildly spread out; the variance is what distinguishes the two, and it is cheap to compute for the sums of indicators that keep appearing in our applications.

The first moment finds one good outcome. The second moment shows that *almost every* outcome is good: if $\mathrm{Var}(X)$ is small compared with $\mathbb{E}[X]^2$, then X is concentrated around $\mathbb{E}[X]$.

From this chapter on the statements are asymptotic, so it is worth fixing once and for all what the limits are taken in.

Notation (Asymptotics)**.** All objects come in sequences indexed by a parameter $n \to \infty$: a probability space $(\Omega_n, \mathbb{P})$, a random variable X_n on it, and events $A_n \subseteq \Omega_n$. Every Landau symbol — o, O, Θ, $\ll$, $\gg$ — refers to $n \to \infty$, and every other parameter (k, ℓ, ε, c, …) is a constant held fixed as n grows. Thus:

- $\mathbb{E}[X_n] = o(1)$ means $\mathbb{E}[X_n] \to 0$ as $n \to \infty$;
- A_n holds **with high probability** if $\mathbb{P}(A_n) \to 1$ as $n \to \infty$;
- $X_n = (1 + o(1))\mathbb{E}[X_n]$ with high probability means: for every fixed $\varepsilon > 0$,

$$\mathbb{P}\Big(|X_n - \mathbb{E}[X_n]| \le \varepsilon\, \mathbb{E}[X_n]\Big) \longrightarrow 1.$$

  Note the order of the quantifiers: ε is chosen first and the probability tends to 1 for that ε; the rate may depend on ε.

As usual we suppress the subscript and write X for X_n, since in the applications the dependence on n is visible anyway — X counts triangles in $G(n,p)$, and $p = p(n)$.

§4.1. Markov's and Chebyshev's Inequalities

§4.1.i. One Tail from the Mean Alone

Lemma 2.1.1 says that $\{X \le \mathbb{E}[X]\}$ has positive probability, so a small expectation forces some outcome to be small. Markov's inequality is the quantitative version: for a non-negative X, knowing only $\mathbb{E}[X]$ already limits how much mass can sit far out on the right.

Theorem 4.1.1 (Markov's inequality)

Let $X \geq 0$ be a random variable. Then for every $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof. Take X discrete. Splitting the defining sum at a and discarding the small values,

$$\mathbb{E}[X] = \sum_x x\,\mathbb{P}(X = x) = \sum_{x<a} x\,\mathbb{P}(X = x) + \sum_{x \geq a} x\,\mathbb{P}(X = x) \;\geq\; \sum_{x<a} 0 \cdot \mathbb{P}(X = x) + \sum_{x \geq a} a\,\mathbb{P}(X = x),$$

where the first sum was dropped using $X \geq 0$ and the second bounded using $x \geq a$. The right-hand side is $a\,\mathbb{P}(X \geq a)$, and dividing by a gives the claim. ∎

Example 4.1.2 (First moment, restated)**.** Let $X \geq 0$ be integer-valued, as it is whenever X counts something. Taking $a = 1$ in Markov's inequality gives

$$\mathbb{P}(X \geq 1) \leq \mathbb{E}[X], \qquad \text{equivalently} \qquad \mathbb{P}(X = 0) \geq 1 - \mathbb{E}[X],$$

and this is how we have been using the first moment all along. The two readings of it differ in exactly the way the chapter is about.

- *One space, no limit.* If $\mathbb{E}[X] < 1$ then $\mathbb{P}(X = 0) > 0$, so there exists an outcome with $X = 0$. This is the existence statement of the previous chapters, and nothing is tending to anything.
- *A sequence, $n \to \infty$.* If $X = X_n$ is a sequence as in the convention above and $\mathbb{E}[X_n] = o(1)$, then $\mathbb{P}(X_n = 0) \geq 1 - \mathbb{E}[X_n] \to 1$, i.e. $X_n = 0$ with high probability. This is a statement about almost all outcomes, not just one.

§4.1.ii. A Large Mean Says Nothing

The other direction fails completely.

Question 4.1.3. If $\mathbb{E}[X]$ is large, is X large with high probability?

Answer. No. For a single space, let

$$X = \begin{cases} 10^6, & \text{with probability } \frac{1}{1000}, \\ 0, & \text{with probability } \frac{999}{1000}, \end{cases}$$

so that $\mathbb{E}[X] = 1000$ while $X = 0$ with probability 0.999: the mean is carried entirely by a rare enormous value.

To say this asymptotically, take the same example with n in place of 1000: let

$$X_n = \begin{cases} n^2, & \text{with probability } \frac{1}{n}, \\ 0, & \text{with probability } 1 - \frac{1}{n}. \end{cases}$$

Then $\mathbb{E}[X_n] = n \to \infty$, and yet $\mathbb{P}(X_n > 0) = \frac{1}{n} \to 0$, i.e. $X_n = 0$ with high probability. ∎

So $\mathbb{E}[X_n] \to \infty$ does *not* give $\mathbb{P}(X_n > 0) \to 1$; the example shows it does not even prevent $\mathbb{P}(X_n > 0) \to 0$. To rule out this behavior we need to know that X_n does not spread out, which is what the second moment measures.

§4.1.iii. Chebyshev's Inequality

Theorem 4.1.4 (Chebyshev's inequality)
Let X be a random variable and let $a > 0$. Then

$$\mathbb{P}\big(|X - \mathbb{E}[X]| \geq a\big) \leq \frac{\operatorname{Var}(X)}{a^2}.$$

Proof. The random variable $(X - \mathbb{E}[X])^2$ is non-negative and has expectation $\operatorname{Var}(X)$, so Markov's inequality applies to it with threshold a^2:

$$\mathbb{P}\big((X - \mathbb{E}[X])^2 \geq a^2\big) \leq \frac{\mathbb{E}\big[(X - \mathbb{E}[X])^2\big]}{a^2} = \frac{\operatorname{Var}(X)}{a^2}.$$

Finally, since $a > 0$, the event on the left is literally the event $|X - \mathbb{E}[X]| \geq a$. ∎

Remark 4.1.5.

(1) Write $\sigma \coloneqq \sqrt{\operatorname{Var}(X)}$ and put $a = \lambda\sigma$, i.e. measure the deviation in units of σ:

$$\mathbb{P}\big(|X - \mathbb{E}[X]| \geq \lambda\sigma\big) \leq \frac{\operatorname{Var}(X)}{(\lambda\sigma)^2} = \frac{1}{\lambda^2}.$$

This holds for every $\lambda > 0$, with no limit taken anywhere; it is the form in which Chebyshev's inequality is usually stated.

To read it asymptotically, let $\lambda = \lambda(n) \to \infty$ arbitrarily slowly. Then $\lambda(n)^{-2} \to 0$, so

$$\mathbb{P}\Big(|X - \mathbb{E}[X]| < \lambda(n)\,\sigma\Big) \longrightarrow 1.$$

That is, a random variable is unlikely to be $\omega\big(\sqrt{\operatorname{Var}(X)}\big)$ away from its expectation.

(2) Chebyshev can also beat Markov on the *upper* tail. For $a \gg \mathbb{E}[X]$,

$$\mathbb{P}(X \geq a) \leq \mathbb{P}\big(|X - \mathbb{E}[X]| \geq a - \mathbb{E}[X]\big) \leq \frac{\operatorname{Var}(X)}{(a - \mathbb{E}[X])^2},$$

which decays like a^{-2} rather than Markov's a^{-1}. Whenever $\operatorname{Var}(X)$ is not much bigger than $\mathbb{E}[X]$ — and Lemma 4.2.2 below says this is the typical situation — the second bound is the stronger one.

§4.2. The Variance of a Sum of Indicators

To use Chebyshev's inequality we must be able to compute, or at least bound, variances. In every application we have met, the random variable was a sum of indicators, and for those the variance is almost as manageable as the expectation.

Notation . For an event E, write $\mathbb{1}_E$ for its **indicator**, the random variable that is 1 if E occurs and 0 otherwise. Thus $\mathbb{E}[\mathbb{1}_E] = \mathbb{P}(E)$ and $\mathbb{1}_E\mathbb{1}_F = \mathbb{1}_{E\cap F}$.

So let $E_1, \ldots, E_m$ be events, let $X_i \coloneqq \mathbb{1}_{E_i}$ and put $p_i \coloneqq \mathbb{P}(E_i)$, so that

$$X \coloneqq \sum_{i=1}^{m} X_i$$

counts how many of the E_i occur. By Theorem 2.1.2, $\mathbb{E}[X] = \sum_{i=1}^{m} p_i$, with no independence needed anywhere. What about $\operatorname{Var}(X)$?

§4.2.i. The Independent Case

If the X_i are pairwise independent, variances add, and each $\mathrm{Var}(X_i) = p_i(1-p_i)$:

$$\mathrm{Var}(X) = \sum_{i=1}^{m} \mathrm{Var}(X_i) = \sum_{i=1}^{m} p_i(1-p_i) \le \sum_{i=1}^{m} p_i = \mathbb{E}[X].$$

This is already a very good sign: $\mathrm{Var}(X) \le \mathbb{E}[X]$ means $\mathrm{Var}(X) = o\big(\mathbb{E}[X]^2\big)$ as soon as $\mathbb{E}[X] \to \infty$, which by Chebyshev's inequality is concentration.

§4.2.ii. The General Case

In general the cross terms survive:

$$\mathrm{Var}(X) = \sum_{i=1}^{m} \mathrm{Var}(X_i) + \sum_{i \ne j} \mathrm{Cov}(X_i, X_j), \qquad \mathrm{Cov}(X_i, X_j) = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\mathbb{E}[X_j].$$

For indicators this covariance is something we can read off, because $X_i X_j = \mathbb{1}_{E_i \cap E_j}$:

$$\mathrm{Cov}(X_i, X_j) = \underbrace{\mathbb{P}(E_i \cap E_j) - \mathbb{P}(E_i)\mathbb{P}(E_j)}_{=0 \text{ if } E_i, E_j \text{ are independent}}.$$

Only the dependent pairs contribute, which suggests the following bookkeeping.

Definition 4.2.1 (Dependency and Δ)**.** Write $i \sim j$ if $i \ne j$ and the events E_i, E_j are *not* independent, and set

$$\Delta \coloneqq \sum_{i \sim j} \mathbb{P}(E_i \cap E_j),$$

the sum being over *unordered* pairs $\{i, j\}$ with $i \sim j$, each counted once. Since $\sim$ is symmetric, the sum over *ordered* pairs counts each such pair twice:

$$\sum_{(i,j)\,:\,i \sim j} \mathbb{P}(E_i \cap E_j) = 2\Delta. \tag{4.1}$$

Lemma 4.2.2

With the notation above, $\mathrm{Var}(X) \le \mathbb{E}[X] + 2\Delta$.

Proof. The diagonal terms give $\sum_i \mathrm{Var}(X_i) = \sum_i p_i(1-p_i) \le \sum_i p_i = \mathbb{E}[X]$. For the cross terms, a pair with $i \not\sim j$ contributes $\mathrm{Cov}(X_i, X_j) = 0$, and a pair with $i \sim j$ contributes

$$\mathrm{Cov}(X_i, X_j) = \mathbb{P}(E_i \cap E_j) - \mathbb{P}(E_i)\mathbb{P}(E_j) \le \mathbb{P}(E_i \cap E_j).$$

The sum $\sum_{i \ne j}$ in $\mathrm{Var}(X)$ runs over ordered pairs, so by (4.1)

$$\sum_{i \ne j} \mathrm{Cov}(X_i, X_j) \le \sum_{(i,j)\,:\,i \sim j} \mathbb{P}(E_i \cap E_j) = 2\Delta. \qquad \blacksquare$$

Intuition . Δ measures how much the events overlap. The smaller the dependencies, the closer X is to a sum of independent indicators, and the independent case was already concentrated. So the whole method reduces to one question: *is Δ small compared with $\mathbb{E}[X]^2$?*

§4.3. The Second Moment Method

In every application X counts something: the number of monochromatic cliques in a coloring, of triangles in a graph, of short cycles. For such an X,

$$X > 0 \Leftrightarrow \text{at least one of } E_1, \dots, E_m \text{ occurs} \Leftrightarrow \text{the structure being counted exists},$$

and $X = 0$ says that it does not. Deciding which of the two happens is what the method is for, and the two moments do opposite halves of the job.

- $X = 0$ with high probability follows from $\mathbb{E}[X] = o(1)$ alone. This is Example 4.1.2, a one-line application of Markov's inequality; no variance is involved.
- $X > 0$ with high probability does *not* follow from $\mathbb{E}[X] \to \infty$ — that is precisely what Question 4.1.3 showed. Here the first moment is not enough, and the variance is what supplies the rest.

The corollary below supplies the missing second case: it says that $\mathbb{E}[X] \to \infty$ does give $X > 0$ once the variance is under control.

Corollary 4.3.1 (Second moment method)

For each n let $E_1, \dots, E_m$ be events, with $m = m(n)$ and the E_i depending on n as well; let $X = X_n$ be the number of them that occur, and let $\Delta = \Delta_n$ be as in Definition 4.2.1. If, as $n \to \infty$,

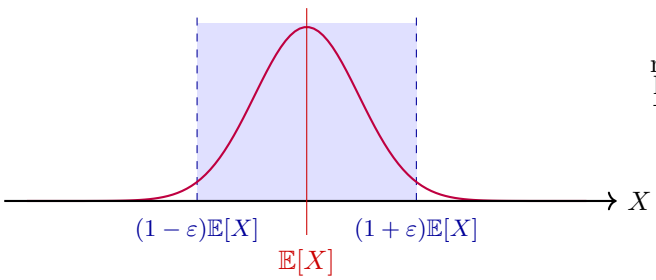
$$\mathbb{E}[X] \to \infty \qquad \text{and} \qquad \Delta = o\big(\mathbb{E}[X]^2\big),$$

then $X = (1 + o(1))\mathbb{E}[X]$ with high probability. In particular $X \geq \frac{1}{2}\mathbb{E}[X] > 0$ with high probability, so at least one E_i occurs.

Proof. Let $\varepsilon > 0$ be a constant. Applying Chebyshev's inequality with $a = \varepsilon\mathbb{E}[X]$ and then Lemma 4.2.2,

$$\mathbb{P}\big(|X - \mathbb{E}[X]| \geq \varepsilon\mathbb{E}[X]\big) \leq \frac{\operatorname{Var}(X)}{\varepsilon^2\mathbb{E}[X]^2} \leq \frac{\mathbb{E}[X] + 2\Delta}{\varepsilon^2\mathbb{E}[X]^2} = \underbrace{\frac{1}{\varepsilon^2\mathbb{E}[X]}}_{\to 0 \text{ as } \mathbb{E}[X]\to\infty} + \underbrace{\frac{2\Delta}{\varepsilon^2\mathbb{E}[X]^2}}_{\to 0 \text{ as } \Delta = o(\mathbb{E}[X]^2)} = o(1).$$

Hence with probability $1 - o(1)$ we have $(1 - \varepsilon)\mathbb{E}[X] \leq X \leq (1 + \varepsilon)\mathbb{E}[X]$. As $\varepsilon > 0$ was an arbitrary constant, this says $X = (1 + o(1))\mathbb{E}[X]$ with high probability. ∎



mass outside the window is at most $\dfrac{\mathbb{E}[X] + 2\Delta}{\varepsilon^2\mathbb{E}[X]^2} = o(1)$

§4.4. Triangles in a Random Graph

Question 4.4.1. For which $p = p(n)$ does $G(n,p)$ contain a triangle?

Recall $G(n,p)$ from Definition 3.4.5. Write X for the number of triangles, so that the question asks when $X > 0$ and when $X = 0$. Which argument settles it depends on how $\mathbb{E}[X]$ behaves:

- if $\mathbb{E}[X] \to 0$, then Markov's inequality gives $X = 0$ with high probability, and $\mathbb{E}[X]$ is all we need;
- if $\mathbb{E}[X] \to \infty$, then Markov's inequality says nothing, and we use Corollary 4.3.1 instead, which additionally requires $\Delta = o\big(\mathbb{E}[X]^2\big)$.

So in either case the work consists of computing

$$\mathbb{E}[X] = \sum_i \mathbb{P}(E_i) \qquad \text{and} \qquad \Delta = \sum_{i \sim j} \mathbb{P}(E_i \cap E_j),$$

and that is the entire content of the section.

§4.4.i. Setting Up and Computing $\mathbb{E}[X]$

Index the triples of vertices as $T_1, \dots, T_{\binom{n}{3}}$, let E_i be the event that all three pairs inside T_i are edges, and let

$$X := \sum_{i=1}^{\binom{n}{3}} \mathbb{1}_{E_i}$$

be the number of triangles. Then $X > 0$ exactly when $G(n,p)$ contains a triangle, so Question 4.4.1 is the question of when $X > 0$ and when $X = 0$.

Each E_i asks for 3 specific edges, so $\mathbb{P}(E_i) = p^3$ and, by Theorem 2.1.2,

$$\mathbb{E}[X] = \binom{n}{3} p^3 = \big(1 + o(1)\big)\frac{(np)^3}{6}.$$

So $\mathbb{E}[X]$ is governed by np, the expected degree of a vertex:

$$p \ll \tfrac{1}{n} \Rightarrow np \to 0 \Rightarrow \mathbb{E}[X] \to 0, \qquad p \gg \tfrac{1}{n} \Rightarrow np \to \infty \Rightarrow \mathbb{E}[X] \to \infty.$$

On the left-hand side $\mathbb{E}[X] \to 0$ and Markov's inequality will finish the job; on the right-hand side $\mathbb{E}[X] \to \infty$ and we shall have to compute Δ. So $p = \frac{1}{n}$ is the one place where the answer can change, and the two sides are treated separately below.

§4.4.ii. Below $\frac{1}{n}$: the First Moment Suffices

Proposition 4.4.2

If $p \ll \frac{1}{n}$, then with high probability $G(n,p)$ has no triangle.

Proof. Here $\mathbb{E}[X] \to 0$, so by Markov's inequality (as in Example 4.1.2),

$$\mathbb{P}(X \geq 1) \leq \mathbb{E}[X] = o(1).$$

■

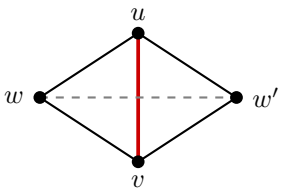
§4.4.iii. Above $\frac{1}{n}$: Computing Δ

Here $\mathbb{E}[X] \to \infty$, which by Question 4.1.3 is not enough on its own. By Corollary 4.3.1 it suffices to show $\Delta = o\big(\mathbb{E}[X]^2\big)$, so we have to compute Δ. By its definition this needs three things: *which* pairs satisfy $i \sim j$, *how many* such pairs there are, and *what* $\mathbb{P}(E_i \cap E_j)$ is for each.

Which pairs. The event E_i depends only on the 3 pairs inside T_i, and distinct pairs of $G(n,p)$ are independent. So if T_i and T_j share at most one vertex, they involve disjoint sets of pairs and

$$\mathbb{P}(E_i \cap E_j) = p^6 = \mathbb{P}(E_i)\mathbb{P}(E_j),$$

i.e. $i \not\sim j$. Hence $i \sim j$ exactly when $|T_i \cap T_j| = 2$, that is, when the two triangles share an edge.



$T_i = \{u, v, w\}$ and $T_j = \{u, v, w'\}$ overlap in the shared edge uv (red), so $E_i \cap E_j$ asks for $3 + 3 - 1 = 5$ edges (the dashed pair ww' plays no role)

How many. Such a pair of triples, taken unordered, is determined by its 4 vertices together with which of the 6 pairs among them is the shared edge, and conversely each such choice gives one. So there are $6\binom{n}{4} = \Theta(n^4)$ unordered pairs $\{i, j\}$ with $i \sim j$.

With what probability. As the picture shows, $E_i \cap E_j$ asks for $3 + 3 - 1 = 5$ distinct edges, so $\mathbb{P}(E_i \cap E_j) = p^5$.

Putting the three together,

$$\Delta = \sum_{i \sim j} \mathbb{P}(E_i \cap E_j) = 6\binom{n}{4} p^5 = \Theta\big(n^4 p^5\big).$$

Proposition 4.4.3

If $p \gg \frac{1}{n}$, then with high probability $G(n,p)$ has $(1+o(1))\binom{n}{3}p^3 \to \infty$ triangles; in particular it has a triangle.

Proof. We have $\mathbb{E}[X] = \Theta\big(n^3 p^3\big) \to \infty$, and with $\Delta = \Theta(n^4 p^5)$ from above,

$$\frac{\Delta}{\mathbb{E}[X]^2} = \Theta\Big(\frac{n^4 p^5}{n^6 p^6}\Big) = \Theta\Big(\frac{1}{n^2 p}\Big) = o(1),$$

the last step because $n^2 p = n \cdot (np) \to \infty$. So $\Delta = o\big(\mathbb{E}[X]^2\big)$ and Corollary 4.3.1 applies. ∎

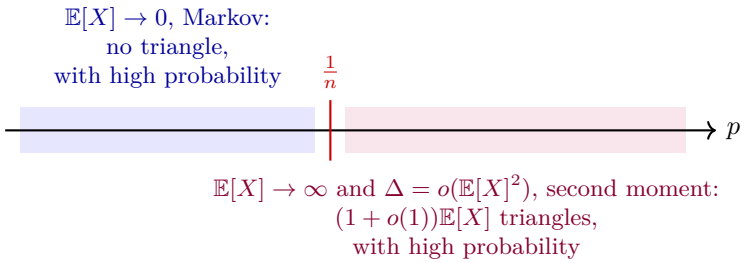
§4.4.iv. Summary

Theorem 4.4.4

Let X be the number of triangles in $G(n,p)$. Then, with high probability,

$$X = \begin{cases} 0, & \text{if } p \ll \frac{1}{n}, \\ (1+o(1))\dfrac{(np)^3}{6} \to \infty, & \text{if } p \gg \frac{1}{n}. \end{cases}$$

So the answer switches at $p = \frac{1}{n}$: below it a triangle is absent with high probability, above it one is present with high probability.



§4.5. Thresholds

Theorem 4.4.4 is one instance of a general phenomenon: for a large class of graph properties, the probability that $G(n,p)$ has the property jumps from 0 to 1 around a single value of p. The class is the following.

Definition 4.5.1 (Monotone increasing property)**.** A **graph property** $\mathcal{P}$ is a family of graphs on the vertex set $[n]$, closed under isomorphism.[1] It is **monotone increasing** if adding edges cannot destroy it: whenever $G \in \mathcal{P}$ and $G \subseteq H$ (with $V(G) = V(H) = [n]$), also $H \in \mathcal{P}$.

For example, *containing a triangle*, *being connected* and *having chromatic number at least k* are monotone increasing; *being triangle-free* is not (it is monotone decreasing).

Lemma 4.5.2 (Monotone coupling)

If $\mathcal{P}$ is monotone increasing, then the function

$$f(p) \coloneqq \mathbb{P}\big(G(n,p) \in \mathcal{P}\big), \qquad p \in [0,1],$$

is non-decreasing: $p \le p'$ implies $f(p) \le f(p')$.

The claim is intuitive, since a larger p gives more edges and edges can only help $\mathcal{P}$. The difficulty is that $G(n,p)$ and $G(n,p')$ are two separate random graphs, so "$G(n,p) \subseteq G(n,p')$" does not even make sense yet. The proof builds both on one probability space, in such a way that the smaller graph is always contained in the larger one; this is called a **coupling**.

Proof. (1) **One space for all p.** For each pair $e \in \binom{[n]}{2}$ take $U_e \sim \mathrm{Unif}[0,1]$, all independent, and for every $p \in [0,1]$ define the graph

$$G_p \coloneqq \big\{e \in \tbinom{[n]}{2} : U_e \le p\big\}.$$

(2) **Each G_p has the right law.** Each e lies in G_p with probability $\mathbb{P}(U_e \le p) = p$, and these events are independent because the U_e are. This is Definition 3.4.5, so $G_p \sim G(n,p)$ and therefore $f(p) = \mathbb{P}(G_p \in \mathcal{P})$.

(3) **The graphs are nested.** Let $p \le p'$. For every outcome, $U_e \le p$ implies $U_e \le p'$, so $G_p \subseteq G_{p'}$ holds *pointwise*, not just on average.

(4) **Apply monotonicity.** By Definition 4.5.1, $G_p \in \mathcal{P}$ and $G_p \subseteq G_{p'}$ give $G_{p'} \in \mathcal{P}$. Hence $\{G_p \in \mathcal{P}\} \subseteq \{G_{p'} \in \mathcal{P}\}$ as events, and taking probabilities,

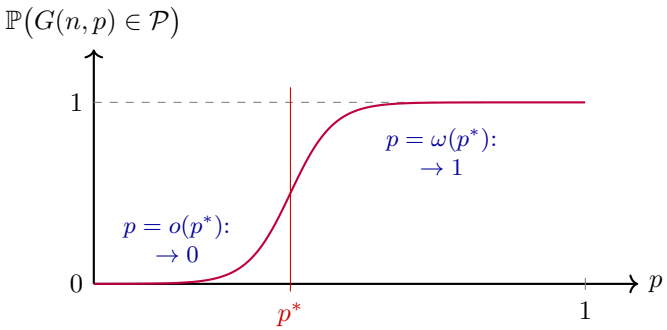
$$f(p) = \mathbb{P}(G_p \in \mathcal{P}) \le \mathbb{P}(G_{p'} \in \mathcal{P}) = f(p'). \qquad \blacksquare$$

[1] That is, $G \in \mathcal{P}$ and $G' \cong G$ imply $G' \in \mathcal{P}$, where $G' \cong G$ means G' is obtained from G by relabelling the vertices via a bijection $\sigma\colon [n] \to [n]$. So $\mathcal{P}$ depends only on the structure of the graph, not on vertex labels: *containing a triangle* qualifies, while $\{G : \{1,2\} \in E(G)\}$ does not.

Think of U_e as the time at which edge e is switched on: raising p from 0 to 1 adds edges one at a time and never removes any, so once $\mathcal{P}$ holds it keeps holding. This is why the curve in the figure below only goes up, and why it makes sense to speak of a single point p^* where it jumps from 0 to 1.

Definition 4.5.3 (Threshold)**.** Let $\mathcal{P}$ be a monotone increasing graph property. We say $p^* = p^*(n)$ is a **threshold** for $\mathcal{P}$ if

$$p = o(p^*) \Rightarrow \mathbb{P}\big(G(n,p) \in \mathcal{P}\big) \to 0, \qquad p = \omega(p^*) \Rightarrow \mathbb{P}\big(G(n,p) \in \mathcal{P}\big) \to 1.$$



Example 4.5.4. Theorem 4.4.4 says exactly that $p^* = \frac{1}{n}$ is a threshold for containing a triangle.

A threshold is determined only up to a constant factor: if p^* is a threshold, so is $c\,p^*$ for any constant $c > 0$, since $o(p^*) = o(c\,p^*)$. In particular Definition 4.5.3 says nothing about $p = c\,p^*$. The stronger notion below pins the transition down to a window of width $o(p^*)$.

Definition 4.5.5 (Sharp threshold)**.** We say p^* is a **sharp threshold** for $\mathcal{P}$ if for every constant $\varepsilon > 0$,

$$p \le (1-\varepsilon)p^* \Rightarrow \mathbb{P}\big(G(n,p) \in \mathcal{P}\big) \to 0, \qquad p \ge (1+\varepsilon)p^* \Rightarrow \mathbb{P}\big(G(n,p) \in \mathcal{P}\big) \to 1.$$

Remark 4.5.6 (Sharp $\Rightarrow$ threshold, but not conversely)**.**

(1) *A sharp threshold is a threshold.* If $p = o(p^*)$, then $p \le \frac{1}{2}p^*$ for all large n, so Definition 4.5.5 with $\varepsilon = \frac{1}{2}$ gives $\mathbb{P}\big(G(n,p) \in \mathcal{P}\big) \to 0$; if $p = \omega(p^*)$, then $p \ge 2p^*$ eventually, and $\varepsilon = 1$ gives $\to 1$.

(2) *A threshold need not be sharp.* Suppose p^* is a sharp threshold for $\mathcal{P}$, and let $q^* := 2p^*$.

- q^* is a threshold: $o(q^*) = o(p^*)$ and $\omega(q^*) = \omega(p^*)$, so the two conditions of Definition 4.5.3 for q^* are exactly those for p^*, which hold by (1).
- q^* is not sharp: take $p = \frac{3}{2}p^*$. Then $p = (1-\frac{1}{4})\,q^*$, so if q^* were sharp, Definition 4.5.5 with $\varepsilon = \frac{1}{4}$ would give $\mathbb{P}\big(G(n,p) \in \mathcal{P}\big) \to 0$. But $p = (1+\frac{1}{2})\,p^*$, so sharpness of p^* with $\varepsilon = \frac{1}{2}$ gives $\mathbb{P}\big(G(n,p) \in \mathcal{P}\big) \to 1$.

So a threshold cannot tell p^* from $2p^*$, whereas a sharp threshold is determined up to a factor $1 + o(1)$.

§4.6. Symmetric Events and Δ^*

Every threshold proof so far used Corollary 4.3.1, whose hypothesis $\Delta = o\big(\mathbb{E}[X]^2\big)$ needs Δ computed. In the applications the events all look alike, and then Δ factors. Return to the

setting of Section 4.2: $X = \sum_{i=1}^{m} \mathbb{1}_{E_i}$, $\mathbb{E}[X] = \sum_i \mathbb{P}(E_i)$. Passing to ordered pairs by (4.1), writing $\mathbb{P}(E_i \cap E_j) = \mathbb{P}(E_i)\,\mathbb{P}(E_j \mid E_i)$ and summing over j first,

$$\Delta = \frac{1}{2} \sum_{(i,j)\,:\,i \sim j} \mathbb{P}(E_i \cap E_j) = \frac{1}{2} \sum_{i=1}^{m} \mathbb{P}(E_i) \sum_{j\,:\,j \sim i} \mathbb{P}(E_j \mid E_i). \tag{4.2}$$

The inner sum in (4.2) depends on i in general. It does not when the events all "look the same", and the way to say this formally is through automorphisms.

Definition 4.6.1 (Automorphism of a probability space)**.** An **automorphism** of $(\Omega, \mathbb{P})$ is a bijection $\sigma\colon \Omega \to \Omega$ that does not change probabilities: $\mathbb{P}\big(\sigma(A)\big) = \mathbb{P}(A)$ for every event A. For finite Ω this just says $\mathbb{P}(\{\sigma(\omega)\}) = \mathbb{P}(\{\omega\})$ for every outcome ω.

Definition 4.6.2 (Symmetric events)**.** The events $E_1, \dots, E_m$ are **symmetric** if for all i, j there are an automorphism σ of $(\Omega, \mathbb{P})$ and a permutation τ of $[m]$ with

$$\sigma(E_l) = E_{\tau(l)} \quad \text{for all } l \in [m], \qquad \text{and} \qquad \tau(i) = j.$$

That is, σ permutes the family $\{E_1, \dots, E_m\}$ and sends E_i to E_j.

Lemma 4.6.3

If $E_1, \dots, E_m$ are symmetric, then $\sum_{l\,:\,l \sim i} \mathbb{P}(E_l \mid E_i)$ is the same for every i. We call this common value

$$\Delta^* := \sum_{l\,:\,l \sim i} \mathbb{P}(E_l \mid E_i).$$

The proof uses the following fact about bijections.

Claim 4.6.1. If $\sigma\colon \Omega \to \Omega$ is a bijection, then $\sigma(A \cap B) = \sigma(A) \cap \sigma(B)$ for all $A, B \subseteq \Omega$. Hence, if σ is an automorphism,

$$\mathbb{P}\big(\sigma(A) \cap \sigma(B)\big) = \mathbb{P}\big(\sigma(A \cap B)\big) = \mathbb{P}(A \cap B).$$

Proof. $\subseteq$: if $\omega \in A \cap B$, then $\sigma(\omega) \in \sigma(A)$ and $\sigma(\omega) \in \sigma(B)$. $\supseteq$: if $\eta \in \sigma(A) \cap \sigma(B)$, then $\eta = \sigma(a) = \sigma(b)$ for some $a \in A$, $b \in B$; injectivity gives $a = b \in A \cap B$, so $\eta \in \sigma(A \cap B)$. The second statement is Definition 4.6.1 applied to the event $A \cap B$. ∎

Proof of Lemma 4.6.3. For each index i write $D_i := \sum_{l\,:\,l \sim i} \mathbb{P}(E_l \mid E_i)$. Fix one index j. We show $D_i = D_j$ for an arbitrary i; then every D_i equals the same number D_j, which is the claim of the lemma.

So let i be arbitrary, and take σ, τ as in Definition 4.6.2 for this pair i, j, so $\sigma(E_l) = E_{\tau(l)}$ and $\sigma(E_i) = E_j$. For every l, Claim 4.6.1 with $A = E_l$, $B = E_i$, together with $\mathbb{P}(\sigma(E)) = \mathbb{P}(E)$, gives

$$\mathbb{P}\big(E_{\tau(l)} \cap E_j\big) = \mathbb{P}(E_l \cap E_i), \qquad \mathbb{P}\big(E_{\tau(l)}\big) = \mathbb{P}(E_l), \qquad \mathbb{P}(E_j) = \mathbb{P}(E_i). \tag{4.3}$$

- *Conditional probabilities are preserved.* By (4.3),

$$\mathbb{P}\big(E_{\tau(l)} \mid E_j\big) = \frac{\mathbb{P}\big(E_{\tau(l)} \cap E_j\big)}{\mathbb{P}(E_j)} = \frac{\mathbb{P}(E_l \cap E_i)}{\mathbb{P}(E_i)} = \mathbb{P}(E_l \mid E_i).$$

- *Dependence is preserved.* By (4.3),

$$\mathbb{P}\big(E_{\tau(l)} \cap E_j\big) - \mathbb{P}\big(E_{\tau(l)}\big)\mathbb{P}(E_j) = \mathbb{P}(E_l \cap E_i) - \mathbb{P}(E_l)\mathbb{P}(E_i),$$

so E_l, E_i are independent iff $E_{\tau(l)}, E_{\tau(i)}$ are. Hence $l \sim i \Leftrightarrow \tau(l) \sim \tau(i) = j$.

Re-indexing the sum by $l' = \tau(l)$ and using the first bullet,

$$D_j = \sum_{l' : l' \sim j} \mathbb{P}(E_{l'} \mid E_j) = \sum_{l : l \sim i} \mathbb{P}\big(E_{\tau(l)} \mid E_j\big) = \sum_{l : l \sim i} \mathbb{P}(E_l \mid E_i) = D_i. \qquad \blacksquare$$

Lemma 4.6.4

If $E_1, \dots, E_m$ are symmetric, then $\Delta = \frac{1}{2}\Delta^* \mathbb{E}[X]$. Consequently

$$\Delta = o\big(\mathbb{E}[X]^2\big) \Leftrightarrow \Delta^* = o\big(\mathbb{E}[X]\big).$$

Proof. By Lemma 4.6.3, the inner sum in (4.2) equals Δ^* for every i, so

$$\Delta = \frac{1}{2}\sum_{i=1}^{m} \mathbb{P}(E_i) \cdot \Delta^* = \frac{1}{2}\Delta^* \sum_{i=1}^{m} \mathbb{P}(E_i) = \frac{1}{2}\Delta^* \mathbb{E}[X].$$

Dividing by $\mathbb{E}[X]^2$ gives $\dfrac{\Delta}{\mathbb{E}[X]^2} = \dfrac{1}{2} \cdot \dfrac{\Delta^*}{\mathbb{E}[X]}$, so one side is $o(1)$ iff the other is. $\blacksquare$

Combining Lemma 4.6.4 with Corollary 4.3.1:

Corollary 4.6.5 (Second moment method, symmetric form)

If $E_1, \dots, E_m$ are symmetric, $\mathbb{E}[X] \to \infty$ and $\Delta^* = o\big(\mathbb{E}[X]\big)$, then $X = (1 + o(1))\,\mathbb{E}[X]$ with high probability.

The gain is that Δ^* only asks about a *single fixed* event E_i: which E_j depend on it, and how likely each is once E_i is known to hold.

In $G(n,p)$ the symmetry comes from relabelling vertices. A **copy** of a graph H is a subgraph of K_n isomorphic to H.

Lemma 4.6.6

Fix a graph H, and for each copy S of H let $E_S := \{S \subseteq G(n,p)\}$. Then the events E_S are symmetric.

Proof. Let S, S' be copies of H. Since $S \cong H \cong S'$, there is a bijection $V(S) \to V(S')$ mapping edges of S onto edges of S'; extend it to a permutation π of $[n]$, so that $\pi(S) = S'$. For a graph G on $[n]$ let $\pi(G)$ be the graph with edges $\{\pi(u), \pi(v)\}$, $\{u, v\} \in E(G)$, and set $\sigma(G) := \pi(G)$.

- σ is an automorphism: it is a bijection with inverse $G \mapsto \pi^{-1}(G)$, and since $e(\pi(G)) = e(G)$,

$$\mathbb{P}(\{\sigma(G)\}) = p^{e(\pi(G))}(1-p)^{\binom{n}{2} - e(\pi(G))} = p^{e(G)}(1-p)^{\binom{n}{2} - e(G)} = \mathbb{P}(\{G\}).$$

- σ permutes the events: for every copy T,

$$G' \in \sigma(E_T) \Leftrightarrow \pi^{-1}(G') \in E_T \Leftrightarrow T \subseteq \pi^{-1}(G') \Leftrightarrow \pi(T) \subseteq G' \Leftrightarrow G' \in E_{\pi(T)},$$

  so $\sigma(E_T) = E_{\pi(T)}$. The map $\tau(T) := \pi(T)$ is a permutation of the copies of H (with inverse $T \mapsto \pi^{-1}(T)$) and $\tau(S) = S'$.

This is Definition 4.6.2. $\blacksquare$

Example 4.6.7 (Triangles, again)**.** Let X count triangles in $G(n,p)$ as in Section 4.4. The triangle events are symmetric by Lemma 4.6.6 with $H = K_3$. Fix a triple T_i. The $T_j \sim T_i$ are those sharing an edge with T_i: 3 choices of the shared edge times $n-3$ choices of the fourth vertex. Given E_i, the event E_j needs only the 2 edges of T_j not in T_i, so $\mathbb{P}(E_j \mid E_i) = p^2$. Hence

$$\Delta^* = 3(n-3)\,p^2 = \Theta(np^2), \qquad \frac{\Delta^*}{\mathbb{E}[X]} = \Theta\Big(\frac{np^2}{n^3p^3}\Big) = \Theta\Big(\frac{1}{n^2p}\Big) = o(1) \quad \text{when } p \gg \tfrac{1}{n},$$

which is Proposition 4.4.3 again. As a check,

$$\frac{1}{2}\Delta^*\,\mathbb{E}[X] = \frac{1}{2}\cdot 3(n-3)p^2 \cdot \frac{n(n-1)(n-2)}{6}p^3 = \frac{n(n-1)(n-2)(n-3)}{4}p^5 = 6\binom{n}{4}p^5,$$

which by Lemma 4.6.4 is Δ, and agrees with the value of Δ computed directly in Section 4.4.

§4.7. Cliques in a Random Graph

> **Question 4.7.1.** For fixed $k \geq 3$, what is the threshold for $G(n,p)$ to contain K_k?

The property $\mathcal{P} = \{G : K_k \subseteq G\}$ is monotone increasing (Definition 4.5.1): if $K_k \subseteq G \subseteq H$, then $K_k \subseteq H$. So Definition 4.5.3 applies, and it makes sense to look for a threshold.

§4.7.i. The First Moment

For each $S \in \binom{[n]}{k}$ let E_S be the event that S spans a clique in $G(n,p)$. This needs all $\binom{k}{2}$ pairs inside S to be edges, each present independently with probability p, so $\mathbb{P}(E_S) = p^{\binom{k}{2}}$. Let

$$X := \sum_{S \in \binom{[n]}{k}} \mathbb{1}_{E_S}$$

be the number of copies of K_k. By Theorem 2.1.2, and since k is fixed while $n \to \infty$,

$$\mathbb{E}[X] = \sum_S \mathbb{P}(E_S) = \binom{n}{k} p^{\binom{k}{2}} = (1+o(1))\frac{n^k}{k!}\,p^{\frac{k(k-1)}{2}} = (1+o(1))\frac{\big(np^{\frac{k-1}{2}}\big)^k}{k!}, \tag{4.4}$$

where $\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k!} = (1+o(1))\frac{n^k}{k!}$ because each of the k factors $n-r$ with $r<k$ fixed is $(1+o(1))n$, and $p^{\binom{k}{2}} = p^{\frac{k(k-1)}{2}} = \big(p^{\frac{k-1}{2}}\big)^k$. So $\mathbb{E}[X]$ is governed by $np^{(k-1)/2} = \big(n^{\frac{2}{k-1}}p\big)^{\frac{k-1}{2}}$, and

$$np^{\frac{k-1}{2}} \to 0 \Leftrightarrow p = o\big(n^{-\frac{2}{k-1}}\big), \qquad np^{\frac{k-1}{2}} \to \infty \Leftrightarrow p = \omega\big(n^{-\frac{2}{k-1}}\big).$$

In the first case (4.4) gives $\mathbb{E}[X] \to 0$, and Markov's inequality gives $\mathbb{P}(X \geq 1) \leq \mathbb{E}[X] \to 0$: no K_k with high probability. In the second case $\mathbb{E}[X] \to \infty$. The events E_S are symmetric by Lemma 4.6.6 with $H = K_k$, so by Corollary 4.6.5 it is enough to show $\Delta^* = o\big(\mathbb{E}[X]\big)$.

§4.7.ii. Computing Δ^*

Fix a k-set S, and let $T \neq S$ be another k-set with $i := |S \cap T|$. Write $P_S := \binom{S}{2}$ for the set of pairs inside S, so $|P_S| = \binom{k}{2}$ and E_S is the event that every pair in P_S is an edge. A pair lies inside both S and T iff it lies inside $S \cap T$, so

$$|P_S \cap P_T| = \binom{i}{2}, \qquad |P_S \cup P_T| = |P_S| + |P_T| - |P_S \cap P_T| = 2\binom{k}{2} - \binom{i}{2}.$$

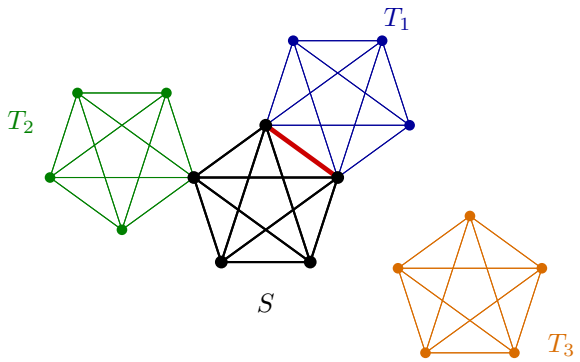
- If $i \le 1$, then $P_S \cap P_T = \varnothing$. The event E_S depends only on the pairs in P_S and E_T only on those in P_T; these are disjoint sets of pairs, which are independent in $G(n,p)$, so E_S and E_T are independent.
- If $i \ge 2$, then the event $E_S \cap E_T$ says that every pair in $P_S \cup P_T$ is an edge, so for $0 < p < 1$

$$\mathbb{P}(E_S \cap E_T) = p^{|P_S \cup P_T|} = p^{2\binom{k}{2} - \binom{i}{2}} > p^{2\binom{k}{2}} = \mathbb{P}(E_S)\,\mathbb{P}(E_T),$$

the inequality because $\binom{i}{2} \ge 1$. So E_S and E_T are not independent.

Hence

$$T \sim S \Leftrightarrow \text{the cliques on } S \text{ and } T \text{ share an edge} \Leftrightarrow |S \cap T| \ge 2.$$



$|S \cap T_1| = 2$: share the red edge, $T_1 \sim S$;
$|S \cap T_2| = 1$: no common edge, $T_2 \not\sim S$;
$|S \cap T_3| = 0$: $T_3 \not\sim S$.

Grouping the dependent T by $i = |S \cap T| \in \{2, \dots, k-1\}$,

$$\Delta^* = \sum_{T\,:\,T \sim S} \mathbb{P}(E_T \mid E_S) = \sum_{i=2}^{k-1} \sum_{T\,:\,|T \cap S| = i} \mathbb{P}(E_T \mid E_S).$$

For each i we need the conditional probability and the number of such T.

Conditional probability. Suppose $|S \cap T| = i \ge 2$. By the second bullet above,

$$\mathbb{P}(E_T \mid E_S) = \frac{\mathbb{P}(E_S \cap E_T)}{\mathbb{P}(E_S)} = \frac{p^{2\binom{k}{2} - \binom{i}{2}}}{p^{\binom{k}{2}}} = p^{\binom{k}{2} - \binom{i}{2}}.$$

Number of T. To build T with $|T \cap S| = i$, choose the i vertices of S that lie in T and then the $k - i$ vertices of T outside S:

$$\#\{T : |T \cap S| = i\} = \underbrace{\binom{k}{i}}_{\Theta(1)} \underbrace{\binom{n-k}{k-i}}_{\Theta(n^{k-i})} = \Theta\big(n^{k-i}\big),$$

where $\binom{n-k}{k-i} = (1 + o(1))\frac{n^{k-i}}{(k-i)!}$ as for $\binom{n}{k}$ above.

Putting the two together and writing $q_i := n^{k-i} p^{\binom{k}{2} - \binom{i}{2}}$,

$$\Delta^* = \sum_{i=2}^{k-1} \binom{k}{i} \binom{n-k}{k-i} p^{\binom{k}{2} - \binom{i}{2}} = \sum_{i=2}^{k-1} \Theta(q_i) = \Theta\Big(\max_{2 \le i \le k-1} q_i\Big),$$

the last step because $\max_i q_i \le \sum_{i=2}^{k-1} q_i \le (k-2) \max_i q_i$ and $k - 2$ is a constant.

§4.7.iii. Which Term Dominates

Lemma 4.7.2

$\max_{2\le i\le k-1} q_i = \max\{q_2, q_{k-1}\}$.

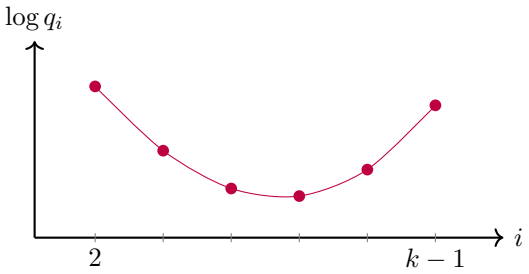
Proof. Consecutive ratios are

$$\frac{q_{i+1}}{q_i} = \frac{n^{k-(i+1)}p^{\binom{k}{2}-\binom{i+1}{2}}}{n^{k-i}p^{\binom{k}{2}-\binom{i}{2}}} = n^{-1}p^{\binom{i}{2}-\binom{i+1}{2}} = n^{-1}p^{-i},$$

using $\binom{i+1}{2} - \binom{i}{2} = i$. Write $r_i := q_{i+1}/q_i = n^{-1}p^{-i}$. Then $r_{i+1}/r_i = p^{-1} \ge 1$, so $r_2 \le r_3 \le \cdots \le r_{k-2}$. Let i_0 be the least $i \in \{2, \dots, k-2\}$ with $r_i > 1$, or $i_0 = k-1$ if there is none. For $i < i_0$ we have $r_i \le 1$, i.e. $q_{i+1} \le q_i$; for $i \ge i_0$ we have $r_i \ge r_{i_0} > 1$, i.e. $q_{i+1} > q_i$. So

$$q_2 \ge q_3 \ge \cdots \ge q_{i_0} < q_{i_0+1} < \cdots < q_{k-1},$$

and every q_i is at most q_2 (if $i \le i_0$) or at most q_{k-1} (if $i \ge i_0$). ∎



ratios $q_{i+1}/q_i = n^{-1}p^{-i}$ increase,
so $\log q_i$ is convex in i
and the maximum is at $i = 2$ or $i = k-1$

So it suffices to compare the two endpoints with $\mathbb{E}[X] = \Theta\big(n^k p^{\binom{k}{2}}\big)$ from (4.4), assuming throughout that $p = \omega\big(n^{-\frac{2}{k-1}}\big)$.

- $i = 2$: here $q_2 = n^{k-2}p^{\binom{k}{2}-1}$, so

$$\frac{q_2}{\mathbb{E}[X]} = \Theta\left(\frac{n^{k-2}p^{\binom{k}{2}-1}}{n^k p^{\binom{k}{2}}}\right) = \Theta\Big(\frac{1}{n^2 p}\Big) = o(1),$$

  because $k \ge 3$ gives $\frac{2}{k-1} \le 1$, hence $p \gg n^{-\frac{2}{k-1}} \ge n^{-1}$ and $n^2 p \gg n \to \infty$.

- $i = k-1$: here $\binom{k}{2} - \binom{k-1}{2} = \frac{(k-1)\big(k-(k-2)\big)}{2} = k-1$, so $q_{k-1} = np^{k-1}$ and, using $\binom{k}{2} - (k-1) = \binom{k-1}{2} = \frac{(k-1)(k-2)}{2}$,

$$\frac{q_{k-1}}{\mathbb{E}[X]} = \Theta\left(\frac{np^{k-1}}{n^k p^{\binom{k}{2}}}\right) = \Theta\Big(n^{-(k-1)}p^{-\frac{(k-1)(k-2)}{2}}\Big) = \Theta\Big(\big(np^{\frac{k-2}{2}}\big)^{-(k-1)}\Big) = o(1),$$

  because raising $p \gg n^{-\frac{2}{k-1}}$ to the power $\frac{k-2}{2} > 0$ gives $p^{\frac{k-2}{2}} \gg n^{-\frac{k-2}{k-1}}$, hence $np^{\frac{k-2}{2}} \gg n^{1-\frac{k-2}{k-1}} = n^{\frac{1}{k-1}} \to \infty$.

By Lemma 4.7.2, $\Delta^* = \Theta(\max\{q_2, q_{k-1}\}) = o\big(\mathbb{E}[X]\big)$, and Corollary 4.6.5 applies. For $k = 3$ the two endpoints coincide, $q_2 = np^2$, matching Example 4.6.7.

Theorem 4.7.3

Fix $k \geq 3$ and let X be the number of copies of K_k in $G(n,p)$. With high probability,

$$X = \begin{cases} 0, & \text{if } p \ll n^{-\frac{2}{k-1}}, \\ (1+o(1))\binom{n}{k}p^{\binom{k}{2}} \to \infty, & \text{if } p \gg n^{-\frac{2}{k-1}}. \end{cases}$$

In particular $p^* = n^{-\frac{2}{k-1}}$ is a threshold for $K_k \subseteq G(n,p)$.

§4.8. Arbitrary Subgraphs

Question 4.8.1. Given a fixed graph H, what is the threshold for $H \subseteq G(n,p)$?

As in Section 4.7, the property $\{G : H \subseteq G\}$ is monotone increasing, since $H \subseteq G \subseteq G'$ gives $H \subseteq G'$, so Definition 4.5.3 applies.

Write $v(H)$ and $e(H)$ for the numbers of vertices and edges of H, and assume $e(H) \geq 1$. As in Lemma 4.6.6, a copy of H is a subgraph of K_n isomorphic to H; let X_H be the number of copies of H contained in $G(n,p)$.

§4.8.i. The First Moment

To compute $\mathbb{E}[X_H]$ we need two things: how many copies of H there are in K_n, and the probability that a given one is present.

Number of copies. An **automorphism** of H is a permutation α of $V(H)$ with $\{u,w\} \in E(H) \Leftrightarrow \{\alpha(u),\alpha(w)\} \in E(H)$, i.e. a way of mapping H onto itself: it rearranges the vertices of H but leaves the edge set unchanged. They form a group $\mathrm{Aut}(H)$, so $|\mathrm{Aut}(H)|$ is the number of such self-maps. The identity is one of them and there are $v(H)!$ permutations in total, so $1 \leq |\mathrm{Aut}(H)| \leq v(H)!$. For an injective map $\varphi\colon V(H) \to [n]$, write $\varphi(H)$ for the graph on $\varphi(V(H))$ with edges $\{\varphi(u),\varphi(w)\}$, $\{u,w\} \in E(H)$; it is a copy of H, and every copy arises this way (take φ to be an isomorphism $H \to S$).

Lemma 4.8.2

With $v = v(H)$,

$$\#\{\text{copies of } H \text{ in } K_n\} = \frac{n(n-1)\cdots(n-v+1)}{|\mathrm{Aut}(H)|} = \Theta\big(n^v\big).$$

Proof. There are $n(n-1)\cdots(n-v+1)$ injective maps $\varphi\colon V(H) \to [n]$, and $\varphi \mapsto \varphi(H)$ maps them onto the copies. We show that each copy is hit exactly $|\mathrm{Aut}(H)|$ times, by showing

$$\varphi'(H) = \varphi(H) \Leftrightarrow \varphi' = \varphi \circ \alpha \text{ for some } \alpha \in \mathrm{Aut}(H).$$

$\Leftarrow$: α maps $V(H)$ onto itself and $E(H)$ onto itself, so $\varphi\circ\alpha$ has the same image vertices and edges as φ. $\Rightarrow$: if $\varphi'(H) = \varphi(H) =: S$, then φ' and φ are bijections $V(H) \to V(S)$, so $\alpha := \varphi^{-1} \circ \varphi'$ is a permutation of $V(H)$ with $\varphi' = \varphi \circ \alpha$, and

$$\begin{aligned} \{u,w\} \in E(H) &\Leftrightarrow \{\varphi'(u),\varphi'(w)\} \in E(S) \\ &\Leftrightarrow \{\varphi(\alpha(u)),\varphi(\alpha(w))\} \in E(S) \Leftrightarrow \{\alpha(u),\alpha(w)\} \in E(H), \end{aligned}$$

so $\alpha \in \mathrm{Aut}(H)$. Distinct α give distinct $\varphi \circ \alpha$ since φ is injective, so each copy has exactly $|\mathrm{Aut}(H)|$ preimages, which gives the formula. For the order of magnitude, $n(n-1)\cdots(n-v+1) = (1+o(1))\,n^v$ as v is fixed, and $|\mathrm{Aut}(H)| \in [1, v!]$ is a constant. ∎

Probability of one copy. A given copy S is in $G(n,p)$ if and only if each of its $e(H)$ edges is, and these appear independently with probability p, so $\mathbb{P}(S \subseteq G(n,p)) = p^{e(H)}$.

Since $X_H = \sum_S \mathbb{1}_{\{S \subseteq G(n,p)\}}$, Theorem 2.1.2 and Lemma 4.8.2 give, with $v = v(H)$ and $e = e(H)$,

$$\mathbb{E}[X_H] = \#\{\text{copies}\} \cdot p^e = \Theta\big(n^v p^e\big) = \Theta\Big(\big(n^{\frac{v}{e}} p\big)^e\Big) \longrightarrow \begin{cases} 0, & \text{if } p \ll n^{-\frac{v}{e}}, \\ \infty, & \text{if } p \gg n^{-\frac{v}{e}}. \end{cases} \tag{4.5}$$

since $\big(n^{v/e}p\big)^e \to 0$ iff $n^{v/e}p \to 0$ iff $p = o(n^{-v/e})$, and likewise for ∞.

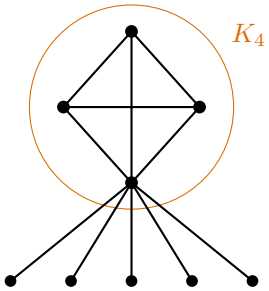
Proposition 4.8.3

If $p \ll n^{-v(H)/e(H)}$, then $\mathbb{P}(X_H \geq 1) \to 0$.

Proof. By Markov's inequality and (4.5), $\mathbb{P}(X_H \geq 1) \leq \mathbb{E}[X_H] \to 0$. ∎

For $H = K_k$ this is the lower half of Theorem 4.7.3, since $v/e = k/\binom{k}{2} = \frac{2}{k-1}$. It is tempting to guess that $n^{-v/e}$ is always the threshold, but the next example shows it is not.

Example 4.8.4. Let H be K_4 with 5 pendant edges attached to one of its vertices, so $v(H) = 9$ and $e(H) = 6 + 5 = 11$.



$v(H) = 9$, $e(H) = 11$, $\dfrac{v(H)}{e(H)} = \dfrac{9}{11}$;

the subgraph K_4 has $\dfrac{v}{e} = \dfrac{4}{6} = \dfrac{2}{3} < \dfrac{9}{11}$

Is $p^* = n^{-9/11}$? Take $p = n^{-3/4}$. Since $\frac{2}{3} < \frac{3}{4} < \frac{9}{11}$,

$$n^{-\frac{9}{11}} \ll p = n^{-\frac{3}{4}} \ll n^{-\frac{2}{3}}.$$

By (4.5), $\mathbb{E}[X_H] \to \infty$. But by Theorem 4.7.3 with $k = 4$, $G(n,p)$ contains no K_4 with high probability, and $H \supseteq K_4$, so it contains no copy of H either. So $n^{-9/11}$ is not the threshold: the many expected copies are concentrated on the rare outcomes that contain a K_4, which then carries a huge number of pendant stars.

The argument in Example 4.8.4 works for any subgraph. If $H' \subseteq H$, every copy of H contains a copy of H', so $\{X_H \geq 1\} \subseteq \{X_{H'} \geq 1\}$, and Proposition 4.8.3 applied to H' gives

$$\mathbb{P}(X_H \geq 1) \leq \mathbb{P}(X_{H'} \geq 1) \to 0 \qquad \text{whenever } p \ll n^{-v(H')/e(H')}.$$

The best bound comes from the densest subgraph.

Definition 4.8.5 (Maximum density)**.** For a graph H with $e(H) \geq 1$, let

$$m(H) := \max_{\substack{H' \subseteq H \\ e(H') \geq 1}} \frac{e(H')}{v(H')}.$$

Proposition 4.8.6
If $p \ll n^{-1/m(H)}$, then $\mathbb{P}\big(H \subseteq G(n,p)\big) \to 0$.

Proof. Take $H' \subseteq H$ attaining the maximum in Definition 4.8.5, so $n^{-1/m(H)} = n^{-v(H')/e(H')}$. Then $p \ll n^{-v(H')/e(H')}$, and the display above gives $\mathbb{P}(X_H \geq 1) \leq \mathbb{P}(X_{H'} \geq 1) \to 0$. ∎

Since $H' = H$ is allowed, $m(H) \geq e(H)/v(H)$, so $n^{-1/m(H)} \geq n^{-v(H)/e(H)}$ and Proposition 4.8.6 is at least as strong as Proposition 4.8.3. In Example 4.8.4, $m(H) = \frac{6}{4} = \frac{3}{2}$, attained by K_4, and $n^{-1/m(H)} = n^{-2/3}$.

Question 4.8.7. If $p \gg n^{-1/m(H)}$, is it true that $\mathbb{P}\big(H \subseteq G(n,p)\big) \to 1$?

§4.8.ii. Balanced Graphs

For some graphs the densest subgraph is H itself, and then the guess $n^{-v(H)/e(H)}$ from Proposition 4.8.3 already equals $n^{-1/m(H)}$.

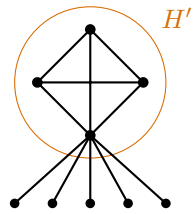
Definition 4.8.8 (Balanced)**.** A graph H is **balanced** if $m(H) = \frac{e(H)}{v(H)}$, i.e. $\frac{e(H')}{v(H')} \leq \frac{e(H)}{v(H)}$ for every $H' \subseteq H$ with $e(H') \geq 1$. It is **strictly balanced** if moreover equality holds only for $H' = H$.

For balanced H, $n^{-1/m(H)} = n^{-v(H)/e(H)}$, so the bound of Proposition 4.8.3 is already the best one available from first moments. Every clique K_k is strictly balanced: a subgraph H' on $j \leq k$ vertices has $e(H') \leq \binom{j}{2}$, so

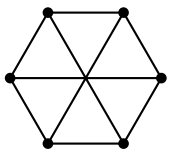
$$\frac{e(H')}{v(H')} \leq \frac{\binom{j}{2}}{j} = \frac{j-1}{2} \leq \frac{k-1}{2} = \frac{e(K_k)}{v(K_k)},$$

with equality throughout only if $j = k$ and $e(H') = \binom{k}{2}$, i.e. $H' = K_k$.

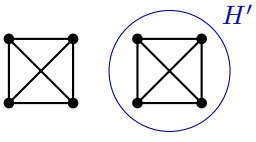
Example 4.8.9. Four graphs, each labelled with its density $\frac{e}{v}$ and, where relevant, a densest proper subgraph H':



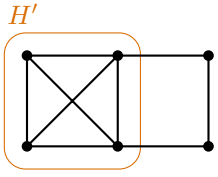
(a) not balanced
$\frac{11}{9} < \frac{6}{4}$



(b) $K_{3,3}$: strictly balanced
$\frac{9}{6}$



(c) $2K_4$: balanced, not strictly
$\frac{12}{8} = \frac{6}{4}$



(d) balanced, not strictly
$\frac{9}{6} = \frac{6}{4}$

(a) The graph of Example 4.8.4: $H' = K_4$ has $\frac{6}{4} > \frac{11}{9}$, so it is not balanced.

(b) $K_{3,3}$ (a hexagon with its three long diagonals) has $\frac{e}{v} = \frac{9}{6}$. A proper subgraph on all 6 vertices has at most 8 edges, so density $\le \frac{8}{6} < \frac{9}{6}$. A subgraph on a vertices of one side and b of the other, with $a + b \le 5$ and $a, b \le 3$, has at most ab edges, and $\frac{ab}{a+b} \le \frac{6}{5} < \frac{9}{6}$ (the maximum is at $\{a, b\} = \{2, 3\}$). So it is strictly balanced.

(c) Two disjoint copies of K_4 have $\frac{12}{8} = \frac{3}{2}$. A subgraph H' splits into parts H'_1, H'_2 in the two copies, each of density $\le \frac{3}{2}$ (as K_4 is balanced), so $\frac{e(H')}{v(H')} = \frac{e(H'_1)+e(H'_2)}{v(H'_1)+v(H'_2)} \le \frac{3}{2}$. Each K_4 on its own attains $\frac{6}{4} = \frac{3}{2}$: balanced, but not strictly.

(d) K_4 plus two vertices joined to each other and to two vertices of the K_4, as drawn: $\frac{e}{v} = \frac{9}{6} = \frac{6}{4}$, equal to the density of the K_4 inside it. A subgraph on $j \le 4$ vertices has density $\le \frac{j-1}{2} \le \frac{3}{2}$; one on 5 vertices misses a vertex of degree ≥ 2, so has ≤ 7 edges and density $\le \frac{7}{5}$; one on 6 vertices has ≤ 9 edges. Balanced, but not strictly.

§4.8.iii. The Threshold for Containing H

The answer to Question 4.8.7 is yes, for every H.

Theorem 4.8.10

Let H be a graph with $e(H) \ge 1$. If $p \gg n^{-1/m(H)}$, then with high probability

$$X_H = (1 + o(1))\,\mathbb{E}[X_H] = \Theta\big(n^{v(H)}p^{e(H)}\big) \to \infty;$$

in particular $\mathbb{P}\big(H \subseteq G(n,p)\big) \to 1$. Together with Proposition 4.8.6, $p^* = n^{-1/m(H)}$ is a threshold for $H \subseteq G(n,p)$. For balanced H this is $n^{-v(H)/e(H)}$, and for $H = K_k$ it recovers Theorem 4.7.3.

Proof. Write $v = v(H)$, $e = e(H)$, $m = m(H)$. The events $E_S = \{S \subseteq G(n,p)\}$, S ranging over copies of H, are symmetric by Lemma 4.6.6, so by Corollary 4.6.5 it suffices to show $\mathbb{E}[X_H] \to \infty$ and $\Delta^* = o\big(\mathbb{E}[X_H]\big)$.

The key inequality. If $J \subseteq H$ has $e(J) \ge 1$, then $\frac{e(J)}{v(J)} \le m$ by Definition 4.8.5, so, as $p \le 1$,

$$n^{v(J)}p^{e(J)} = \big(n\,p^{e(J)/v(J)}\big)^{v(J)} \ge \big(n\,p^m\big)^{v(J)}. \tag{4.6}$$

Here $n\,p^m \to \infty$, since $p \gg n^{-1/m}$. Taking $J = H$ and using (4.5), $\mathbb{E}[X_H] = \Theta(n^v p^e) \to \infty$.

Now fix a copy S, let $T \ne S$ be another copy, and let J be the graph of common edges, $E(J) = E(S) \cap E(T)$, on the endpoints of those edges.

- *Which T.* If $e(J) = 0$, then E_S, E_T are determined by disjoint sets of pairs and are independent. If $e(J) \ge 1$, then

  $$\mathbb{P}(E_S \cap E_T) = p^{2e-e(J)} > p^{2e} = \mathbb{P}(E_S)\,\mathbb{P}(E_T)$$

  for $0 < p < 1$. So $T \sim S$ iff $e(J) \ge 1$, and then $i := |V(S) \cap V(T)| \ge 2$, as a common edge has both ends in $V(S) \cap V(T)$.

- *Conditional probability.* By the definition of conditional probability and the previous bullet,

  $$\mathbb{P}(E_T \mid E_S) = \frac{\mathbb{P}(E_S \cap E_T)}{\mathbb{P}(E_S)} = \frac{p^{2e-e(J)}}{p^e} = p^{e-e(J)}.$$

Since $J \subseteq S \cong H$, J is isomorphic to a subgraph of H with $e(J) \geq 1$, and $V(J) \subseteq V(S) \cap V(T)$ gives $v(J) \leq i$. Hence, using $n^i \geq n^{v(J)}$, then (4.6) together with $(n\,p^m)^{v(J)} \geq n\,p^m$ (true for large n, as $n\,p^m \geq 1$ and $v(J) \geq 1$),

$$\mathbb{P}(E_T \mid E_S) = p^e \cdot \frac{1}{p^{e(J)}} = n^v p^e \cdot \frac{1}{n^{v-i}} \cdot \frac{1}{n^i\, p^{e(J)}} \leq n^v p^e \cdot \frac{1}{n^{v-i}} \cdot \frac{1}{n^{v(J)} p^{e(J)}} \leq n^v p^e \cdot \frac{1}{n^{v-i}} \cdot \frac{1}{n\,p^m}.$$

- *Number of T with $|V(S) \cap V(T)| = i$.* Choose the i shared vertices among the v vertices of S, the other $v - i$ vertices among $[n]$, and a copy of H on the resulting v-set (at most $v!$ of them):
$$\#\big\{T : |V(S) \cap V(T)| = i\big\} \leq \binom{v}{i} \cdot n^{v-i} \cdot v! = O\big(n^{v-i}\big).$$

Grouping the $T \sim S$ by i, bounding each $\mathbb{P}(E_T \mid E_S)$ by the second bullet and the number of terms by the third,

$$\begin{aligned}
\Delta^* &= \sum_{i=2}^{v} \sum_{\substack{T \sim S \\ |V(S)\cap V(T)|=i}} \mathbb{P}(E_T \mid E_S) \leq \sum_{i=2}^{v} \#\big\{T : |V(S) \cap V(T)| = i\big\} \cdot \max_{\substack{T \sim S \\ |V(S)\cap V(T)|=i}} \mathbb{P}(E_T \mid E_S) \\
&\leq \sum_{i=2}^{v} O\big(n^{v-i}\big) \cdot n^v p^e \cdot \frac{1}{n^{v-i}} \cdot \frac{1}{n\,p^m} = \sum_{i=2}^{v} O\Big(\frac{n^v p^e}{n\,p^m}\Big) = O\Big(\frac{n^v p^e}{n\,p^m}\Big) \\
&= o\big(n^v p^e\big) = o\big(\mathbb{E}[X_H]\big),
\end{aligned}$$

where the sum has only $v - 1$ terms, $n\,p^m \to \infty$, and $\mathbb{E}[X_H] = \Theta(n^v p^e)$ by (4.5). ■